CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-501: Introduction to DBMS

**Session:** December 2013 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Define Database and Database Management System.**

**Definition:** A Database is an organized collection of logically related data. A DBMS is the software interface that controls data storage, retrieval, and access permissions.

**Q2 (3M) What is Data Redundancy?**

**Definition:** Unnecessary duplication of identical data items across multiple database files, causing wasted storage and risk of inconsistency.

**Q3 (3M) Define Primary Key.**

**Definition:** A unique attribute chosen to identify tuples in a relation. Cannot accept duplicate values or NULL.

**Q4 (3M) What is Relational Data Model?**

**Definition:** A data model representing data in terms of mathematical relations (2D tables) composed of rows (tuples) and columns (attributes).

**Q5 (3M) What is a Transaction?**

**Definition:** A logical unit of database processing that includes one or more database access operations, executed as an indivisible atomic action.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) Explain the Three-Level ANSI/SPARC Database Architecture.**

**Architecture:** External, Conceptual, and Internal levels providing logical and physical data independence.

**Q7 (7.5M) Explain Hashing and all collision resolution techniques in detail.**

**Collision Resolution:** Detailed explanation of Linear Probing, Quadratic Probing, Chaining with overflow buckets, and Extendible Hashing.

**Q8 (7.5M) What do you understand by Transaction System? Explain ACID properties.**

**Transaction Management:** Atomicity, Consistency, Isolation, and Durability guaranteed through concurrency control and recovery.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M) What is Normalization? Explain in detail 1st, 2nd, and 3rd Normal Form with importance.**

**Complete Normalization:** 1NF (atomic values), 2NF (elimination of partial dependencies), and 3NF (elimination of transitive dependencies) with complete table decomposition walkthroughs.

**Q10 (15M)** **Explain Relational Algebra Operations with expressions and examples.**

**Relational Algebra:** Complete coverage of Selection, Projection, Cartesian Product, Union, Difference, and Natural Join.

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**Q11 (15M)** **Explain ER Diagram concepts: Entity Sets, Weak Entities, Attributes, Relationships, and Cardinality. Draw ER diagram for Banking System.**

**ER Model:** Detailed conceptual design for Customer, Account, Branch, and Loan entities with conversion to relational tables.

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**Q12 (15M)** **Explain File Organization Techniques: ISAM, B-Tree, and B+ Tree with diagrams.**

**File Organization:** Detailed structural differences and access speeds between Indexed Sequential, B-Tree, and B+ Tree structures.

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**Q13 (15M)** **Explain Concurrency Control Protocols: Two-Phase Locking (2PL) and Timestamp Ordering.**

**Concurrency Control:** Strict 2PL, Rigorous 2PL, Thomas Write Rule, and timestamp-based conflict resolution.

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## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Differentiate between Sequential and Random Access in files.**

**Differences:** Sequential access reads records consecutively in order from the beginning of the file (O(N)). Random (direct) access jumps immediately to any record block by its calculated address or key index (O(1)).

**Q2 (3M) What do you mean by Relational Algebra?**

**Definition:** A procedural mathematical query language consisting of a collection of operations that take one or two relations as input and produce a new relation as output.

**Q3 (3M) What are the various Relational Algebra operators?**

**Operators:** Selection (\sigma), Projection (\pi), Union (\cup), Set Difference (-), Cartesian Product (\times), Rename (\rho), Join (\bowtie), and Division (\div).

**Q4 (3M) Define Primary Key and Foreign Key.**

**Definitions:** Primary Key uniquely identifies tuples in a table. Foreign Key references the primary key in another table, enforcing referential integrity.

**Q5 (3M) What is Data Inconsistency?**

**Definition:** Different duplicate copies of the same data item holding contradictory, out-of-sync values across multiple files.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) Explain DBMS Architecture in detail.**

**Architecture:** Complete ANSI/SPARC 3-schema architecture with internal, conceptual, and external levels.

**Q7 (7.5M) Differentiate between Association and Aggregation with suitable examples.**

**Differences:** Association represents a general structural relationship between entities (e.g., Student *enrolled in* Course). Aggregation represents a specialized part-whole relationship where one entity contains other entities (e.g., University *aggregates* Departments).

**Q8 (7.5M) Explain B-Tree and B+ Tree File Organization.**

**Explanation:** Self-balancing tree indexes where B+ Trees store all data pointers exclusively in linked leaf nodes, maximizing branching factor and range query performance.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M) Given tables: DEPT(Deptno, Dname, Loc), PROJECT(Projno, Deptno, Projname). Write SQL queries for project allocations and department summaries.**

**SQL Solutions:** Complete SQL join queries, aggregate counts per department, and filter criteria.

**Q10 (15M)** **What do you mean by Normalization? Why do we normalize? Explain 1NF, 2NF, 3NF, and BCNF.**

**Comprehensive Normalization Guide:** Full mathematical criteria, functional dependencies, partial and transitive dependency elimination, and BCNF decompositions.

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**Q11 (15M)** **Explain Concurrency Control: Locking Protocols (2PL) and Serializability.**

**Concurrency Control:** In-depth coverage of Conflict Serializability, Precedence Graphs, Shared/Exclusive locks, and 2-Phase Locking.

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**Q12 (15M)** **Draw an ER Diagram for a Hospital Management Enterprise.**

**ER Model:** Complete diagram for Doctors, Patients, Beds, and Medical Records with mapping to relational schema.

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**Q13 (15M)** **Explain Log-Based Recovery and Checkpoint Mechanisms in DBMS.**

**Database Recovery:** Deferred update, immediate update, write-ahead logging (WAL), and checkpoint-based fast recovery.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-501: Introduction to DBMS

**Session:** December 2015 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Define Database Management System (DBMS).**

**Definition:** A DBMS is software designed to manage, store, query, and maintain structured data securely and reliably, guaranteeing data independence and concurrent multi-user access.

**Q2 (3M) Differentiate between Schema and Instance.**

**Comparison:** A Schema is the overall structural blueprint designed once, while an Instance is the actual operational collection of data tuples present at any given moment.

**Q3 (3M) What is Data Independence?**

**Definition:** The capacity to change the data schema at one architectural level without altering schemas at higher levels.

**Q4 (3M) What do you understand by Hashing? Explain with example.**

**Definition:** Direct access technique mapping search keys to physical bucket addresses via hash function h(k) = k \pmod m for O(1) lookups.

**Q5 (3M) What are Nested SQL Queries? Explain with example.**

**Definition:** A query enclosed inside another SQL query, passed via `WHERE` or `HAVING` clauses using operators like `IN`, `EXISTS`, `ALL`, `ANY`.

```
SELECT EmpName FROM Employee
WHERE Salary > (SELECT AVG(Salary) FROM Employee);
```

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) Explain the Three-Level Architecture of DBMS and its primary objectives.**

**Architecture:** View Level → Conceptual Level → Internal Storage Level. Primary objective is achieving program-data independence and providing customized views to multiple users.

**Q7 (7.5M) Explain BCNF (Boyce-Codd Normal Form) and compare with 3NF.**

**Explanation:** BCNF requires every determinant in a non-trivial functional dependency to be a super key. Stricter than 3NF, eliminating all redundancy based on functional dependencies.

**Q8 (7.5M) Explain Concurrency Control and ACID properties.**

**Explanation:** Mechanisms ensuring Atomicity, Consistency, Isolation, and Durability when multiple transactions access the database simultaneously.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain ER Modeling: Entities, Attributes, Relationships, and Cardinality Constraints. Draw complete ER diagram for College Administration.**

**Full Analysis:** Complete conceptual ER model with Student, Faculty, Department, and Course entity sets, relationships, and cardinalities.

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**Q10 (15M)** **Define Normalization. Why do we need to normalize? Explain 1NF, 2NF, and 3NF with examples.**

**Full Normalization Process:** Comprehensive breakdown of insertion, deletion, and update anomalies, followed by step-by-step normalization through 1NF, 2NF, and 3NF.

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**Q11 (15M)** **Explain Relational Algebra Operations with expressions and SQL equivalents.**

**Operations:** Selection (\sigma), Projection (\pi), Cartesian Product (\times), Union (\cup), Set Difference (-), and Natural Join (\bowtie).

---

**Q12 (15M)** **Explain Indexing and Hashing: Primary vs Secondary Index, Dense vs Sparse Index.**

**Indexing Schemes:** Clustered/Primary indexing on ordered files vs Non-clustered/Secondary indexing on unordered candidate keys, dense vs sparse index trade-offs.

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**Q13 (15M)** **Explain Log-Based Recovery and Checkpoints.**

**Recovery:** Detailed explanation of Undo/Redo protocols, Write-Ahead Logging (WAL), system checkpoints, and crash restart recovery algorithms.

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# BCA-501: Introduction to DBMS

**Session:** December 2016 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M)** **Differentiate between Database System and File System.**

**Differences:** DBMS offers centralized control, eliminated data inconsistency, program-data independence, and built-in transaction recovery, which file systems lack.

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**Q2 (3M)** **Explain Foreign Key with ON DELETE CASCADE.**

**Explanation:** A foreign key links a child record to a parent record. With `ON DELETE CASCADE`, deleting a parent row automatically cascades and deletes all corresponding child rows, maintaining referential integrity.

---

**Q3 (3M)** **What is a Weak Entity Set? How is it identified?**

**Definition:** An entity set lacking sufficient attributes to form a primary key. Identified using a *discriminator (partial key)* combined with the primary key of its identifying strong entity set.

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**Q4 (3M)** **Differentiate between Natural Join and Theta Join.**

**Differences:** Theta Join (R \bowtie_\theta S) combines rows based on any arbitrary condition. Natural Join (R \bowtie S) automatically equates attributes having identical names and eliminates duplicate columns.

---

**Q5 (3M)** **Explain COMMIT and ROLLBACK commands in SQL.**

**TCL Commands:**

- `COMMIT` : Permanently saves all transaction modifications to disk.
- `ROLLBACK` : Reverts all uncommitted changes back to the start of the transaction or savepoint.

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## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Explain BCNF (Boyce-Codd Normal Form) with example.**

**Explanation:** A relation is in BCNF if for every non-trivial functional dependency X \to Y, X is a Super Key. Eliminates anomalies arising from overlapping candidate keys.

---

**Q7 (7.5M)** **Compare Dense Index and Sparse Index in database indexing.**

**Comparison:**

- **Dense Index:** Contains an index record for *every single search-key value* in the data file. Faster search, larger index size.
- **Sparse Index:** Contains index records for only some search-key values (typically one per disk block). Consumes less space, but requires scanning within blocks.

---

**Q8 (7.5M)** **Explain Deadlock Detection and Prevention methods in DBMS.**

**Methods:** Deadlock prevention uses timestamp ordering (Wait-Die and Wound-Wait schemes). Deadlock detection periodically constructs a Wait-For Graph (WFG) and aborts a cycle-causing transaction.

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Draw a detailed ER Diagram for a Banking Enterprise.**

**Entities:** `CUSTOMER`, `ACCOUNT`, `BRANCH`, `LOAN`.
**Relationships:** Customer deposits in Account, Customer borrows Loan, Branch maintains Accounts and Loans.

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**Q10 (15M)** **Explain Relational Algebra operators: Selection, Projection, Cartesian Product, Join, and Division.**

**Relational Algebra:** Complete explanation of fundamental algebraic query operators with mathematical notations and practical examples.

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**Q11 (15M)** **Explain Nested Subqueries and Correlated Subqueries in SQL with examples.**

**Subqueries:**

- **Nested Subquery:** Inner query executes once, and its result is used by outer query (e.g., `WHERE Salary > (SELECT AVG(Salary) FROM Emp)`).
- **Correlated Subquery:** Inner query references columns of the outer query and re-executes for each row evaluated by outer query.

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**Q12 (15M)** **Explain ACID Properties and Serializability in transaction management.**

**ACID & Serializability:** In-depth coverage of Atomicity, Consistency, Isolation, Durability, conflict serializability, and conflict equivalence of schedules.

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**Q13 (15M)** **Explain ISAM and B+ Tree File Organizations with comparative analysis.**

**File Organization:** Detailed architectural comparison between static ISAM hierarchy and dynamic self-balancing B+ Trees.

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## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

### Q1 (3M) Define Metadata and Data Dictionary.

**Definition: Metadata** is "data about data" describing database structures, constraints, and data types. A **Data Dictionary** is the repository that permanently stores this metadata.

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### Q2 (3M) Define Tuple, Relation, Attribute, and Domain.

**Definitions:**

- **Relation:** A two-dimensional table of records.
- **Tuple:** A single row in a relation representing an individual entity instance.
- **Attribute:** A named column describing a specific property.
- **Domain:** The set of allowable atomic values for an attribute.

---

### Q3 (3M) What are Cardinality and Participation Constraints in ER Modeling?

**Definitions:**

- **Cardinality Ratio:** Specifies the maximum number of relationship instances that an entity can participate in (1:1, 1:N, M:N).
- **Participation Constraint:** Specifies whether existence of an entity depends on its being related to another entity: *Total* (double line) or *Partial* (single line).

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### Q4 (3M) Differentiate between DDL and DML.

**DDL:** Defines database schemas and tables ( `CREATE` , `ALTER` , `DROP` ).
**DML:** Manipulates and retrieves data instances ( `INSERT` , `UPDATE` , `DELETE` , `SELECT` ).

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### Q5 (3M) What is a View in SQL?

**Definition:** A **View** is a virtual table whose contents are defined by an underlying SQL query, providing security and abstraction without replicating stored data.

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## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

### Q6 (7.5M) Explain 1NF and 2NF with examples.

**1NF:** Eliminates repeating groups and non-atomic values.
**2NF:** In 1NF and contains no partial functional dependencies (all non-key attributes depend fully on the primary key).

---

**Q7 (7.5M)** **What is Query Optimization? Explain heuristic optimization.**

**Explanation:** Query optimization is the process of choosing the most efficient execution plan for evaluating a query. Heuristic optimization applies algebraic rewrite rules:

- Perform `Selection (σ)` as early as possible (reduces intermediate relation sizes).
- Perform `Projection (π)` early to eliminate unneeded attributes.
- Execute the most restrictive selections before performing Joins.

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**Q8 (7.5M)** **Explain Log-Based Recovery with WAL (Write-Ahead Logging) and Checkpoints.**

**Log-Based Recovery:** Records all database modifications in an append-only log file before writing changes to disk (Write-Ahead Logging). During crash recovery, log records are used to UNDO incomplete transactions and REDO committed ones. Checkpoints reduce recovery overhead by flushing dirty buffers periodically.

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain the Three-Level ANSI/SPARC Database Architecture with detailed diagram.**

**Architecture:** External Schema (Views) → Conceptual Schema (Logical tables) → Internal Schema (Physical disk blocks and access paths). Ensures physical and logical data independence.

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**Q10 (15M)** **Given tables VENDOR(VCode, Name, Address, Phone) and PRODUCT(PCode, PName, Price, VCode). Write SQL queries for:**
**(i) Display all vendors from 'Delhi'.**
**(ii) Display products with price greater than 500.**
**(iii) Display vendor name who supplies product 'Laptop'.**
**(iv) Count total products supplied by each vendor.**

**SQL Solutions:**

```
-- (i) Vendors from Delhi
SELECT * FROM VENDOR WHERE Address LIKE '%Delhi%';

-- (ii) Products with price > 500
SELECT PCode, PName, Price FROM PRODUCT WHERE Price > 500;

-- (iii) Vendor supplying 'Laptop'
SELECT V.Name
FROM VENDOR V
JOIN PRODUCT P ON V.VCode = P.VCode
WHERE P.PName = 'Laptop';

-- (iv) Count total products per vendor
SELECT V.VCode, V.Name, COUNT(P.PCode) AS TotalProducts
FROM VENDOR V
LEFT JOIN PRODUCT P ON V.VCode = P.VCode
GROUP BY V.VCode, V.Name;
```

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**Q11 (15M)** **Explain Relational Algebra vs Relational Calculus.**

**Comparison:** Relational Algebra is procedural (specifies operations step-by-step), while Relational Calculus (Tuple & Domain) is non-procedural/declarative (expresses what is required via mathematical predicate logic). Codd's Theorem proves they are equivalent in expressive power.

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**Q12 (15M)** **Explain BCNF (Boyce-Codd Normal Form) and compare with 3NF.**

**BCNF:** Stricter form of 3NF where every determinant in a non-trivial functional dependency must be a super key. Eliminates anomalies that 3NF allows when there are overlapping composite candidate keys.

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**Q13 (15M)** **Given tables: PLAY(PlayCode, PlayName, DirNum), DIRECTOR(DirNum, DirName, DirDOB). Write SQL queries to find director names and plays directed.**

**SQL Solutions:**

```
-- Director of play 'Hamlet'
SELECT D.DirName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
WHERE P.PlayName = 'Hamlet';

-- All plays directed by each director
SELECT D.DirName, P.PlayName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
ORDER BY D.DirName;
```

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## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M)** **Define Super Keys, Candidate Keys, and Primary Keys.**

**Definitions:**

- **Super Key:** A set of one or more attributes that uniquely identifies each tuple in a relation (may contain redundant attributes).
- **Candidate Key:** A minimal super key having no extraneous/redundant attributes.
- **Primary Key:** The specific candidate key chosen by the database designer as the principal tuple identifier. Cannot contain `NULL`.

---

**Q2 (3M)** **Explain the term "Data Independence".**

**Definition: Data Independence** is the capacity to change the schema at one level of the database system without having to alter the schema at the next higher level. It includes Physical Data Independence (modifying internal storage without affecting conceptual schema) and Logical Data Independence (modifying conceptual schema without altering external views).

---

**Q3 (3M)** **What is Hashing? Explain.**

**Definition: Hashing** is a direct addressing technique where a mathematical algorithm (hash function) calculates the exact physical block address of a record from its search key value, enabling O(1) constant-time retrieval.

---

**Q4 (3M)** **What do you mean by Relational Database? Describe.**

**Definition:** A **Relational Database** is a collection of data items organized into formal tables (relations) of rows and columns based on first-order predicate logic, where relations are linked together via primary and foreign key constraints.

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**Q5 (3M)** **Briefly explain the terms Security and Authorization.**

**Definitions:**

- **Database Security:** Protection of the database against unauthorized access, malicious attacks, corruption, or destruction.
- **Authorization:** The mechanism that defines and enforces what actions (e.g., SELECT, INSERT, UPDATE, DROP) a validated user or role is permitted to perform on specific database objects.

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## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **What is a Referential Integrity Constraint? Why do we need it? Explain with a suitable example.**

**Explanation:** Referential integrity requires that a foreign key value in a referencing table must match an existing primary key value in the referenced table, or be `NULL`.
**Need:** Prevents orphan records and inconsistent states.
**Example:**

```
-- Parent Table
CREATE TABLE Department (DeptNo INT PRIMARY KEY, DName VARCHAR(30));

-- Child Table with Foreign Key
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    DeptNo INT,
    CONSTRAINT fk_emp_dept FOREIGN KEY (DeptNo) REFERENCES Department(DeptNo)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

If department 10 is deleted, `ON DELETE CASCADE` automatically removes all employees assigned to department 10, preventing dangling references.

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**Q7 (7.5M)** **List four significant differences between a File-Processing System and a DBMS.**

**Differences:**

| Aspect | File-Processing System | DBMS |
|---|---|---|
| Data Redundancy | High redundancy across duplicate files. | Controlled and normalized. |
| Data Independence | None (tight coupling with code). | Logical & Physical Data Independence. |
| Concurrency | Prone to race conditions and lost updates. | ACID compliant concurrency control. |
| Crash Recovery | Manual restore; inconsistent state. | Automated log-based rollback and redo. |

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**Q8 (7.5M)** **Explain Third Normal Form (3NF) with example.**

**Definition:** A relation is in **3NF** if it is in 2NF and has no transitive functional dependencies for non-prime attributes. Formally, for every non-trivial FD X \to Y, either X is a super key or Y is a prime attribute.
**Example:** `Student(RollNo, Name, DeptID, DeptHead)`.
RollNo \to DeptID and DeptID \to DeptHead (Transitive!).
Decompose into: `Student(RollNo, Name, DeptID)` and `Department(DeptID, DeptHead)`.

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain the different locking techniques with examples for concurrency control.**

**Locking Techniques:**

1. **Shared Lock (S-Lock / Read Lock):** Multiple transactions can simultaneously hold shared locks to read data without conflict.
2. **Exclusive Lock (X-Lock / Write Lock):** Only one transaction can hold an exclusive lock to read and write. No other transaction can hold any lock on that item.
3. **Two-Phase Locking (2PL):** Growing Phase (acquiring locks) followed by Shrinking Phase (releasing locks). Guarantees conflict serializability.
4. **Strict 2PL:** Exclusive locks held until transaction commits, eliminating cascading rollbacks.

---

**Q10 (15M)** **Explain Indexed Sequential, B-Tree, and B+ Tree file organizations with examples.**

**Comparison of File Organizations:**

- **Indexed Sequential (ISAM):** Records stored in sequential key order with a multilevel static index. Good for sequential and direct access, but degrades with many overflow records.
- **B-Tree:** Self-balancing dynamic multi-way search tree. Keys and data record pointers reside in both internal nodes and leaf nodes.
- **B+ Tree:** Modern standard for relational DBMS storage engines. Internal nodes store only search keys to guide navigation, while **all actual data records/pointers reside in leaf nodes** connected via a doubly linked list, enabling superior range scans.

---

**Q11 (15M)** **Explain Super Class, Sub Class, Inheritance, and Generalization with examples.**

**Enhanced ER (EER) Concepts:**

- **Superclass:** A generic entity set that includes distinct subgroups (e.g., `PERSON`).
- **Subclass:** A specialized sub-entity set that inherits all attributes and relationships from its superclass, while adding specialized attributes (e.g., `STUDENT` with GPA, `TEACHER` with Rank).
- **Attribute Inheritance:** Subclass entities automatically inherit the primary key and all general attributes of their superclass.
- **Generalization:** Bottom-up synthesis where lower-level entity sets are combined into a generalized superclass entity.

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**Q12 (15M)** **Write short notes on: (i) Strong and Weak Entity, (ii) Different Levels of Data Abstraction, (iii) Database Recovery Techniques.**

**(i) Strong vs Weak Entity:** A Strong entity has a primary key of its own. A Weak entity does not have a primary key and depends on the existence of an identifying strong entity set (identified by discriminator + strong entity PK).
**(ii) Data Abstraction Levels:** Physical level (low-level block storage), Logical level (community tables and relations), and View level (user-specific abstractions).
**(iii) Database Recovery Techniques:** Log-based recovery (Write-Ahead Logging), Deferred/Immediate update schemes, and Checkpoints to restore consistency following system failures.

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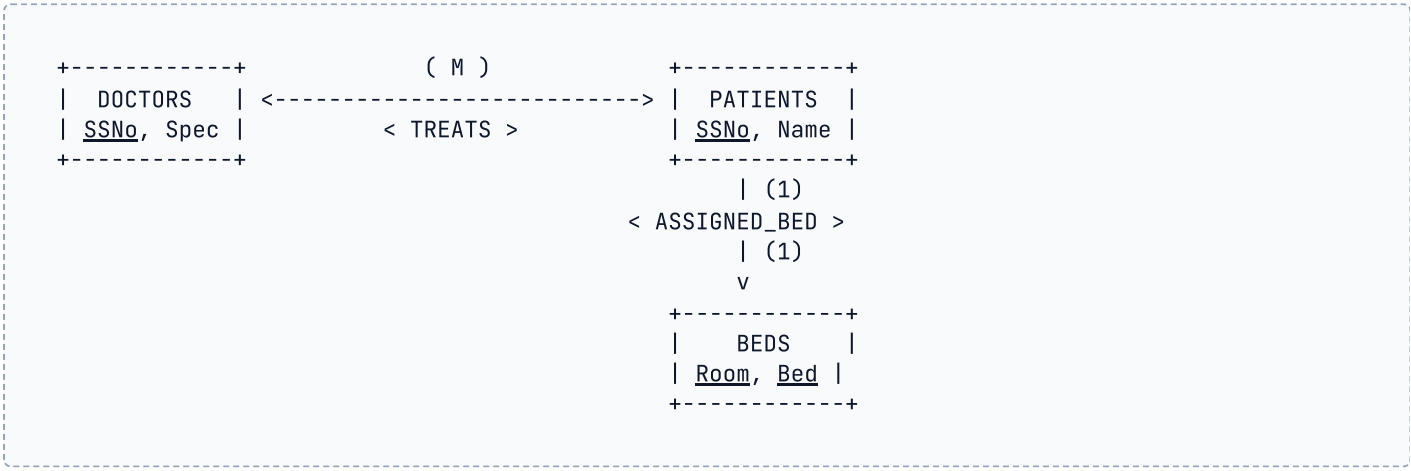
**Q13 (15M)** **A hospital maintains data about entities: PATIENTS, DOCTORS, BEDS, ACCOUNTS. Construct an E-R diagram specifying keys, mapping cardinalities, and participation constraints. Show table mapping.**

**Entities & Attributes:**

- `PATIENTS(SSNo, LastName, FirstName, HomePhone, Sex, DOB, Age, Street, City, State, Zip)`
- `DOCTORS(SSNo, LastName, FirstName, OfficePhone, Pager, Specialty)`
- `BEDS(RoomNumber, BedNumber, Type, Status, PricePerHour)`
- `ACCOUNTS(AccountNo, DateIn, DateOut, TotalAmount, Status)`

**Relationships:**

- `TREATS` (M:N between DOCTORS and PATIENTS)
- `ASSIGNED_BED` (1:1 between PATIENTS and BEDS)
- `HAS_ACCOUNT` (1:1 between PATIENTS and ACCOUNTS)



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# BCA-501: Introduction to DBMS

**Session:** November 2019 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M)** **Name the three levels of Data Abstraction.**

**Three Levels:**

1. **Physical Level (Lowest):** Describes how data is stored on disk blocks.
2. **Logical / Conceptual Level (Middle):** Describes what data is stored and relationships among data.
3. **View / External Level (Highest):** Describes customized views for specific user groups.

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**Q2 (3M)** **Differentiate between Entity Integrity and Referential Integrity.**

**Differences:**

- **Entity Integrity:** States that no primary key attribute value can be `NULL`, ensuring every tuple can be uniquely identified.
- **Referential Integrity:** States that a foreign key value must either match an existing primary key value in the parent relation or be `NULL`.

---

**Q3 (3M)** **Differentiate between Candidate Key and Primary Key.**

**Explanation:** A relation may possess multiple candidate keys (minimal super keys). The database designer selects one candidate key as the **Primary Key**, while all remaining candidate keys are designated as **Alternate Keys**.

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**Q4 (3M)** **List any four disadvantages of File Processing Systems.**

**Disadvantages:** Data redundancy and inconsistency, lack of data independence, concurrent access anomalies, and poor security enforcement.

---

**Q5 (3M)** **Distinguish between Functional Dependency and Multivalued Dependency.**

**Comparison:**

- **Functional Dependency (X → Y):** For any two tuples with identical X values, their Y values must be identical (single-valued dependency). Basis of 2NF, 3NF, BCNF.
- **Multivalued Dependency (X ↠ Y):** The presence of (x, y_1, z_1) and (x, y_2, z_2) implies the presence of (x, y_1, z_2) and (x, y_2, z_1), representing independent multi-valued facts. Basis of 4NF.

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## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Explain Second Normal Form (2NF) and Third Normal Form (3NF) with examples.**

**2NF:** In 1NF and no partial dependencies (every non-prime attribute depends on the whole candidate key).
**3NF:** In 2NF and no transitive dependencies (for every non-trivial FD X \to Y, X is a super key or Y is prime).

---

**Q7 (7.5M)** **Compare B-Tree and B+ Tree Index Structures.**

**Comparison:**

| Feature | B-Tree | B+ Tree |
|---|---|---|
| Data Storage | Keys and data records/pointers stored in both internal and leaf nodes. | Data pointers stored **only in leaf nodes**; internal nodes store only routing keys. |
| Leaf Node Linking | Leaf nodes are not linked. | Leaf nodes are connected via a **doubly linked list**. |
| Range Queries | Slow (requires in-order tree traversal). | Extremely fast (scan linked list across leaf nodes). |
| Node Capacity | Fewer keys per node due to data pointers. | High fan-out, fewer levels, reduced disk I/O. |

---

**Q8 (7.5M)** **Explain Conflict Serializability and Testing for Serializability using Precedence Graph.**

**Precedence Graph Method:**

1. Create a node for every committed transaction T_i.
2. Draw a directed edge T_i \to T_j if T_i performs a conflicting operation before T_j on the same data item (Write-Read, Read-Write, or Write-Write).
3. If the graph contains **NO cycle**, the schedule is conflict serializable. Topological sort gives the equivalent serial schedule.

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain the Three-Tier Client/Server Architecture of DBMS with Data Independence.**

**Three-Tier Architecture:**

- **Presentation Tier (User Interface):** Web browser or desktop client GUI (React, HTML).
- **Application Tier (Business Logic):** Web/Application server executing business validation and query processing (Java, Node.js, Python).
- **Database Tier (Data Storage):** Enterprise RDBMS managing database engine, query processing, transactions, and tables.

Provides enhanced security, scalability, and clean modular separation.

---

**Q10 (15M)** **Draw an ER Diagram for a Library Management System with full entities and relational schema.**

**Entities:** `BOOK`, `MEMBER`, `PUBLISHER`, `BORROWS`.

```
[MEMBER] ----- (M:N via BORROWS with IssueDate, ReturnDate) ----- [BOOK]
                                                                    | (N:1)
                                                                [PUBLISHER]
```

---

**Q11 (15M)** **Discuss SQL Characteristics and explain five Aggregate Functions with complete query examples.**

**SQL Queries for Aggregate Functions:**

```
-- 1. COUNT: Total employees in IT
SELECT COUNT(*) FROM Employee WHERE Dept = 'IT';

-- 2. SUM: Total payroll for Sales
SELECT SUM(Salary) FROM Employee WHERE Dept = 'Sales';

-- 3. AVG: Average salary in company
SELECT AVG(Salary) FROM Employee;

-- 4. MIN and MAX: Salary bounds
SELECT MIN(Salary) AS MinSalary, MAX(Salary) AS MaxSalary FROM Employee;
```

---

**Q12 (15M)** **Discuss Selection, Projection, and Join operators of Relational Algebra with detailed expressions.**

**Detailed Operators:**

- \sigma_{City='Noida'}(Customer) selects rows where City is Noida.
- \pi_{Name, Email}(Customer) projects distinct Name and Email attributes.
- Customer \bowtie_{Customer.ID = Order.CustID} Order performs equijoin between Customer and Order relations.

---

**Q13 (15M)** **Explain Database Recovery Techniques: Log-Based Recovery, Immediate Update, Deferred Update, and Checkpoints.**

**Log-Based Recovery:**

- **Deferred Update (NO-UNDO / REDO):** Updates are recorded only in the log until commit. During recovery, only committed transactions are REDONE; uncommitted transactions require NO UNDO.
- **Immediate Update (UNDO / REDO):** Updates can be written to disk before commit. Active uncommitted transactions at crash must be UNDONE; committed ones are REDONE.
- **Checkpoints:** Periodically flushes dirty log buffers to disk. Recovery scans backward only to the most recent checkpoint, dramatically shortening crash recovery time.

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# BCA-501: Introduction to DBMS

**Session:** December 2020 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

### Q1 (3M) Define DML (Data Manipulation Language).

**Definition: Data Manipulation Language (DML)** provides constructs for accessing, inserting, updating, and querying stored database records. Classified into *Procedural DML* (user specifies what data and how to retrieve it) and *Non-Procedural / Declarative DML* (SQL, where user specifies only what data is needed).

### Q2 (3M) Give examples of Simple and Composite Attributes of an entity.

**Examples:**

- **Simple (Atomic) Attribute:** Cannot be divided into subparts (e.g., `RollNo = 101`, `Age = 21`, `Salary = 50000`).
- **Composite Attribute:** Can be divided into smaller sub-components with independent meanings (e.g., `Address` divided into `HouseNo`, `Street`, `City`, `Pincode`; or `Name` into `FirstName`, `LastName`).

### Q3 (3M) What do you mean by Referential Integrity Constraint?

**Definition: Referential Integrity** requires that any value of a foreign key attribute in a referencing relation must either match an existing primary key value in the referenced relation or be `NULL`. Enforces valid relationships and prevents "orphan records".

### Q4 (3M) What is a Data Dictionary (System Catalog)?

**Definition:** A **Data Dictionary** is a reserved set of read-only system tables maintained by the DBMS that stores metadata (data about data), including schemas, table descriptions, column domains, integrity constraints, indexes, user privileges, and storage allocations.

### Q5 (3M) Define Relational Schema.

**Definition:** A **Relational Schema** denotes the design structure of a relation, symbolized as R(A_1, A_2, \dots, A_n), where R is the relation name and A_i are attribute names defined over specific domains dom(A_i).

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

### Q6 (7.5M) Explain Lossless-Join Decomposition and Dependency Preservation in relational design.

**1. Lossless-Join Decomposition:**
A decomposition of relation R into R_1 and R_2 is lossless if and only if natural join of the projections equals the original relation without generating spurious tuples:

```
R_1 ⋈ R_2 = R ⇔ (R_1 ∩ R_2 → R_1) OR (R_1 ∩ R_2 → R_2)
```

**2. Dependency Preservation:**
A decomposition preserves dependencies if the union of functional dependencies enforceable within individual decomposed relations covers all functional dependencies of the original relation: (F_1 \cup F_2)^+ = F^+.

**Q7 (7.5M)** **Compare 3NF and BCNF with an illustrative example.**

**Key Differences:**

- **3NF:** For every non-trivial FD X \to Y, either X is a super key *OR* Y is a prime attribute (part of any candidate key). Allows dependency preservation always.
- **BCNF:** For every non-trivial FD X \to Y, X **must strictly be a super key**. Eliminates all FD redundancy, but may not always preserve all functional dependencies.

---

**Q8 (7.5M)** **Draw an ER Diagram for a University Examination System with entities, attributes, and relationships.**

**Entities:** `STUDENT` (RollNo, Name), `COURSE` (CourseCode, Title), `EXAM_CENTER` (CenterID, Location), `ADMIT_CARD` (AdmitCardNo).

```
[STUDENT] ----- (1:1) ----- [ADMIT_CARD]
    | (M)
    |
 < ENROLLS >
    |
    v (N)
 [COURSE] ----- (N:1) ----- [EXAM_CENTER]
```

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Discuss Relational Algebra fundamental operations with examples.**

**Operations:**

1. **Selection (σ):** \sigma_{Marks \ge 60}(Student)
2. **Projection (π):** \pi_{RollNo, Name}(Student)
3. **Cartesian Product (×):** Student \times Course
4. **Union (∪):** SectionA \cup SectionB
5. **Set Difference (−):** AllStudents - PassedStudents
6. **Rename (ρ):** \rho_{S}(Student)

---

**Q10 (15M)** **Explain File Organization Techniques: Sequential, Indexed Sequential (ISAM), and Hashed File Organization.**

**1. Sequential File Organization:** Records stored consecutively ordered by a search key. Fast for full table scans, slow O(N) for insertions/random lookups.
**2. Indexed Sequential Access Method (ISAM):** Combines sequential ordering with hierarchical index blocks (Cylinder, Track index). Enables rapid binary search.
**3. Hashed File Organization:** Uses hash function on search key for direct bucket addressing. Offers O(1) lookups, but poor for range queries.

---

**Q11 (15M)** **Explain Database Security and Authorization. Discuss GRANT and REVOKE commands in SQL.**

**SQL Security Commands:**

```
-- Granting SELECT and INSERT privileges with delegation rights
GRANT SELECT, INSERT ON Employee TO user_ravi WITH GRANT OPTION;

-- Granting role-based access
CREATE ROLE clerk;
GRANT SELECT, UPDATE(Salary) ON Employee TO clerk;
GRANT clerk TO user_amit;

-- Revoking privileges
REVOKE INSERT ON Employee FROM user_ravi CASCADE;
```

---

**Q12 (15M)** **Explain Two-Phase Locking (2PL) and describe Deadlock Prevention and Detection mechanisms.**

**Deadlock Handling:**

- **Deadlock Prevention:** Uses transaction timestamps to avoid circular wait:
  - *Wait-Die (Non-preemptive):* Older waits for younger; younger dies immediately.
  - *Wound-Wait (Preemptive):* Older wounds (preempts) younger; younger waits for older.
- **Deadlock Detection:** Builds periodic *Wait-For Graph (WFG)*; cycles in the graph indicate deadlocks, resolved by aborting the lowest-cost victim transaction.

---

**Q13 (15M)** **Explain Views in SQL. What are the advantages of Views? When is a View updatable?**

**Views in SQL:** A **View** is a virtual table defined by an underlying stored SQL query.
**Advantages:** Data security (restricts column access), simplified queries, and logical data independence.
**Updatable View Criteria:**

- Query contains only one base table in `FROM` clause.
- No aggregate functions ( `SUM` , `AVG` , `COUNT` ).
- No `GROUP BY` , `HAVING` , `DISTINCT` , or `UNION` .
- Primary key of base table must be included in the view columns.

---

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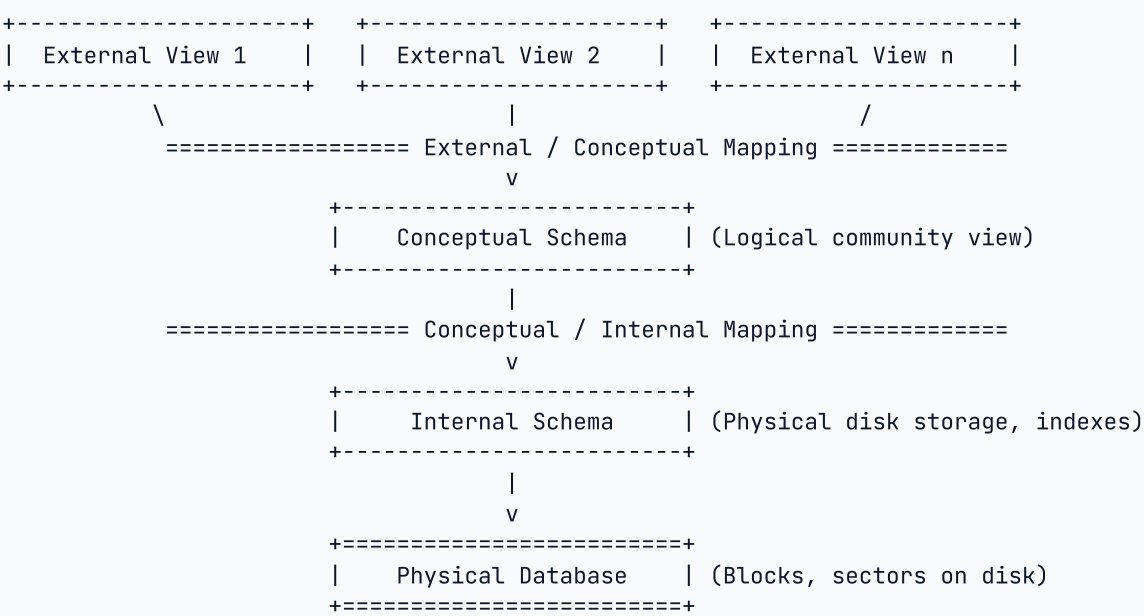
# BCA-501: Introduction to DBMS

**Session:** December 2021 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

**Q1 (7.5M)** **Explain the Three-Level Architecture of DBMS with a neat block diagram.**

**Three-Level ANSI/SPARC Architecture:**




**1. External Level (View Level):** User-specific customized views presenting only relevant subsets of data while hiding confidential fields.
**2. Conceptual Level (Logical Level):** Describes *what data* is stored in the entire database and the relationships among data without physical storage details.
**3. Internal Level (Physical Level):** Describes *how data* is physically stored on storage media (record formats, B+ trees, hashing).

---

**Q2 (7.5M)** **Define Primary Key, NOT NULL Key, and UNIQUE Key with differences.**

**Definitions & Differences:**

| Feature | PRIMARY KEY | UNIQUE KEY | NOT NULL |
|---|---|---|---|
| Purpose | Uniquely identifies each record. | Ensures distinct non-duplicate values. | Prevents missing or null values. |
| NULL Allowed | **Never** allowed (strictly 0 nulls). | Allowed (usually 1 or more nulls per column). | Does not enforce uniqueness, only disallows null. |
| Count per table | Only **1** per table. | Multiple unique constraints allowed. | Can be applied to every column. |
| Index created | Automatically creates Primary Clustered Index. | Creates Secondary Unique Non-Clustered Index. | No index created by default. |

---

**Q3 (7.5M)** **Explain Hashing in brief with its collision handling techniques.**

**Definition: Hashing** is an efficient direct search technique where a mathematical hash function h(k) maps a search key k directly to the disk address/bucket number where the record resides, providing average O(1) time complexity.
**Collision:** Occurs when two distinct search keys hash to the same bucket address: h(k_1) = h(k_2).
**Collision Handling Techniques:**

- **Open Addressing (Linear Probing):** Searches the next contiguous sequential bucket until an empty slot is located.
- **Chaining / Separate Overflow Buckets:** Maintains a linked list of overflow buckets chained to the primary bucket.
- **Extendible Hashing (Dynamic Hashing):** Uses a directory of pointers with local and global depths to double directories and split buckets dynamically without rehashing all records.

---

**Q4 (7.5M)** **What are the pitfalls of Lock-Based Protocols in concurrency control?**

**Pitfalls:**

1. **Deadlock:** Occurs when two or more transactions are waiting indefinitely for locks held by each other (e.g., T_1 holds Lock on A and waits for B, while T_2 holds Lock on B and waits for A). Requires deadlock detection (Wait-For Graph) and victim rollback.
2. **Starvation:** A transaction waits indefinitely because other higher-priority transactions repeatedly acquire locks on the required data item.
3. **Cascading Rollback:** In non-strict 2PL, if a transaction aborts, all dependent uncommitted transactions that read its dirty data must also be rolled back in a cascade.
4. **Concurrency Bottleneck:** Excessive long-duration exclusive locks reduce system throughput and degrade response times.

---

**Q5 (7.5M)** **Why is Concurrency Control needed in DBMS? Explain with concurrency anomalies.**

**Need for Concurrency Control:** When multiple transactions execute concurrently without coordination, interference causes severe anomalies:

- **Lost Update Problem:** Two transactions overwrite each other's updates without reading the intermediate result.
- **Dirty Read (Uncommitted Dependency):** A transaction reads modified data from an uncommitted transaction that subsequently aborts.
- **Inconsistent Analysis (Non-Repeatable Read):** A transaction reads data while another transaction modifies it midway, producing skewed sums.

---

## SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

**Q6 (15M)** **Given the Relational Schema:**
**EMPLOYEE(EmpID, EmpName, Age, Gender, Salary, City, DeptID, Designation, HireDate)**
**DEPARTMENT(DeptID, DeptName, Location)**
**Write SQL Queries for:**
**1. Display the age of 'Male' employees.**
**2. Display the name of the highest salary employee.**
**3. Display the name of department of employee 'Aman'.**
**4. Display all employees belonging to the same address (city).**
**5. Which employee is the oldest manager in the company?**

**SQL Queries:**

```
-- Query 1: Age of Male employees
SELECT EmpName, Age
FROM EMPLOYEE
WHERE Gender = 'Male';

-- Query 2: Name of the highest salary employee
SELECT EmpName, Salary
FROM EMPLOYEE
WHERE Salary = (SELECT MAX(Salary) FROM EMPLOYEE);

-- Query 3: Department name of employee 'Aman'
SELECT D.DeptName
FROM DEPARTMENT D
JOIN EMPLOYEE E ON D.DeptID = E.DeptID
WHERE E.EmpName = 'Aman';

-- Query 4: Display all employees grouped by identical city
SELECT City, COUNT(EmpID) AS TotalEmps,
       GROUP_CONCAT(EmpName) AS EmployeeNames
FROM EMPLOYEE
GROUP BY City
HAVING COUNT(EmpID) > 1;

-- Query 5: Oldest manager in the company
SELECT EmpName, Age, Designation, HireDate
FROM EMPLOYEE
WHERE Designation = 'Manager'
ORDER BY Age DESC
LIMIT 1;
```

---

## SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

**Q7 (22.5M)** **Explain Two-Phase Locking (2PL) Protocol. Compare Strict 2PL and Rigorous 2PL with serializability guarantees.**

**Two-Phase Locking (2PL):** A concurrency control protocol guaranteeing conflict serializability by dividing a transaction into two distinct phases:

1. **Growing Phase:** Transaction may obtain locks (Shared or Exclusive) but cannot release any lock.
2. **Shrinking Phase:** Transaction may release locks but cannot acquire any new lock.

**Variants of 2PL:**

- **Basic 2PL:** Guaranteed conflict serializability, but vulnerable to cascading aborts and deadlocks.
- **Strict 2PL:** All Exclusive (X) locks held by a transaction must be retained until the transaction commits or aborts. *Guarantees strict schedules and eliminates cascading aborts!*
- **Rigorous 2PL:** Both Shared (S) and Exclusive (X) locks must be retained until transaction commit/abort. Ensures serializability in commit order.

---

**Q8 (22.5M)** **Define Normalization. Explain 1NF, 2NF, 3NF, and BCNF with a complete enterprise example.**

**Comprehensive Enterprise Example:**
Consider unnormalized table: `STUDENT_COURSE_REG(StudentID, StudentName, CourseID, CourseName, Instructor, InstructorOffice)`.
Functional Dependencies:

- StudentID \to StudentName
- CourseID \to CourseName, Instructor
- Instructor \to InstructorOffice

Candidate Key: (StudentID, CourseID).

**Step 1: Convert to 2NF (Eliminate Partial Dependencies):**
- `STUDENT(StudentID, StudentName)`
- `COURSE(CourseID, CourseName, Instructor, InstructorOffice)`
- `ENROLLMENT(StudentID, CourseID)`

**Step 2: Convert to 3NF (Eliminate Transitive Dependency in COURSE):**
In `COURSE`, Instructor \to InstructorOffice is transitive.
- `COURSE(CourseID, CourseName, Instructor)` (FK: Instructor)
- `INSTRUCTOR(Instructor, InstructorOffice)`

All relations are now in 3NF and BCNF with zero redundancy!

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# BCA-501: Introduction to DBMS

**Session:** December 2023 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

### Q1 (3M) Define Database Management System (DBMS).

**Definition:** A **Database Management System (DBMS)** is a computerized system software suite that enables users to define, create, maintain, query, and control access to a shared database. It serves as an intelligent interface between end-user applications and physical data storage.

---

### Q2 (3M) List any five applications of DBMS.

**Applications:**

1. **Banking:** Customer accounts, loans, ATM transactions, fund transfers.
2. **Airlines & Railways:** Real-time reservation systems, flight schedules, ticketing.
3. **Universities & Colleges:** Student admissions, course registrations, grades, fee accounts.
4. **E-Commerce & Retail:** Online product catalogs, shopping carts, billing, inventory control.
5. **Telecommunications:** Call records, monthly billing, subscriber information, network routing.

---

### Q3 (3M) What are the disadvantages of a File Processing System?

**Disadvantages:**

- **Data Redundancy & Inconsistency:** Same data stored in multiple files leading to conflicting copies.
- **Difficulty in Accessing Data:** Requires writing a new custom program for every ad-hoc query.
- **Data Isolation:** Data scattered across disparate file formats cannot be easily joined.
- **Integrity Problems:** Hard to enforce consistency constraints across independent files.
- **Atomicity Failures:** Incomplete updates during crashes leave data in corrupt states.

---

### Q4 (3M) Define Data Model.

**Definition:** A **Data Model** is an abstract framework providing a collection of conceptual concepts for describing data, relationships, semantics, and constraints in a database. Major categories: Conceptual/High-level (ER Model), Representational/Logical (Relational Model), and Physical/Low-level models.

---

### Q5 (3M) What are Attributes? Give examples.

**Definition:** An **Attribute** is a descriptive property or characteristic that describes an entity in an entity set.
**Examples for Entity STUDENT:**

- *Simple Attribute:* `RollNo`, `Age`
- *Composite Attribute:* `FullName` (composed of `FirstName`, `LastName`)
- *Multivalued Attribute:* `PhoneNumbers`, `Hobbies`
- *Derived Attribute:* `Age` (computed from `DateOfBirth` and current date)

---

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Compare Generalization and Specialization with examples.**

**Comparison:**

| Criteria | Generalization | Specialization |
|---|---|---|
| Approach | **Bottom-up** design process | **Top-down** design process |
| Mechanism | Combines common features of multiple lower-level entities into a single higher-level entity. | Splits a higher-level entity into specialized sub-entities based on distinct features. |
| Goal | Minimizes redundancy by factoring out common attributes. | Identifies distinguishing characteristics of specific sub-categories. |
| Example | Entities `CAR` and `TRUCK` generalized into higher entity `VEHICLE`. | Entity `EMPLOYEE` specialized into `SALARIED_EMP` and `HOURLY_EMP`. |

---

**Q7 (7.5M)** **Explain Primary Key, Super Key, and Foreign Key with examples.**

**Definitions & Relationships:**

- **Super Key:** A set of one or more attributes that uniquely identifies a tuple within a relation. May contain redundant attributes (e.g., `{EmpID, Name}`).
- **Primary Key:** A strictly minimal candidate key selected by the database designer to uniquely identify each tuple. No duplicates, no `NULL` allowed.
- **Foreign Key:** An attribute that establishes a referential relationship by referencing the primary key of another table.

```
Relation: EMPLOYEE(EmpID, EmpName, Email, DeptNo)
Relation: DEPARTMENT(DeptNo, DeptName, Building)

Super Keys: {EmpID}, {EmpID, EmpName}, {Email}, {EmpID, Email}
Candidate Keys: {EmpID}, {Email}
Primary Key: {EmpID}
Foreign Key: {DeptNo} in EMPLOYEE references DeptNo in DEPARTMENT
```

---

**Q8 (7.5M)** **What is Normalization? Explain the importance of Normalization in database design.**

**Definition: Normalization** is the systematic process of organizing relation schemas based on functional dependencies to eliminate anomalies and avoid redundant storage.

**Importance:**

- Eliminates Insertion, Deletion, and Modification anomalies.
- Reduces storage space by ensuring each fact is recorded in only one place.
- Prevents data corruption by enforcing referential integrity constraints across smaller, well-structured tables.

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

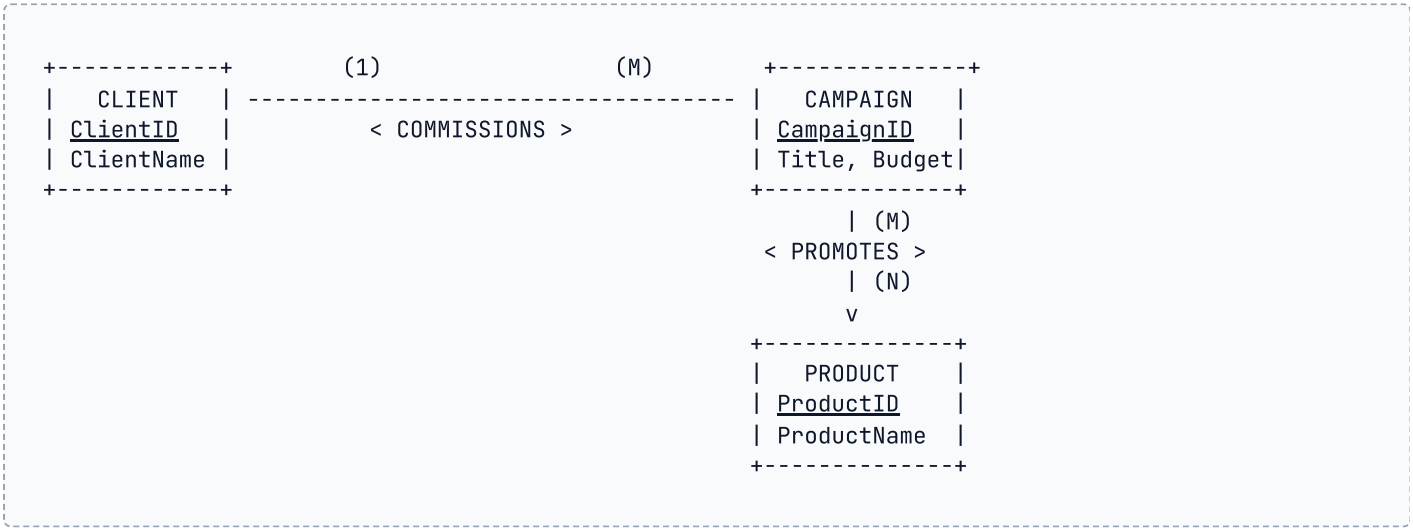
**Q9 (15M)** **Draw an ER Diagram for a small Marketing Company database, assuming your own data requirements.**

**Enterprise Requirements:**

- **CAMPAIGN:** CampaignID, Title, Budget, StartDate, EndDate.
- **CLIENT:** ClientID, ClientName, ContactPerson, Email, Phone.
- **PRODUCT:** ProductID, ProductName, Category.
- **MARKETING_AGENT:** AgentID, AgentName, CommissionRate.

**Relationships:**

- A Client commissions one or more Campaigns (1:N).
- Each Campaign promotes one or more Products (M:N).
- Agents are assigned to Campaigns (M:N).




---

**Q10 (15M)** **Describe the classification of Database Languages. Which type of language is SQL?**

**Classification of Database Languages:**

1. **Data Definition Language (DDL):** Specifies the database schema, storage structure, and constraints ( `CREATE` , `ALTER` , `DROP` , `TRUNCATE` ).
2. **Data Manipulation Language (DML):** Retrieves, inserts, updates, and deletes stored data. Can be procedural (relational algebra) or non-procedural/declarative (relational calculus, SQL).
3. **Data Control Language (DCL):** Controls permissions and access authorization ( `GRANT` , `REVOKE` ).
4. **Transaction Control Language (TCL):** Manages the logical unit of work ( `COMMIT` , `ROLLBACK` , `SAVEPOINT` ).

**Classification of SQL:** SQL is a **comprehensive declarative (non-procedural) language** that integrates DDL, DML, DCL, and TCL in a single standardized unified language.

---

**Q11 (15M)** **Explain DDL Commands and DML Commands with complete SQL syntax and executable examples.**

**1. DDL Commands (Data Definition Language):**

```
-- 1. CREATE TABLE
CREATE TABLE Student (
    RollNo INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Branch VARCHAR(20) DEFAULT 'BCA',
    FeePaid DECIMAL(10,2) CHECK (FeePaid ⩾ 0)
);

-- 2. ALTER TABLE: Add and modify column
ALTER TABLE Student ADD COLUMN Email VARCHAR(100);
ALTER TABLE Student MODIFY COLUMN Name VARCHAR(80);

-- 3. DROP & TRUNCATE
TRUNCATE TABLE Student; -- Deletes all rows, preserves table structure
DROP TABLE Student;     -- Deletes rows and schema permanently
```

**2. DML Commands (Data Manipulation Language):**

```
-- 1. INSERT
INSERT INTO Student (RollNo, Name, Branch, FeePaid, Email)
VALUES (101, 'Aman Sharma', 'BCA', 25000.00, 'aman@example.com');

-- 2. UPDATE
UPDATE Student
SET FeePaid = 30000.00
WHERE RollNo = 101;

-- 3. DELETE
DELETE FROM Student WHERE RollNo = 101;

-- 4. SELECT
SELECT RollNo, Name, FeePaid FROM Student WHERE Branch = 'BCA';
```

---

**Q12 (15M)** **Explain Data Independence with its types: Physical and Logical Data Independence. Why is it significant?**

**Definition: Data Independence** is the ability to modify schema definitions at one level of the 3-level ANSI/SPARC architecture without altering schemas at the higher level.

**1. Logical Data Independence:**

- The ability to modify the **Conceptual Schema** (e.g., adding/deleting attributes, splitting tables) without having to rewrite external schemas or existing application programs.
- *How achieved:* Using SQL Views that present the original logical schema to old applications.
- *Difficulty:* Harder to achieve because applications are closely tied to logical data models.

**2. Physical Data Independence:**

- The ability to modify the **Internal/Physical Schema** (e.g., changing file organizations, building B+ Tree indexes, moving to new storage media) without changing the Conceptual Schema.
- *Ease:* Easier to achieve because logical structure remains completely decoupled from disk-level storage.

---

**Q13 (15M) Who are Data Administrators (DA)? What are the functions of a Database Administrator (DBA)? Differentiate between DA and DBA.**

**Data Administrator (DA) vs Database Administrator (DBA):**

| Aspect | Data Administrator (DA) | Database Administrator (DBA) |
|---|---|---|
| Role Level | Managerial and business level | Technical and operational level |
| Focus | Business data policies, enterprise data dictionary, data governance. | Physical database design, database server installation, query tuning, security. |
| Scope | Decides *what data* needs to be collected across the enterprise. | Decides *how data is stored, indexed, secured, and recovered*. |

**Key Functions of DBA:**

1. Hardware/software installation and database upgrades.
2. Creating database users, granting roles, and monitoring access privileges.
3. Performance tuning: Analyzing execution plans, optimizing SQL queries, re-indexing.
4. Automating database backups and executing disaster recovery drills.

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# BCA-501: Introduction to DBMS

**Session:** December 2024 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

### Q1 (3M) What is Schema in DBMS?

**Definition:** A **Database Schema** is the formal structural blueprint and design description of the entire database. It specifies:

- Table names, columns, data types, and domains.
- Integrity constraints (primary keys, foreign keys, check constraints).
- Relationships between entities and storage representations.

It is stored in the system catalog and does not change frequently.

---

### Q2 (3M) What are the advantages of using DBMS?

**Key Advantages:**

1. **Minimal Redundancy:** Normalization ensures data is stored in one primary location.
2. **Data Integrity:** Automatic enforcement of domain and referential integrity constraints.
3. **Concurrency Control:** Multi-user simultaneous access without collisions via locking protocols.
4. **Crash Recovery:** WAL (Write-Ahead Logging) and checkpoints guarantee durability after hardware failure.
5. **Data Independence:** Separates user views from low-level storage modifications.

---

### Q3 (3M) Who is a DBA? Mention their key functions.

**Definition:** A **Database Administrator (DBA)** is an IT professional who manages the database environment, maintains performance, controls user permissions, schedules backups, and enforces security policies.

---

### Q4 (3M) What is a Database?

**Definition:** A **Database** is an integrated and logically related collection of records, designed to meet the information processing needs of multiple users across an organization.

---

### Q5 (3M) What is RDBMS?

**Definition:** A **Relational Database Management System (RDBMS)** is a DBMS based on E.F. Codd's Relational Model (1970). Data is organized into 2D tables (relations) consisting of rows (tuples) and columns (attributes), manipulated using SQL and relational algebra. Examples: Oracle, MySQL, PostgreSQL, SQL Server.

---

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Find the minimum number of tables required for the given ER diagram in the relational model. Explain table conversion rules.**

**Rules for ER to Relational Reduction:**

1. **Strong Entity Sets:** Each strong entity converts into a separate independent table.
2. **Weak Entity Sets:** Converts into a table containing its discriminator (partial key) combined with the primary key of the identifying strong entity as composite primary key.
3. **1:1 Relationships:** Combine both entities into 1 table, or place the primary key of one table as a foreign key in the other table.
4. **1:N Relationships:** Place the primary key of the '1' side entity into the table of the 'N' side entity as a foreign key (no new table needed).
5. **M:N Relationships:** Requires a new separate relationship table containing primary keys of both participating entities as foreign keys.

**Result:** For an ER model with two strong entities A and B in 1:N relationship, exactly **2 tables** are needed. If the relationship is M:N, exactly **3 tables** are required.

---

**Q7 (7.5M)** **What are UNION, MINUS, and INTERSECT commands in DBMS? Explain with syntax and examples.**

**Conditions:** Both queries must be *Union-Compatible* (same number of columns with compatible data types).

- `UNION` : Combines results of two SELECT queries and eliminates duplicate tuples.
- `INTERSECT` : Returns only the common tuples present in both result sets.
- `MINUS / EXCEPT` : Returns tuples from the first query that are not present in the second query.

**SQL Examples:**

```
-- UNION: All students from CS or IT
SELECT RollNo, Name FROM Student_CS
UNION
SELECT RollNo, Name FROM Student_IT;

-- INTERSECT: Students enrolled in both Math and Physics clubs
SELECT StudentID FROM MathClub
INTERSECT
SELECT StudentID FROM PhysicsClub;

-- MINUS: Registered students who have NOT paid exam fees
SELECT StudentID FROM RegisteredStudents
EXCEPT
SELECT StudentID FROM PaidStudents;
```

---

**Q8 (7.5M)** **Define the following terms: (a) Data Model, (b) Database Schema.**

**(a) Data Model:** A conceptual collection of tools, constructs, and notations used to describe data, data relationships, data semantics, and consistency constraints. Major models include:

- **Relational Model:** Tables of tuples and attributes (SQL).
- **Entity-Relationship (ER) Model:** High-level conceptual modeling.
- **Object-Oriented & Document (NoSQL) Models:** JSON, key-value, graph.

**(b) Database Schema:** The overall structural skeleton of the database defined during the design phase, distinct from the dynamic instances/states stored inside it.

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Define Normal Forms. Explain 1NF, 2NF, 3NF, and BCNF with suitable examples.**

**Detailed Normal Forms Analysis:**

| Normal Form | Requirement / Condition | Anomaly Eliminated |
|---|---|---|
| 1NF | All attribute values must be atomic (no multivalued / nested attributes). | Repeating groups and unstructured data arrays |
| 2NF | In 1NF and no non-prime attribute is partially dependent on any candidate key. | Partial Functional Dependencies |
| 3NF | In 2NF and for every X → Y, either X is a Super Key or Y is a Prime Attribute. | Transitive Functional Dependencies |
| BCNF | For every non-trivial functional dependency X → Y, X must strictly be a Super Key. | All functional dependency based redundancies |

**BCNF Example:**
Suppose `STUDENT_ADVISOR(StudentID, Subject, Advisor)` where each student has an advisor per subject, and each advisor teaches only one subject: FD_1: (StudentID, Subject) \to Advisor, FD_2: Advisor \to Subject.
Candidate Key is (StudentID, Subject) or (StudentID, Advisor).
In FD_2, `Advisor` is not a super key, violating BCNF.
Decompose into: `Advisor_Subject(Advisor, Subject)` and `Student_Advisor(StudentID, Advisor)`. Both are now in BCNF!

---

**Q10 (15M)** **Explain Normalization with its advantages. What anomalies arise in unnormalized databases?**

**Anomalies in Unnormalized Tables:**

1. **Insertion Anomaly:** Cannot record information about an entity without having to insert another unrelated entity (e.g., cannot register a new department until at least one employee is hired).
2. **Deletion Anomaly:** Deleting a record of one entity unintentionally erases information about another entity (e.g., deleting the last employee in a department deletes all department details).
3. **Update / Modification Anomaly:** Changing an entity's attribute requires updating hundreds of duplicate rows; if one row is missed, data becomes contradictory.

**Advantages of Normalization:**

- Eliminates data redundancy and conserves disk storage.
- Maintains referential integrity automatically across tables.
- Speeds up row updates, inserts, and index lookups.

---

**Q11 (15M)** **Explain SQL with its various characteristics and features.**

**Structured Query Language (SQL):** The ANSI/ISO standard declarative language for relational databases.
**Key Characteristics:**

- **Declarative / Non-Procedural:** Users specify *what data* they want, leaving the query optimizer to determine *how* to retrieve it.
- **Comprehensive Capabilities:** Encompasses DDL (structure), DML (manipulation), DCL (security/privileges), and TCL (transactions).
- **Portability & Standards:** Works uniformly across major vendors (Oracle, MySQL, SQL Server, PostgreSQL).
- **Integrity Constraints Support:** Native enforcement of `PRIMARY KEY`, `FOREIGN KEY`, `CHECK`, `UNIQUE`, and `DEFAULT`.
- **View & Subquery Support:** Enables nested queries, common table expressions (CTEs), and dynamic virtual tables (Views).

---

**Q12 (15M)** **What is Relational Algebra? Discuss its basic and extended operations with expressions.**

**Overview:** Relational algebra provides the formal theoretical mathematical foundation for query evaluation and execution plan optimization in RDBMS.
**Operations Summary:**

- **Selection (σ):** Horizontal subset of rows based on conditions.
- **Projection (π):** Vertical subset of columns, duplicate-free.
- **Cartesian Product (×):** All pairs of concatenated rows.
- **Set Union (∪), Intersection (∩), Difference (−):** Set-theoretic operations on union-compatible relations.
- **Theta Join ($\bowtie_\theta$):** Cross product followed by condition filtering.
- **Division (÷):** Used for queries involving "for all" / universal quantification.

---

**Q13 (15M)** **Explain the reduction of ER Schema to Relational Tables with a complete case study.**

**Case Study: Company Database:**
Entities: `EMPLOYEE`, `DEPARTMENT`, `PROJECT`.
Relationships:

- `WORKS_FOR` (N:1 from Employee to Department)
- `MANAGES` (1:1 from Employee to Department with `StartDate`)
- `WORKS_ON` (M:N between Employee and Project with `Hours`)

**Relational Reduction:**

```
1. DEPARTMENT(DeptNo, DeptName, MgrEmpID, MgrStartDate)
   -- 1:1 MANAGES folded into DEPARTMENT with MgrEmpID as FK
2. EMPLOYEE(EmpID, EmpName, Salary, DeptNo)
   -- N:1 WORKS_FOR represented via DeptNo as FK referencing DEPARTMENT
3. PROJECT(ProjNo, ProjName, DeptNo)
   -- N:1 CONTROLS represented via DeptNo as FK
4. WORKS_ON(EmpID, ProjNo, Hours)
   -- M:N relationship becomes a separate junction table with composite PK
   -- FK EmpID → EMPLOYEE, FK ProjNo → PROJECT
```

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-501: Introduction to DBMS

**Session:** December 2025 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

### Q1 (3M) What is a Database?

**Definition:** A **Database** is an organized, logically coherent collection of related data modeled to represent real-world entities and enterprise operations. Key properties include:

- **Self-describing nature:** Stores both data and metadata (system catalog/data dictionary).
- **Program-data independence:** Structure of data files is stored separately from user access programs.
- **Multi-user transaction processing:** Guarantees concurrent access without collision or corruption.

---

### Q2 (3M) State the key advantages of DBMS over File Processing Systems.

**Advantages:**

| Aspect | Traditional File System | Database Management System (DBMS) |
|---|---|---|
| Data Redundancy | High redundancy across duplicate files | Controlled and minimized via normalization |
| Data Inconsistency | High risk due to isolated updates | Eliminated via centralized repository |
| Data Independence | None; changes in files break code | Physical and Logical Data Independence achieved |
| Concurrent Access | Prone to race conditions and lost updates | ACID compliant concurrency control (2PL, timestamp) |
| Security & Integrity | OS-level passwords, hard to enforce | Fine-grained role-based access & constraints |

---

### Q3 (3M) Who is a Database Administrator (DBA)? Mention key duties.

**Definition:** A **Database Administrator (DBA)** is the central authority responsible for supervising and administering the entire database environment.
**Primary Functions:**

1. **Schema Definition:** Designing and altering conceptual, external, and internal schemas.
2. **Storage & Access Tuning:** Specifying storage structures, file organizations, and primary/secondary indexes.
3. **Security & Authorization:** Granting and revoking user privileges, managing roles, and enforcing access control.
4. **Backup & Recovery:** Scheduling periodic backups and executing disaster recovery strategies.
5. **Integrity Enforcement:** Defining domain, entity, and referential integrity constraints.

---

### Q4 (3M) Differentiate between Schema and Instance in DBMS.

**Comparison:**

- **Database Schema:** The overall logical blueprint, structure, and design of the database (e.g., table names, column data types, constraints). It is specified during system design and changes very infrequently.
- **Database Instance (State):** The actual data content populated within the database at a specific moment in time. It changes dynamically whenever records are inserted, updated, or deleted.

*Analogy:* In programming languages, a Schema corresponds to a Class/Data Type, whereas an Instance corresponds to an Object/Value of that type.

---

**Q5 (3M)** **What do you mean by Data Inconsistency in File Management Systems? Explain.**

**Explanation: Data Inconsistency** occurs when multiple copies of the exact same data across different files contain conflicting, contradictory values.

**Example:** In a university file system, a student's address is recorded in the *Library File* and also in the *Fee Accounting File*. If the student changes their residential address and only the Library office updates their local file while the Fee office does not, the university now holds two contradictory addresses for the same student. This occurs due to uncoordinated duplicate storage without centralized integrity control.

---

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Explain ER Diagram. Describe all standard ER notations and symbols with diagrams.**

**Definition:** An **Entity-Relationship (ER) Diagram** is a graphical semantic modeling tool that represents the logical structure of a database in terms of entities, their attributes, and relationships.

**Standard Symbols & Notations:**

```
[ Rectangle ]           → Entity Set (e.g., STUDENT, COURSE)
[[ Double Rectangle ]]  → Weak Entity Set (e.g., DEPENDENT, INSTALLMENT)
( Ellipse / Oval )      → Attribute (e.g., Age, Name)
( Underlined Text )     → Primary Key Attribute (e.g., RollNo)
(( Double Ellipse ))    → Multivalued Attribute (e.g., PhoneNumbers, Skills)
( Dashed Ellipse )      → Derived Attribute (e.g., Age calculated from DOB)
{ Diamond }             → Relationship Set (e.g., ENROLLED_IN)
{{ Double Diamond }}    → Identifying Relationship for Weak Entity Set
======== Double Line    → Total Participation (Every entity must participate)
-------- Single Line    → Partial Participation
```

---

**Q7 (7.5M)** **Explain Primary Key, Foreign Key, and Candidate Key with suitable SQL examples.**

**1. Candidate Key:** A minimal super key — an attribute or minimal set of attributes that uniquely identifies each tuple in a relation without any redundant attributes.

**2. Primary Key:** The specific candidate key chosen by the database designer to uniquely identify tuples in a table. It cannot accept duplicate values and cannot contain `NULL`.

**3. Foreign Key:** An attribute in a child table whose values must match the primary key of a referenced parent table (or be `NULL`), enforcing referential integrity.

**SQL Implementation Example:**

```sql
-- Parent Table: DEPARTMENT
CREATE TABLE Department (
    DeptID INT PRIMARY KEY,              -- Primary Key
    DeptName VARCHAR(50) NOT NULL UNIQUE -- Candidate Key
);

-- Child Table: EMPLOYEE
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,               -- Primary Key
    EmpName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE,           -- Candidate Key
    DeptID INT,                          -- Foreign Key
    CONSTRAINT fk_dept FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

---

**Q8 (7.5M)** **Differentiate between DDL and DML commands with syntax and examples.**

**Comparison:**

| Feature | Data Definition Language (DDL) | Data Manipulation Language (DML) |
|---|---|---|
| Purpose | Defines and modifies the schema structure of the database. | Manages, accesses, and manipulates the data within the tables. |
| Commit Nature | Auto-committed permanently (cannot be rolled back in most RDBMS). | Transactional (can be committed or rolled back using TCL). |
| Commands | `CREATE`, `ALTER`, `DROP`, `TRUNCATE`, `RENAME` | `INSERT`, `UPDATE`, `DELETE`, `SELECT` |

**SQL Examples:**

```
-- DDL Commands:
CREATE TABLE Student (RollNo INT PRIMARY KEY, Name VARCHAR(50));
ALTER TABLE Student ADD COLUMN City VARCHAR(30);

-- DML Commands:
INSERT INTO Student (RollNo, Name, City) VALUES (101, 'Aman', 'Meerut');
UPDATE Student SET City = 'Delhi' WHERE RollNo = 101;
```

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain the ACID properties of a transaction with practical examples. Why are they essential?**

**Definition:** A **Transaction** is a logical unit of database processing that includes one or more database access operations. To preserve database consistency and reliability, every DBMS must guarantee the four **ACID Properties**:

**1. Atomicity ("All or Nothing"):**

- Either all operations of the transaction are committed successfully, or none take effect. If failure occurs mid-transaction, all changes are rolled back.
- *Example:* Bank transfer of \$500 from Account A to Account B. If \$500 is debited from A and power fails before crediting B, the recovery subsystem rolls back A's debit.

**2. Consistency (Preservation of Invariants):**

- Execution of a transaction in isolation preserves database integrity constraints (e.g., total balance of A + B must remain unchanged before and after the transfer).

**3. Isolation (Independence of Concurrent Executions):**

- Even if multiple transactions execute concurrently, the intermediate uncommitted state of one transaction is hidden from others, producing a serializable execution schedule. Enforced via locking (2PL) and timestamp ordering.

**4. Durability (Survivability of Committed Data):**

- Once a transaction commits, its updates persist permanently in secondary storage and will survive subsequent power failures, crashes, or OS reboots via Write-Ahead Logging (WAL) and checkpointing.

---

**Q10 (15M)** **What is Normalization? Explain 1NF, 2NF, 3NF, and BCNF with suitable relational examples and anomaly reduction.**

**Definition: Normalization** is a formal process of decomposing relations with anomalies into structured relations to eliminate insertion, deletion, and update anomalies while avoiding redundant data storage.

**1. First Normal Form (1NF):**

- A relation is in 1NF if and only if all attribute values are atomic (no multivalued, composite, or nested attributes).
- *Violation:* `Employee(EmpID, Skills)` where `Skills = "Java, Python, C++"`.
- *Decomposition:* Split into atomic rows: `(101, 'Java')`, `(101, 'Python')`, `(101, 'C++')`.

**2. Second Normal Form (2NF):**

- Relation must be in 1NF and have **no partial dependency** (no non-prime attribute should depend on a proper subset of any candidate key).
- *Example:* `Enrollment(StudentID, CourseID, CourseFee, Grade)`. Here `CourseFee` depends only on `CourseID` (partial).
- *Decomposition:* `StudentCourses(StudentID, CourseID, Grade)` and `CourseDetails(CourseID, CourseFee)`.

**3. Third Normal Form (3NF):**

- Relation must be in 2NF and have **no transitive dependency** (for every non-trivial FD X \to Y, either X is a super key or Y is a prime attribute).
- *Example:* `Employee(EmpID, EmpName, DeptNo, DeptHead)`. Here `EmpID → DeptNo` and `DeptNo → DeptHead`, creating transitive dependency.
- *Decomposition:* `Emp(EmpID, EmpName, DeptNo)` and `Dept(DeptNo, DeptHead)`.

**4. Boyce-Codd Normal Form (BCNF):**

- A stricter version of 3NF where for every non-trivial FD X \to Y, determinant X **must strictly be a Super Key**.

---

**Q11 (15M) Explain SQL Aggregate Functions (COUNT, SUM, AVG, MIN, MAX) with GROUP BY and HAVING clauses. Provide full query demonstrations.**

**Aggregate Functions:** Functions that perform calculations on a set of column values and return a single scalar summary value.

- `COUNT(column)` : Returns total count of non-null values. `COUNT(*)` counts all rows.
- `SUM(column)` : Calculates total numeric sum of values.
- `AVG(column)` : Calculates arithmetic mean of non-null values.
- `MIN(column)` / `MAX(column)` : Returns the minimum / maximum value.

**Role of GROUP BY and HAVING:**

- `GROUP BY` divides records into distinct groups based on identical column values.
- `HAVING` filters grouped rows after aggregation (whereas `WHERE` filters individual tuples prior to grouping).

**Comprehensive SQL Example:**

```
-- Schema: Employee (EmpID, EmpName, Salary, DeptID, City)
-- Query: Find DeptID, total employees, average salary, and max salary
-- for departments having more than 3 employees and average salary > 45000:

SELECT
    DeptID,
    COUNT(EmpID) AS TotalEmployees,
    AVG(Salary) AS AverageSalary,
    MAX(Salary) AS HighestSalary,
    MIN(Salary) AS LowestSalary
FROM Employee
WHERE City IN ('Delhi', 'Noida', 'Gurugram')
GROUP BY DeptID
HAVING COUNT(EmpID) > 3 AND AVG(Salary) > 45000
ORDER BY AverageSalary DESC;
```

---

**Q12 (15M)** **What is Relational Algebra? Discuss its fundamental operations (Select, Project, Union, Set Difference, Cartesian Product, Rename) and derived Join operation.**

**Definition: Relational Algebra** is a formal procedural query language that takes one or two relations as input and produces a new relation as output without modifying the underlying stored tables.

**Fundamental Operators:**

1. **Select (σ):** Unary operator that filters rows satisfying a predicate condition P.

```
σ_{Salary > 50000 AND Dept = 'IT'} (Employee)
```

2. **Project (π):** Unary operator that selects specified columns and eliminates duplicate tuples.

```
π_{EmpName, Salary} (Employee)
```

3. **Union (∪):** Binary operator combining tuples from two union-compatible relations.

```
Student_CS ∪ Student_IT
```

4. **Set Difference (−):** Returns tuples present in relation R but not in S (union compatible).

```
All_Students - Placed_Students
```

5. **Cartesian Product (×):** Cross product pairing every tuple of R with every tuple of S.

```
Employee × Department
```

6. **Rename (ρ):** Renames relation or attributes: \rho_{S(A_1, A_2)} (R).

**Derived Operation — Natural Join (⋈):** Combines tuples of two relations matching on identical common attribute names:

```
Employee ⋈ Department = π_{...} (σ_{Employee.DeptID = Department.DeptID} (Employee × Department))
```

**Q13 (15M)** **Draw an ER Diagram for a Hospital Management System specifying entities, attributes, primary keys, relationships, and cardinalities. Show conversion to Relational Schema.**

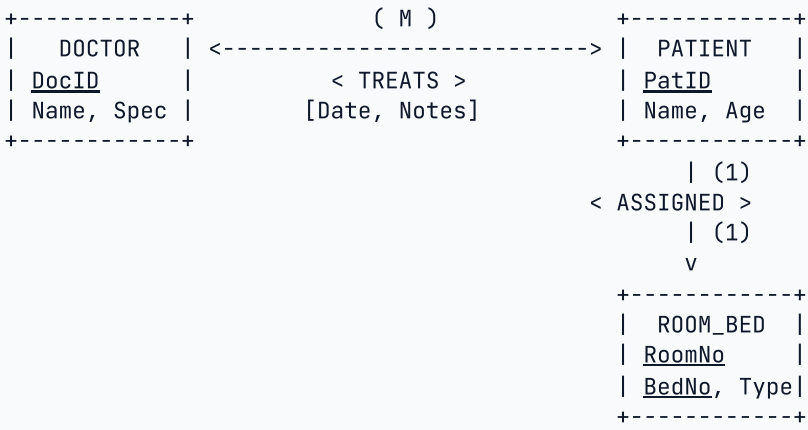
**1. Entity Sets & Attributes:**

- **DOCTOR**: DocID, Name, Specialty, Phone
- **PATIENT**: PatID, Name, Age, Gender, Address
- **ROOM / BED**: RoomNo, BedNo, RoomType, DailyCharge
- **BILL / ACCOUNT**: BillID, BillDate, TotalAmount, PaymentStatus

**2. Relationships & Cardinalities:**

- `TREATS` : Many-to-Many between DOCTOR and PATIENT with attribute `DateOfTreatment` .
- `ASSIGNED_TO` : Many-to-One between PATIENT and BED (A patient is assigned to one bed; a bed holds one patient at a time).
- `GENERATES` : One-to-One between PATIENT and BILL.

**3. ASCII ER Diagram Representation:**




**4. Relational Schema Reduction:**

1. Doctor(DocID, Name, Specialty, Phone)
2. Patient(PatID, Name, Age, Gender, Address, RoomNo, BedNo)
   FOREIGN KEY (RoomNo, BedNo) REFERENCES Room_Bed(RoomNo, BedNo)
3. Room_Bed(RoomNo, BedNo, RoomType, DailyCharge)
4. Treats(DocID, PatID, TreatmentDate, Prescription)
   FOREIGN KEY (DocID) REFERENCES Doctor(DocID),
   FOREIGN KEY (PatID) REFERENCES Patient(PatID)
5. Bill(BillID, PatID, BillDate, TotalAmount, Status)
   FOREIGN KEY (PatID) REFERENCES Patient(PatID)

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-502: Java Programming & Dynamic Webpage Design

**Session:** December 2016 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) What is Byte-code in JAVA? Explain.**

**Definition:** The intermediate machine instruction set generated by `javac`, executed by the JVM to provide platform independence.

**Q2 (3M) What is Garbage Collection in Java?**

**Definition:** Automatic memory reclamation mechanism of the JVM destroying unreferenced heap objects.

**Q3 (3M) Differentiate between Class and Interface.**

**Differences:** A class contains full implementation and state; an interface contains method specifications and constants.

**Q4 (3M) What is an Applet?**

**Definition:** A small Java program designed to be embedded in an HTML web page and executed inside a web browser.

**Q5 (3M) What is Thread in Java?**

**Definition:** A lightweight sub-process representing an independent execution path in a program.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) What do you mean by Constructor? Explain Constructor Overloading.**

**Constructor Overloading:** Defining multiple constructors in a single class with different parameter lists.

**Q7 (7.5M) Explain Packages in Java and how to create user-defined packages.**

**Packages:** Directory-based namespaces created with `package` statement.

**Q8 (7.5M) Explain Exception Handling in Java with try, catch, and finally.**

**Handling:** Graceful recovery from runtime program faults.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M) Explain Applet Life Cycle in detail with an Applet code to draw shapes.**

**Applet Shapes Code:** Complete applet drawing rectangles, lines, and ovals using `Graphics`.

**Q10 (15M) Explain Multithreading and Thread Synchronization with examples.**

**Multithreading & Synchronization:** Producer-Consumer problem and synchronized locks.

**Q11 (15M) Explain JDBC Architecture and steps to connect Java to a database.**

**JDBC:** Complete connectivity steps and SQL operations.

**Q12 (15M)** **Explain Servlets, HttpServlet, and its life cycle methods.**

**Servlets:** Complete server-side Java web request dispatching.

---

**Q13 (15M)** **Explain Client-Server Socket Programming in Java with complete program.**

**Socket Programming:** `ServerSocket` and `Socket` communication code over TCP.

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-502: Java Programming & Dynamic Webpage Design

**Session:** December 2017 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M)** **How will you create hyperlinks in a web document? Write the code also.**

**HTML Hyperlink:**

```
<a href="https://www.ccsuniversity.ac.in" target="_blank">Visit CCSU Official Portal</a>
```

**Q2 (3M)** **What is a Constructor? List its characteristics.**

**Characteristics:** Same name as class, no return type (not even void), automatically called on `new`, can be overloaded.

**Q3 (3M)** **Differentiate between Method Overloading and Method Overriding.**

**Overloading:** Same method name, different signatures, within same class (compile-time).
**Overriding:** Same signature, in subclass, overriding parent behavior (runtime).

**Q4 (3M)** **What is Bytecode in Java?**

**Definition:** Architecture-neutral intermediate instructions executed by JVM.

**Q5 (3M)** **What is the purpose of the 'this' keyword?**

**Definition:** Refers to the current class instance to differentiate shadowed instance fields from parameters.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Explain the Life Cycle of an Applet with diagram.**

**Applet Lifecycle:** Born ( `init` ) → Running ( `start` , `paint` ) → Idle ( `stop` ) → Dead ( `destroy` ).

**Q7 (7.5M)** **Explain Packages and Access Protection in Java.**

**Access Protection:** Detailed comparison of private, default, protected, and public scopes.

**Q8 (7.5M)** **Explain Interface in Java with code example.**

**Interface:** Blueprint of a class containing abstract methods and constants.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain Exception Handling in Java: try, catch, finally, throw, and throws with custom exception.**

**Custom Exception:**

```
class InvalidAgeException extends Exception {
    public InvalidAgeException(String msg) { super(msg); }
}
```

**Q10 (15M)** **Explain Multithreading in Java: Thread class, Runnable interface, and Thread Priorities.**

**Multithreading:** Full comparison of extending Thread vs implementing Runnable, lifecycle states, and priority scheduling.

---

**Q11 (15M)** **What is JDBC? Explain the 4 types of JDBC drivers and steps for database connectivity.**

**JDBC Drivers:** Type-1, Type-2, Type-3, and Type-4 drivers with architectural diagrams.

---

**Q12 (15M)** **Explain Servlets vs JSP. Write a complete HttpServlet program handling GET requests.**

**Servlet Architecture:** Complete HttpServlet implementation serving dynamic web responses.

---

**Q13 (15M)** **Explain Layout Managers in AWT: FlowLayout, BorderLayout, and GridLayout.**

**Layout Managers:** Visual layout structures and coordinate placements.

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-502: Java Programming & Dynamic Webpage Design

**Session:** December 2018 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

### Q1 (3M) What is a Final Variable? Give an example.

**Definition:** A variable declared with the `final` keyword becomes a constant whose value cannot be reassigned once initialized:

```
final double PI = 3.14159;
// PI = 3.14; // Compile-time error: cannot assign a value to final variable
```

### Q2 (3M) What is Garbage Collection in Java?

**Definition:** The automatic process of finding and freeing heap memory allocated to objects that are no longer referenced by any active thread.

### Q3 (3M) What is an Abstract Method?

**Definition:** A method declared without a body (implementation) using the `abstract` keyword, terminating with a semicolon. Must be implemented by subclasses.

### Q4 (3M) What is a Layout Manager in Java AWT?

**Definition:** An interface/class that manages the automatic positioning and sizing of GUI components within a container window.

### Q5 (3M) What is the purpose of the 'super' keyword?

**Definition:** References the immediate superclass object to call parent constructors ( `super()` ) or access hidden parent members.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

### Q6 (7.5M) What are Packages? How are packages created in Java? Explain and give example.

**Packages:** Namespaces organizing classes and avoiding naming collisions. Created using the `package` keyword at the top of a source file:

```
package com.ccsu.bca;
public class Student {
    public void display() { System.out.println("BCA Student"); }
}
```

### Q7 (7.5M) Explain Interface in Java with an example of Multiple Inheritance.

**Explanation:** Pure abstraction tool allowing classes to implement multiple interfaces simultaneously, overcoming single class inheritance limits.

**Q8 (7.5M)** **Explain Applet Life Cycle methods.**

**Lifecycle:** `init()`, `start()`, `paint(Graphics g)`, `stop()`, and `destroy()`.

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain Multithreading in Java. Write a program demonstrating Thread synchronization using synchronized methods.**

**Thread Synchronization:** Coordinates thread access to shared mutable resources to prevent race conditions.

```
class Table {
    synchronized void printTable(int n) {
        for(int i = 1; i ⩽ 5; i++) {
            System.out.println(n * i);
            try { Thread.sleep(300); } catch(Exception e) {}
        }
    }
}
```

---

**Q10 (15M)** **What is Exception Handling? Explain Checked vs Unchecked Exceptions with try, catch, finally, and throws.**

**Detailed Exception Model:** Checked exceptions verified at compile-time (e.g. `IOException`) vs Unchecked exceptions occurring at runtime (e.g. `NullPointerException`).

---

**Q11 (15M)** **Explain JDBC Architecture and steps to connect Java to a database. Write a program to insert a record.**

**Complete JDBC Program:** Register driver, establish connection, execute SQL insert using `Statement`, and close resources.

---

**Q12 (15M)** **What is JSP? Explain its life cycle, advantages over Servlets, and scriptlet tags.**

**JSP vs Servlets:** Translation, compilation, and execution lifecycle; comparison of presentation layer benefits.

---

**Q13 (15M)** **Explain Event Handling in Java AWT with Delegation Event Model.**

**Delegation Event Model:** Event sources generate event objects delivered to registered listener interfaces.

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-502: Java Programming & Dynamic Webpage Design

**Session:** November 2019 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) What is Bytecode and Java Virtual Machine (JVM)?**

**Explanation:** Bytecode is compiled intermediate instructions for the virtual JVM processor, enabling cross-platform portability.

**Q2 (3M) List the four fundamental principles of Object-Oriented Programming.**

**OOP Principles:** Encapsulation, Abstraction, Inheritance, and Polymorphism.

**Q3 (3M) What is Garbage Collection in Java?**

**Explanation:** Automatic reclamation of heap memory occupied by dereferenced objects.

**Q4 (3M) What is Final variable, Final method, and Final class?**

**Final keyword:** Variable becomes a constant, method cannot be overridden, class cannot be inherited (subclassed).

**Q5 (3M) Define Constructor in Java.**

**Definition:** A special method called automatically when an object is instantiated, having the same name as the class and no return type.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) Explain Interface in Java with code demonstrating multiple inheritance.**

**Interface:** Contract with method declarations implemented by classes to achieve multiple inheritance.

**Q7 (7.5M) Explain Applet Life Cycle and how Applets are embedded into HTML.**

**Applet in HTML:** Embedded using `<applet code="MyApplet.class" width="300" height="200"></applet>`.

**Q8 (7.5M) Explain Exception Handling using try, catch, and finally blocks.**

**Exception Handling:** Structured handling preventing runtime crashes and releasing resources safely.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M) What is JSP? Explain advantages of JSP and main components. How is JSP different from Servlet?**

**JSP vs Servlet:** Comprehensive comparison, JSP compilation lifecycle, directives, actions, and scriptlets.

**Q10 (15M) Explain Multithreading in Java: Thread Life Cycle, States, and Synchronization.**

**Thread Life Cycle:** New → Runnable → Running → Blocked/Waiting → Terminated.

**Q11 (15M)** **Explain JDBC Architecture, Statement vs PreparedStatement, and write a program to insert student records.**

**JDBC Program:** Complete program inserting records using PreparedStatement to prevent SQL injection.

---

**Q12 (15M)** **Explain Event Handling in Java AWT/Swing with MouseListener and ActionListener.**

**Event Handling:** Source → Event Object → Listener interface implementation.

---

**Q13 (15M)** **Write HTML and CSS code to design a responsive institutional web page.**

**Web Design:** Clean HTML markup with external/internal CSS styling, tables, lists, and navigation bar.

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-502: Java Programming & Dynamic Webpage Design

**Session:** December 2020 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Explain why Java is considered the best language for Internet Applications.**

**Reasons:** Platform independence via bytecode/JVM, built-in security (sandbox security manager), robust memory management, native multithreading, and extensive networking APIs ( `java.net` ).

**Q2 (3M) Discuss various primitive data types supported by Java.**

**Primitive Types:** 8 primitive types: `byte` , `short` , `int` , `long` , `float` , `double` , `char` , `boolean` with explicit fixed sizes across all platforms.

**Q3 (3M) What is the purpose of the 'static' keyword in Java?**

**Explanation:** Marks class-level members (variables, methods, blocks) shared by all instances, callable without creating an object of the class (e.g., `ClassName.staticMethod()` ).

**Q4 (3M) What are Wrapper Classes in Java?**

**Explanation:** Classes that wrap primitive data types into object representations ( `Integer` , `Double` , `Boolean` ) enabling autoboxing and unboxing for collections.

**Q5 (3M) Define Interface in Java.**

**Definition:** A reference type containing pure abstract method declarations and static constants, defining a functional contract for implementing classes.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) Explain Applet Life Cycle and write an Applet to display a greeting message.**

**Applet Lifecycle:** Complete coverage of `init()` , `start()` , `paint()` , `stop()` , and `destroy()` .

**Q7 (7.5M) Explain Thread Synchronization and the 'synchronized' keyword in Java.**

**Synchronization:** Prevents thread interference and race conditions when multiple threads access shared resources simultaneously using monitor locks.

**Q8 (7.5M) Differentiate between Throw and Throws in Java Exception Handling.**

**Throw:** Explicitly triggers an exception instance inside a method body.
**Throws:** Declares exceptions thrown by a method in its header signature.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M) What is JDBC? Explain JDBC Architecture and execute a database SELECT query.**

**JDBC Architecture:** Application → JDBC API → DriverManager → JDBC Driver → Database. Complete sample SELECT query program.

**Q10 (15M)** **Write HTML code for a complete Web Form with Table formatting and Validation.**

**HTML Implementation:** Form markup with input types, select dropdowns, radio buttons, and table alignments.

---

**Q11 (15M)** **Discuss JSP Architecture, Scripting Elements (<%! %>, <% %>, <%= %>), and Directives.**

**JSP Scripting Elements:**

- **Declaration ( `<%! %>` ):** Declares class variables and methods.
- **Scriptlet ( `<% %>` ):** Java code placed inside the `_jspService()` method.
- **Expression ( `<%= %>` ):** Directly prints evaluated values to the output stream.

---

**Q12 (15M)** **Explain Servlets, HttpServletRequest, and HttpServletResponse with complete GET/POST demonstration.**

**Servlet Processing:** Complete explanation of HTTP request parsing, header inspection, and HTML output streaming.

---

**Q13 (15M)** **What are JavaBeans? Explain Bean properties and deployment in JSP using jsp:useBean.**

**JavaBeans:** Reusable software components adhering to conventions: zero-arg constructor, private fields, and public getters/setters.

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-502: Java Programming & Dynamic Webpage Design

**Session:** December 2021 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

**Q1 (7.5M)** **Explain destroy() method, a stage in the life cycle of an Applet.**

**Explanation:** `destroy()` is the final callback method in the Applet lifecycle, invoked by the browser or AppletViewer before the applet is completely removed from memory and garbage collected. It is called only once after the `stop()` method to release non-memory resources (threads, sockets, database handles).

**Q2 (7.5M)** **What do you mean by a Package in Java? State the different Access Specifiers available in Java.**

**Access Specifiers:**

| Access Specifier | Same Class | Same Package | Subclass (Diff Pkg) | World (Everywhere) |
|---|---|---|---|---|
| `private` | Yes | No | No | No |
| `default` (package-private) | Yes | Yes | No | No |
| `protected` | Yes | Yes | Yes (via inheritance) | No |
| `public` | Yes | Yes | Yes | Yes |

**Q3 (7.5M)** **What is Exception? Give different types of exceptions that occur during runtime. Why handle exceptions?**

**Explanation:** An **Exception** is an abnormal runtime event that disrupts the normal flow of program instructions. Types: Checked ( `IOException` , `SQLException` ) and Unchecked/Runtime ( `ArithmeticException` , `NullPointerException` , `ArrayIndexOutOfBoundsException` ). Handled to maintain graceful execution and avoid abrupt program termination.

**Q4 (7.5M)** **What are the life-cycle methods of a JSP? Differences between JSP Custom Tags and JavaBeans.**

**JSP Lifecycle:** `jspInit()` → `_jspService(HttpServletRequest, HttpServletResponse)` → `jspDestroy()` .
**Differences:** JavaBeans encapsulate reusable business data and logic via getter/setter methods. Custom Tags encapsulate presentation logic and HTML formatting within custom XML-style tags.

**Q5 (7.5M)** **Explain Thread Priority. What are default values? Write a program to define two threads where one prints 1 to 10 and the other prints 10 to 1.**

**Thread Priorities:** Integer scale from 1 ( `MIN_PRIORITY` ) to 10 ( `MAX_PRIORITY` ), with default 5 ( `NORM_PRIORITY` ).
**Program:**

```
class ForwardPrinter extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 10; i++) {
            System.out.println("Forward: " + i);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

class BackwardPrinter extends Thread {
    public void run() {
        for(int j = 10; j ⩾ 1; j--) {
            System.out.println("Backward: " + j);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

public class PriorityThreadDemo {
    public static void main(String[] args) {
        ForwardPrinter t1 = new ForwardPrinter();
        BackwardPrinter t2 = new BackwardPrinter();
        t1.setPriority(Thread.MAX_PRIORITY);
        t2.setPriority(Thread.MIN_PRIORITY);
        t1.start();
        t2.start();
    }
}
```

## SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

**Q6 (15M)** **Explain Client-Server Socket Programming in Java with complete code.**

**Socket Programming:** `ServerSocket` on server side and `Socket` on client side transmitting data over TCP streams using `DataInputStream` and `DataOutputStream` .

## SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

**Q7 (22.5M)** **Explain JDBC Architecture and write a program to update employee salaries based on department.**

**Program:**

```
import java.sql.*;

public class UpdateSalaryDemo {
    public static void main(String[] args) {
        try {
            Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/db", "root",
            PreparedStatement ps = con.prepareStatement("UPDATE Employee SET Salary = Salary * 1.1
            ps.setString(1, "IT");
            int rows = ps.executeUpdate();
            System.out.println("Salaries updated for " + rows + " employees.");
            con.close();
        } catch(Exception e) { e.printStackTrace(); }
    }
}
```

---

**Q8 (22.5M)** **Explain AWT Event Handling with Delegation Event Model. Write a program to handle Button click events.**

**Delegation Event Model:** An event source generates an event and sends it to registered event listeners implementing listener interfaces (e.g., `ActionListener`).

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-502: Java Programming & Dynamic Webpage Design

**Session:** December 2023 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

### Q1 (3M) How is an Object created in Java?

**Object Creation:** Uses the `new` operator:

```
ClassName obj = new ClassName();
```

Steps: (1) *Declaration* (associates variable name with object type), (2) *Instantiation* (`new` allocates memory on the Heap), (3) *Initialization* (invokes the constructor).

### Q2 (3M) Compare JDK, JRE, and JVM.

**Relationship:**

- **JVM (Java Virtual Machine):** The runtime engine that interprets and executes bytecode.
- **JRE (Java Runtime Environment):** Contains JVM + Java Standard Class Libraries; sufficient to run Java programs.
- **JDK (Java Development Kit):** Contains JRE + Development Tools (`javac`, `javadoc`, `jar`, `debugger`); needed to write and compile programs.

### Q3 (3M) Difference between JDBC and ODBC.

**Differences:** ODBC is a C/C++ platform-dependent API designed originally for Windows. JDBC is a platform-independent pure Java API tailored for Java applications and cross-platform RDBMS connectivity.

### Q4 (3M) Write differences between JSP and Servlet.

**Differences:** Servlets are Java code with embedded HTML strings (ideal for business logic and controllers). JSP is HTML markup with embedded Java scriptlets (ideal for presentation and views in MVC).

### Q5 (3M) What is Method Overloading vs Method Overriding?

**Comparison:** Overloading occurs in the same class with identical method names but different parameter lists (compile-time polymorphism). Overriding occurs in subclasses with identical method signatures and return types (runtime dynamic dispatch).

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

### Q6 (7.5M) What is an Applet? Describe the complete Life Cycle of an Applet.

**Applet Life Cycle:** `init()` → `start()` → `paint(Graphics g)` → `stop()` → `destroy()`.

**Q7 (7.5M)** **Explain Interface in Java. How is multiple inheritance achieved using interfaces?**

**Explanation:** A class can implement multiple interfaces simultaneously:

```
interface Printable { void print(); }
interface Showable { void show(); }
class Document implements Printable, Showable {
    public void print() { System.out.println("Printing..."); }
    public void show() { System.out.println("Showing..."); }
}
```

---

**Q8 (7.5M)** **Explain Exception Handling in Java with try-catch-finally block.**

**Handling:** Code capable of raising runtime exceptions is placed inside `try`, recovered in `catch`, and resource deallocation performed in `finally`.

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **What are Packages in Java? How is a user-defined package created and imported? Explain with complete code example.**

**Creating & Importing User-Defined Package:**

```
// File: mypack/Calculator.java
package mypack;

public class Calculator {
    public int add(int a, int b) { return a + b; }
    public int multiply(int a, int b) { return a * b; }
}

// File: TestPackage.java
import mypack.Calculator;

public class TestPackage {
    public static void main(String[] args) {
        Calculator calc = new Calculator();
        System.out.println("Sum: " + calc.add(10, 20));
        System.out.println("Product: " + calc.multiply(10, 20));
    }
}
```

---

**Q10 (15M)** **Define Thread. How to create a thread in Java? Write a program that executes two threads simultaneously.**

**Complete Dual Thread Program:**

```
class ThreadOne extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 5; i++) {
            System.out.println("Thread 1 executing count: " + i);
            try { Thread.sleep(400); } catch(Exception e) {}
        }
    }
}

class ThreadTwo extends Thread {
    public void run() {
        for(int j = 10; j ⩽ 15; j++) {
            System.out.println("Thread 2 executing count: " + j);
            try { Thread.sleep(400); } catch(Exception e) {}
        }
    }
}

public class DualThreadDemo {
    public static void main(String[] args) {
        ThreadOne t1 = new ThreadOne();
        ThreadTwo t2 = new ThreadTwo();
        t1.start();
        t2.start();
    }
}
```

---

**Q11 (15M)** **Explain JDBC Architecture and steps involved in connecting a Java application to a Database. Write complete program.**

**Steps:** (1) Load Driver, (2) Create Connection, (3) Create Statement, (4) Execute Query, (5) Process ResultSet, (6) Close Connection.

---

**Q12 (15M)** **Explain HTML form controls and write code for an Admission Form.**

**Admission Form:** Includes text fields, dropdown selects, checkboxes, text areas, and submit buttons.

---

**Q13 (15M)** **Explain JSP implicit objects and session tracking in Web Applications.**

**Implicit Objects:** `request`, `response`, `session`, `application`, `out`, `config`, `pageContext`, `page`, `exception`. Session tracking via Cookies, URL Rewriting, and HttpSession.

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-502: Java Programming & Dynamic Webpage Design

**Session:** December 2024 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

### Q1 (3M) Discuss Primitive Data Types in Java.

**Java Data Types:**

- **Integer Types:** `byte` (1 byte), `short` (2 bytes), `int` (4 bytes), `long` (8 bytes).
- **Floating-Point:** `float` (4 bytes, IEEE 754), `double` (8 bytes).
- **Character:** `char` (2 bytes, 16-bit Unicode character).
- **Boolean:** `boolean` (represents true or false).

---

### Q2 (3M) Explain TCP/IP Server Socket in Java.

**Explanation:** Java provides the `java.net.ServerSocket` class to implement a TCP server that listens for incoming connection requests on a designated port.

```
ServerSocket server = new ServerSocket(8080);
Socket client = server.accept(); // Blocks until connection arrives
```

---

### Q3 (3M) What are Packages? How do you call a package in Java?

**Definition:** A **Package** is a namespace mechanism that organizes related classes and interfaces into modular directory structures, preventing naming conflicts and providing access protection.
**Calling a Package:** Uses the `import` keyword:

```
import java.util.ArrayList; // Imports single class
import java.io.*;           // Imports all classes in package
```

---

### Q4 (3M) Explain the Anchor and Table tags in HTML.

**HTML Tags:**

- `<a href="url">Link Text</a>` : Creates a hyperlink to navigate between web pages or download files.
- `<table><tr><th>...</th><td>...</td></tr></table>` : Renders structured tabular data using rows ( `<tr>` ), headers ( `<th>` ), and data cells ( `<td>` ).

---

### Q5 (3M) Briefly explain Internet Concepts: URL, HTTP, and Web Server.

**Concepts:**

- **URL (Uniform Resource Locator):** Global address of a resource on the web (e.g., `https://www.example.com/index.html` ).
- **HTTP:** HyperText Transfer Protocol, a stateless client-server protocol running over TCP port 80/443 for web communication.
- **Web Server:** Hardware/software that serves web resources to clients (Apache, Nginx, Tomcat).

---

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Explain Applet Life Cycle. Write a program to demonstrate a simple Java applet to display an image. Compare Applets over HTML.**

**Applet Image Display Code:**

```
import java.applet.*;
import java.awt.*;

/* <applet code="ImageApplet.class" width="400" height="300"></applet> */
public class ImageApplet extends Applet {
    Image img;
    public void init() {
        img = getImage(getDocumentBase(), "college_logo.png");
    }
    public void paint(Graphics g) {
        g.drawImage(img, 20, 20, this);
        g.drawString("CCSU BCA V Semester", 20, 260);
    }
}
```

**Comparison over HTML:** Applets provide dynamic client-side computation, graphical interactivity, and local state management that static HTML cannot natively provide.

---

**Q7 (7.5M)** **Explain TCP/IP Client Socket. Write constructors and methods used to create client sockets.**

**Client Socket (java.net.Socket):**

- **Constructors:**
  - `Socket(String host, int port)` — Establishes TCP stream connection.
  - `Socket(InetAddress address, int port)`
- **Methods:**
  - `getInputStream()` — Returns InputStream for reading server responses.
  - `getOutputStream()` — Returns OutputStream for sending data.
  - `close()` — Gracefully closes the TCP connection.

---

**Q8 (7.5M)** **Explain the concepts of Inheritance in Java and its various types.**

**Inheritance:** The mechanism where a subclass derives attributes and behaviors from a superclass using `extends`.
**Types in Java:**

1. **Single Inheritance:** Class B extends Class A.
2. **Multilevel Inheritance:** Class C extends Class B, which extends Class A.
3. **Hierarchical Inheritance:** Class B and Class C both extend Class A.
4. **Multiple Inheritance:** Achieved through **Interfaces** (Java does not support multiple class inheritance to avoid the Diamond Problem).

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **What is JDBC? Write a complete JDBC program to insert and display records of employees using PreparedStatement.**

**Complete PreparedStatement JDBC Program:**

```
import java.sql.*;

public class EmployeeJDBC {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/company_db";
        String user = "root";
        String password = "password";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(url, user, password);

            // 1. Insert record using PreparedStatement
            String insertSql = "INSERT INTO Employee(EmpID, Name, Salary) VALUES(?, ?, ?)";
            PreparedStatement psInsert = conn.prepareStatement(insertSql);
            psInsert.setInt(1, 101);
            psInsert.setString(2, "Rahul Sharma");
            psInsert.setDouble(3, 55000.00);
            int rows = psInsert.executeUpdate();
            System.out.println("Inserted rows: " + rows);

            // 2. Display all employee records
            String selectSql = "SELECT EmpID, Name, Salary FROM Employee";
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery(selectSql);

            System.out.println("\n--- EMPLOYEE RECORDS ---");
            while(rs.next()) {
                System.out.println(rs.getInt("EmpID") + "\t" +
                                   rs.getString("Name") + "\t" +
                                   rs.getDouble("Salary"));
            }
            conn.close();
        } catch(Exception e) {
            e.printStackTrace();
        }
    }
}
```

---

**Q10 (15M)** **What are Packages? Explain Java API Packages. List any five methods of the String class with examples.**

**String Class Methods:**

1. `length()` : Returns character count → `"Hello".length() == 5` .
2. `charAt(int index)` : Returns character at index → `"Hello".charAt(1) == 'e'` .
3. `substring(int begin, int end)` : Extracts substring → `"Java".substring(0, 2) == "Ja"` .
4. `toUpperCase()` : Converts to uppercase → `"bca".toUpperCase() == "BCA"` .
5. `equals(Object another)` : Compares string contents → `"Java".equals("Java") == true` .

---

**Q11 (15M)** **Explain Servlets with their Life Cycle. How is it different from the Life Cycle of JSP?**

**Comparison:**

- **Servlet:** Java class compiled into bytecode; container directly executes `init()`, `service()`, and `destroy()`.
- **JSP:** Text-based document with HTML + embedded Java code; container first translates JSP into a temporary Java Servlet source file, compiles it to bytecode, and invokes `_jspInit()`, `_jspService()`, and `_jspDestroy()`.

---

**Q12 (15M)** **Discuss JSP in detail. What are JSP Directives? Explain various types of directives with examples.**

**JSP Directives ( `<%@ directive attribute="value" %>` ):**

1. **page Directive:** Controls page-level settings: `<%@ page import="java.util.*, java.sql.*" contentType="text/html" errorPage="error.jsp" %>`
2. **include Directive:** Inserts static files at translation time: `<%@ include file="header.jsp" %>`
3. **taglib Directive:** Imports custom tag libraries (JSTL): `<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>`

---

**Q13 (15M)** **What are Layout Managers? Explain five Layout Managers supported by AWT/Swing.**

**Five Layout Managers:**

1. **FlowLayout:** Arranges components row-by-row like words in a paragraph.
2. **BorderLayout:** Positions in North, South, East, West, Center regions.
3. **GridLayout:** Rectangular grid of equal-sized cells.
4. **CardLayout:** Stacks components like a deck of cards (one visible at a time).
5. **GridBagLayout:** Most flexible layout manager using `GridBagConstraints` for variable cell spans and weights.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-502: Java Programming & Dynamic Webpage Design

**Session:** December 2025 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) What is the difference between Process-based and Thread-based multitasking in Java?**

**Comparison:**

| Feature | Process-Based Multitasking | Thread-Based Multitasking |
|---|---|---|
| Granularity | Heavyweight; each process has its own separate address space. | Lightweight; threads share the same address space in a single process. |
| Context Switching | High overhead; managed by operating system kernel. | Low overhead; switches quickly within the JVM. |
| Communication | Requires Inter-Process Communication (IPC, sockets, pipes). | Direct shared memory, wait(), notify(), synchronized blocks. |

**Q2 (3M) Explain the 'super' keyword in Java with usages.**

**Usage of 'super':** A reference variable used in a subclass to refer directly to its immediate parent class object:

1. **Invoke Parent Constructor:** `super(args);` — must be the first line in subclass constructor.
2. **Access Parent Variables:** `super.variableName` — accesses shadowed parent instance fields.
3. **Call Parent Methods:** `super.methodName()` — invokes overridden parent class methods.

**Q3 (3M) What is Bytecode in Java? Why does it enable platform independence?**

**Explanation:** Java source code ( `.java` ) is compiled by `javac` into an intermediate machine-independent format called **Bytecode** (stored in `.class` files). Any platform equipped with a standard Java Virtual Machine (JVM) interprets and executes this bytecode natively, achieving "Write Once, Run Anywhere" (WORA).

**Q4 (3M) Explain Garbage Collection in Java.**

**Definition:** Java features automatic memory management via the Garbage Collector (GC), a background daemon thread that periodically scans the Heap memory and deallocates memory occupied by unreachable (unreferenced) objects, preventing memory leaks without manual pointer freeing.

**Q5 (3M) What is the purpose of the 'finally' block in Exception Handling?**

**Definition:** The `finally` block always executes regardless of whether an exception is thrown, caught, or unhandled. Used to perform critical resource clean-up actions such as closing file streams, database connections, and network sockets.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Differentiate between Abstract Class and Interface in Java with examples.**

**Comparison:**

| Aspect | Abstract Class | Interface |
|---|---|---|
| Methods | Can have abstract and concrete methods with state. | Contains abstract methods (and default/static in Java 8+). |
| Variables | Can have instance variables (any access modifier). | All fields are implicitly `public static final` constants. |
| Inheritance | Single inheritance (a class `extends` only 1 abstract class). | Multiple inheritance (a class can `implements` multiple interfaces). |

---

**Q7 (7.5M)** **Explain Exception Handling mechanism in Java with keywords: try, catch, finally, throw, and throws.**

**Keywords & Mechanism:**

- `try` : Encloses code that may potentially throw an exception.
- `catch` : Catches and handles specific exception types.
- `finally` : Guaranteed cleanup block executed after try/catch.
- `throw` : Explicitly throws an exception object (e.g., `throw new ArithmeticException("Divide by zero");`).
- `throws` : Declares exceptions in method signatures for callers to handle.

---

**Q8 (7.5M)** **Explain the Applet Life Cycle and its core callback methods.**

**Lifecycle Methods:**

1. `init()` : Called once when the applet is first loaded into the browser/appletviewer to initialize variables.
2. `start()` : Invoked every time the applet becomes visible or browser tab is opened.
3. `paint(Graphics g)` : Renders shapes, text, and images on the screen surface.
4. `stop()` : Invoked when the user leaves the page or minimizes the browser.
5. `destroy()` : Called before the applet is completely reclaimed from memory.

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Write an HTML code to create a User Registration Web Form containing: Username, Date of Birth, Address, Gender (Radio buttons), Email-ID, Mobile Number, and Submit/Reset buttons.**

**HTML Code:**

```
<!DOCTYPE html>
<html>
<head>
    <title>User Registration Form</title>
</head>
<body>
    <h2>Student / User Registration Form</h2>
    <form action="register" method="POST">
        <table cellpadding="6">
            <tr>
                <td>User Name:</td>
                <td><input type="text" name="username" required /></td>
            </tr>
            <tr>
                <td>Date of Birth:</td>
                <td><input type="date" name="dob" required /></td>
            </tr>
            <tr>
                <td>Address:</td>
                <td><textarea name="address" rows="3" cols="30"></textarea></td>
            </tr>
            <tr>
                <td>Gender:</td>
                <td>
                    <input type="radio" name="gender" value="Male" /> Male
                    <input type="radio" name="gender" value="Female" /> Female
                </td>
            </tr>
            <tr>
                <td>Email-ID:</td>
                <td><input type="email" name="email" required /></td>
            </tr>
            <tr>
                <td>Mobile Number:</td>
                <td><input type="tel" name="mobile" pattern="[0-9]{10}" required /></td>
            </tr>
            <tr>
                <td colspan="2" align="center">
                    <input type="submit" value="Register Now" />
                    <input type="reset" value="Clear Form" />
                </td>
            </tr>
        </table>
    </form>
</body>
</html>
```

**Q10 (15M)** **What is JDBC? Explain the Four Types of JDBC Drivers with architectural diagrams and characteristics.**

**Definition: Java Database Connectivity (JDBC)** is an industry-standard Java API that enables Java programs to interact with relational databases using SQL queries.

**Four Types of JDBC Drivers:**

1. **Type-1: JDBC-ODBC Bridge Driver:** Translates JDBC calls to ODBC calls. Requires ODBC driver installation on every client machine. Obsolete and slow.
2. **Type-2: Native-API / Partially Java Driver:** Converts JDBC calls directly into native C/C++ vendor database client libraries (e.g., Oracle OCI). Requires client-side installation.
3. **Type-3: Network Protocol / All-Java Middleware Driver:** Translates JDBC calls into an intermediate server middleware protocol, which then talks to multiple databases. Flexible for 3-tier architectures.
4. **Type-4: Thin Driver / Direct-to-Database Pure Java Driver:** Converts JDBC calls directly into vendor-specific database network protocols via TCP/IP sockets. 100% pure Java, zero client installation, highest performance (Modern standard!).

---

**Q11 (15M)** **Explain Servlet Life Cycle and compare it with JSP Life Cycle. Write a complete HttpServlet code example.**

**Servlet Life Cycle Methods:**

1. `init(ServletConfig config)` : Loads servlet class, creates instance, and initializes parameters (called once).
2. `service(HttpServletRequest req, HttpServletResponse res)` : Dispatches to `doGet()` or `doPost()` for every incoming HTTP request.
3. `destroy()` : Invoked when container unloads the servlet, releasing database and memory resources.

**Servlet Implementation Example:**

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class WelcomeServlet extends HttpServlet {
    public void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String user = request.getParameter("username");
        out.println("<html><body>");
        out.println("<h2>Welcome, " + user + "!</h2>");
        out.println("<p>Server Response Time: " + new java.util.Date() + "</p>");
        out.println("</body></html>");
    }
}
```

---

**Q12 (15M)** **Define Thread in Java. Explain how to create threads using (a) Thread class and (b) Runnable interface. Write a complete multithreaded program.**

**Multithreading in Java:**

```
// Approach 1: Extending Thread class
class NumberPrinter extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 5; i++) {
            System.out.println("Thread 1: " + i);
            try { Thread.sleep(500); } catch(Exception e) {}
        }
    }
}

// Approach 2: Implementing Runnable interface
class LetterPrinter implements Runnable {
    public void run() {
        for(char c = 'A'; c ⩽ 'E'; c++) {
            System.out.println("Thread 2: " + c);
            try { Thread.sleep(500); } catch(Exception e) {}
        }
    }
}

public class MultiThreadDemo {
    public static void main(String[] args) {
        NumberPrinter t1 = new NumberPrinter();
        Thread t2 = new Thread(new LetterPrinter());
        t1.start(); // Spawns new thread
        t2.start();
    }
}
```

---

**Q13 (15M)** **What are Layout Managers in AWT/Swing? Explain BorderLayout, FlowLayout, and GridLayout with examples.**

**Layout Managers:** Automatically position, size, and rearrange GUI components inside containers regardless of screen resolution or OS window scaling.

- **FlowLayout:** Arranges components sequentially from left to right, wrapping to the next line when space runs out. Default for `Panel` and `Applet`.
- **BorderLayout:** Divides container into 5 regions: `NORTH`, `SOUTH`, `EAST`, `WEST`, and `CENTER`. Default for `Frame` and `JFrame`.
- **GridLayout:** Arranges components into a rectangular grid of equal-sized rows and columns.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-503: Computer Networks

**Session:** December 2016 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M)** **Which device is used for Modulation and Demodulation? Explain its function.**

**Answer:** A **MODEM (Modulator-Demodulator)**. Converts digital computer pulses to analog signals for transmission over analog lines (modulation) and translates received analog waves back into digital bits (demodulation).

---

**Q2 (3M)** **What is Network Topology? List advantages and disadvantages of Mesh Topology.**

**Mesh Topology:**

- *Advantages:* Dedicated point-to-point links eliminate traffic congestion; robust fault tolerance (link failure affects only two nodes); excellent privacy.
- *Disadvantages:* Requires massive cabling (N(N-1)/2 links) and expensive multi-port hardware; complex installation.

---

**Q3 (3M)** **Differentiate between Simplex, Half-Duplex, and Full-Duplex Transmission.**

**Differences:** Simplex is unidirectional, Half-duplex is alternate bidirectional, Full-duplex is simultaneous bidirectional.

---

**Q4 (3M)** **What is Bandwidth of a channel?**

**Definition:** The range of frequencies contained in a signal or passed by a transmission channel: B = f_{max} - f_{min} (measured in Hertz, or bps in digital communication).

---

**Q5 (3M)** **What is a Protocol in Computer Networks?**

**Definition:** A formal set of rules and conventions governing the format, timing, sequencing, and error control of data transmission between communicating entities (Syntax, Semantics, Timing).

---

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Explain Seven Layers of ISO/OSI Reference Model.**

**OSI Model:** Physical, Data Link, Network, Transport, Session, Presentation, Application layers with primary duties.

---

**Q7 (7.5M)** **Explain Sliding Window Protocol: Stop-and-Wait vs Sliding Window.**

**Comparison:** Stop-and-Wait sends 1 frame at a time; Sliding Window allows sending W frames before receiving acknowledgments, multiplying channel utilization.

---

**Q8 (7.5M)** **Explain Distance Vector Routing Algorithm with Bellman-Ford equation.**

**Algorithm:** Path updating using D_x(y) = \min_v \{ c(x, v) + D_v(y) \} across neighboring nodes.

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain Error Control using CRC with an example bit sequence.**

**CRC Division:** Modulo-2 binary arithmetic and frame checksum verification.

---

**Q10 (15M)** **Explain Guided Media: Twisted Pair, Coaxial Cable, and Optical Fiber.**

**Guided Media:** Full analysis of transmission mechanisms, attenuation, and physical connectors.

---

**Q11 (15M)** **Explain Switching Techniques: Circuit Switching, Message Switching, and Packet Switching.**

**Switching Methods:** Architectural breakdown, delay calculations, and applications.

---

**Q12 (15M)** **Explain Congestion Control: Leaky Bucket and Token Bucket Algorithms.**

**Traffic Shaping:** Detailed algorithms, diagrams, and burst handling.

---

**Q13 (15M)** **Explain IP Addressing (IPv4 Classes A, B, C, D, E) and Subnetting.**

**IP Addressing:** Address format, network IDs, host IDs, default masks, and subnet mask calculation.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-503: Computer Networks

**Session:** December 2017 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Repeater operates in which layer of the OSI model? Explain its function.**

**Explanation:** A **Repeater** operates at the **Physical Layer (Layer 1)** of the OSI model. It receives weak or distorted electrical/optical signals from one cable segment, cleans, amplifies, regenerates the signal to its original strength and retimes it before transmitting it over the next segment, overcoming transmission attenuation.

**Q2 (3M) Differentiate between FDM and TDM Multiplexing.**

**Differences:** FDM divides available channel bandwidth into distinct non-overlapping frequency bands (analog). TDM allocates the entire channel bandwidth to each source for dedicated repeating time slots (digital).

**Q3 (3M) A local telephone network is an example of which switching technology?**

**Answer: Circuit Switching**. A dedicated physical communication circuit is established between the calling and called parties through telephone exchanges for the entire duration of the call.

**Q4 (3M) What is Guided Media? List examples.**

**Definition:** Physical solid transmission media that provide a physical conduit for electromagnetic signals: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.

**Q5 (3M) Compare TCP and UDP at the Transport Layer.**

**Comparison:** TCP is connection-oriented, reliable (acknowledgments, retransmissions), with flow and congestion control (HTTP, FTP). UDP is connectionless, unreliable, lightweight, and low-latency (DNS, VoIP, video streaming).

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) Explain Seven Layers of ISO/OSI Reference Model.**

**OSI Architecture:** Physical, Data Link, Network, Transport, Session, Presentation, Application layers with primary duties.

**Q7 (7.5M) Explain Stop-and-Wait Flow Control Protocol and calculate its efficiency.**

**Efficiency:** \eta = \frac{T_t}{T_t + 2T_p} = \frac{1}{1 + 2a} where a = \frac{\text{Propagation Time}}{\text{Transmission Time}}.

**Q8 (7.5M) Explain Dijkstra's Algorithm for Shortest Path Routing.**

**Dijkstra's Algorithm:** Iteratively selects minimum-cost unvisited nodes to build the shortest-path routing tree.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M) Explain Error Control using CRC with complete mathematical division example.**

**CRC Division:** Modulo-2 polynomial division with verification at receiver side.

**Q10 (15M)** **Explain Sliding Window Protocols: Go-Back-N and Selective Repeat.**

**Sliding Window:** In-depth operation, sliding window mechanics, and timeout handling.

---

**Q11 (15M)** **Explain Transmission Media: UTP, STP, Coaxial, and Optical Fiber.**

**Detailed Media:** Physical construction, connectors (RJ45, BNC, SC/ST), bandwidth, and propagation speed.

---

**Q12 (15M)** **Explain Congestion Control Algorithms: Leaky Bucket and Token Bucket.**

**Traffic Shaping:** Comparison of uniform leak rate vs burst absorption using tokens.

---

**Q13 (15M)** **Explain Switching Techniques: Circuit, Packet, and Message Switching with detailed diagrams.**

**Switching Architectures:** Latency, store-and-forward delay, and buffer utilization.

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-503: Computer Networks

**Session:** December 2018 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) What is the difference between TCP and OSI Model?**

**Differences:** OSI is a 7-layer theoretical conceptual model developed by ISO. TCP/IP is a 4-layer practical protocol suite developed by DoD that forms the basis of the modern Internet. OSI strictly separates session and presentation layers, whereas TCP/IP combines them into the Application layer.

**Q2 (3M) What is Network Topology? Mention any two.**

**Topologies:** Star topology (all nodes connect to a central hub/switch) and Bus topology (all devices share a single backbone cable terminated at both ends).

**Q3 (3M) Explain Transmission Modes: Simplex, Half-Duplex, and Full-Duplex.**

**Modes:** Simplex is strictly one-way (keyboard to monitor). Half-Duplex is bidirectional but one direction at a time (walkie-talkie). Full-Duplex is simultaneous two-way transmission (telephone conversation).

**Q4 (3M) What is Modulation?**

**Definition:** The process of varying one or more properties (amplitude, frequency, phase) of a high-frequency carrier wave with a data-bearing modulating signal for long-distance transmission.

**Q5 (3M) What is Bit Rate vs Baud Rate?**

**Definitions:** Bit rate is the number of bits transmitted per second (bps). Baud rate is the number of signal unit changes (symbols) per second. \text{Bit Rate} = \text{Baud Rate} \times \text{Bits per Baud}.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) Explain any one protocol used for flow control in noisy channel: Stop-and-Wait ARQ.**

**Stop-and-Wait ARQ:** Sender sends frame with sequence number (0 or 1) and starts timer. If positive ACK received, next frame is sent. If ACK lost or timer expires, frame is retransmitted. Handles lost frames and duplicate frames using 1-bit sequence numbering.

**Q7 (7.5M) Explain Fiber Distributed Data Interface (FDDI) operation.**

**FDDI:** High-speed 100 Mbps token-passing optical fiber network based on dual counter-rotating token rings (Primary and Secondary). If the primary ring breaks, the dual ring self-heals by wrapping around, ensuring fault-tolerant operation.

**Q8 (7.5M) Explain Distance Vector Routing Algorithm.**

**Explanation:** Routers periodically advertise cost estimates to all known destinations to immediate neighbors. Uses Bellman-Ford algorithm to compute least-cost paths.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain ISO/OSI Reference Model in detail with all seven layers and functions.**

**OSI Model:** Complete 7-layer functional architecture, services, and protocols.

---

**Q10 (15M)** **Explain Sliding Window Protocols: Go-Back-N and Selective Repeat with comparisons.**

**Protocols:** Window sizing, ACK mechanism (cumulative vs individual), buffering requirements, and network utilization efficiency.

---

**Q11 (15M)** **Explain Error Detection using CRC (Cyclic Redundancy Check) with complete mathematical example.**

**CRC Algorithm:** Bit addition, modulo-2 division, remainder computation, and receiver verification.

---

**Q12 (15M)** **Explain Guided Media: Twisted Pair, Coaxial Cable, and Optical Fiber with advantages and limitations.**

**Guided Media:** Full physical characteristics, bandwidth capacity, attenuation rates, and shielding.

---

**Q13 (15M)** **Explain Circuit Switching, Packet Switching, and Message Switching with detailed comparison.**

**Switching Methods:** Complete comparison of connection setup, transmission delays, and resource utilization.

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-503: Computer Networks

**Session:** November 2019 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Differentiate between Bridges and Repeaters.**

**Differences:** Repeaters operate at the Physical Layer (Layer 1) to regenerate electrical signals without looking at frames. Bridges operate at the Data Link Layer (Layer 2) to inspect MAC addresses and filter traffic between segments.

**Q2 (3M) What are the services provided by the Transport Layer?**

**Services:** Connection-oriented delivery (TCP) vs connectionless datagram delivery (UDP), port-based process addressing, segmentation/reassembly, end-to-end flow control, and error recovery.

**Q3 (3M) Define Modulation and Demodulation.**

**Explanation:** Modulation transforms digital bitstreams into analog signals for transmission across physical media. Demodulation recovers original digital bits from incoming modulated analog waves.

**Q4 (3M) What is Network Topology? Name two topologies with advantages.**

**Topologies:** Star topology (central hub, easy fault isolation) and Mesh topology (point-to-point dedicated links, highest fault tolerance).

**Q5 (3M) What is Shannon Capacity Theorem?**

**Theorem:** C = B \log_2(1 + SNR) gives the theoretical upper bound on data transmission speed in bits per second over a noisy channel.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) Explain Dijkstra's Algorithm with an example network.**

**Dijkstra's Algorithm:** Finds single-source shortest path to all destinations by maintaining tentative distance labels and iteratively selecting the minimum-cost unvisited node.

**Q7 (7.5M) Explain Distance Vector Routing and the Count-to-Infinity problem.**

**Distance Vector Routing:** Uses Bellman-Ford algorithm; slow reaction to link failures causes loops (count-to-infinity), resolved via split horizon.

**Q8 (7.5M) Explain Sliding Window Protocol for Flow Control.**

**Sliding Window:** Permits sending multiple frames within a window without waiting for immediate individual acknowledgments.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M) Explain ISO/OSI Reference Model in detail with all seven layers.**

**OSI Architecture:** Full 7-layer functional breakdown and data encapsulation process.

**Q10 (15M)** **Explain Guided Media: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.**

**Transmission Media:** Construction, signal propagation, attenuation characteristics, and network cabling standards (Cat5e, Cat6, Single-mode/Multi-mode fiber).

---

**Q11 (15M)** **Explain Error Detection using CRC (Cyclic Redundancy Check) with numerical problem.**

**CRC Problem:** Generator polynomial division, bit stuffing, and frame validation on receiver side.

---

**Q12 (15M)** **Explain Circuit Switching, Packet Switching, and Message Switching with comparison.**

**Switching Techniques:** Comprehensive comparison of latency, bandwidth efficiency, and modern telecom implementations.

---

**Q13 (15M)** **Explain Congestion Control in Computer Networks: Leaky Bucket and Token Bucket Algorithms.**

**Traffic Shaping:** Deep dive into burst mitigation, token accumulation, and packet scheduling.

---

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-503: Computer Networks

**Session:** December 2020 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Four channels, each with 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between channels?**

**Solution:** N = 4 channels of 100 KHz each.
Number of guard bands required between 4 channels = 4 - 1 = 3.
Each guard band = 10 KHz.
Total minimum bandwidth:
B = (4 \times 100) + (3 \times 10) = 400 + 30 = \mathbf{430 \text{ KHz}}

**Q2 (3M) Discuss Point-to-Point Protocol (PPP).**

**PPP:** A data link layer protocol used to establish a direct connection between two nodes over dial-up, serial cable, or broadband (PPPoE). Provides authentication via PAP/CHAP, framing, and dynamic IP configuration (IPCP).

**Q3 (3M) Explain the functions of the Presentation Layer.**

**Functions:** Data syntax translation, character code conversion, data compression (lossless/lossy), and data encryption/decryption (cryptography).

**Q4 (3M) Differentiate between LAN, MAN, and WAN.**

**Comparison:** LAN covers a room or building (high speed, private ownership). MAN covers a city (cable TV network). WAN covers a country or globe (telecom lines, satellite, Internet).

**Q5 (3M) What is a Collision Domain vs Broadcast Domain?**

**Definitions:** A Collision Domain is a network segment where simultaneous transmissions collide (divided by switches). A Broadcast Domain is a segment receiving all broadcast frames (divided by routers).

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) How does DTE-DCE interface work? Discuss the essence of Modems.**

**DTE-DCE Interface:** DTE (Data Terminal Equipment, e.g. computer/router) generates digital data. DCE (Data Circuit-terminating Equipment, e.g. modem) converts signals for transmission across the physical telephone link. Standards include RS-232 / EIA-232.

**Q7 (7.5M) Explain Error Detection using CRC with generator polynomial $x^3 + x + 1$.**

**CRC Calculation:** Complete modulo-2 polynomial division with bit representation `1011`.

**Q8 (7.5M) Explain Distance Vector Routing Algorithm.**

**Algorithm:** Periodic exchange of routing tables between adjacent neighbors based on Bellman-Ford algorithm.

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain OSI Reference Model with detailed layer descriptions.**

**7-Layer Model:** Detailed functions, protocols, and data units (Bits, Frames, Packets, Segments, Data).

---

**Q10 (15M)** **Explain Sliding Window Protocols: Stop-and-Wait, Go-Back-N, and Selective Repeat.**

**Flow Control:** Efficiency equations, window buffering, and timeout retransmission mechanisms.

---

**Q11 (15M)** **Explain Transmission Media: UTP, STP, Coaxial, and Fiber Optic Cable.**

**Transmission Media:** Attenuation, propagation velocity, bandwidth, and EMI immunity analysis.

---

**Q12 (15M)** **Explain Congestion Control Algorithms: Leaky Bucket and Token Bucket.**

**Traffic Shaping:** Smooth constant-rate leakage vs bursty-tolerant token generation.

---

**Q13 (15M)** **Explain Switching Techniques: Circuit, Message, and Packet Switching.**

**Switching:** Detailed comparison of setup time, propagation delay, transmission delay, and queuing overhead.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-503: Computer Networks

**Session:** December 2021 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

**Q1 (7.5M)** **Explain the functions of the Session Layer in the OSI model.**

**Session Layer (Layer 5):** Responsible for establishing, maintaining, synchronizing, and terminating communication sessions between applications. Controls dialog (simplex/half-duplex/full-duplex) and inserts synchronization checkpoints into data streams to facilitate restart upon failure.

---

**Q2 (7.5M)** **Write short notes on: (a) Throughput & Wavelength, (b) Synchronous Protocols, (c) Routers.**

**Notes:**

- **(a) Throughput & Wavelength:** Throughput is the actual measurement of data transferred across a network per second. Wavelength (\lambda = v / f) is the physical distance between consecutive signal wave peaks in optical communication.
- **(b) Synchronous Protocols:** Continuous transmission of data blocks synchronized to a shared clock, using sync bytes/flags (e.g. HDLC).
- **(c) Routers:** Network Layer (Layer 3) internetworking devices that inspect destination IP addresses and make packet forwarding decisions based on routing tables.

---

**Q3 (7.5M)** **Discuss ISDN, its services, and layer structure.**

**ISDN (Integrated Services Digital Network):** Transmits voice, data, and video simultaneously over standard telephone copper lines. Channels: B-channel (64 kbps, user data), D-channel (16 or 64 kbps, control signaling). BRI provides 2B+D (144 kbps); PRI provides 23B+D (1.544 Mbps) or 30B+D (2.048 Mbps).

---

**Q4 (7.5M)** **Explain Flow Control and Error Control in Computer Networks.**

**Mechanisms:** Flow control ensures a fast sender does not overwhelm a slow receiver (Sliding Window). Error control detects corrupted frames via CRC and retransmits them via ARQ schemes (Stop-and-Wait, Go-Back-N, Selective Repeat).

---

**Q5 (7.5M)** **Explain TCP Packet Format (Header Structure) in detail.**

**TCP Header Fields:**

- Source Port (16 bits) and Destination Port (16 bits)
- Sequence Number (32 bits) and Acknowledgment Number (32 bits)
- Header Length / Data Offset (4 bits) and Reserved bits (6 bits)
- Control Flags (6 bits: URG, ACK, PSH, RST, SYN, FIN)
- Window Size (16 bits) for flow control
- Checksum (16 bits) and Urgent Pointer (16 bits)

---

## SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

**Q6 (15M)** **Explain Seven Layers of ISO/OSI Reference Model with their functions.**

**Complete OSI Model:** Detailed explanation of Physical, Data Link, Network, Transport, Session, Presentation, and Application layers.

---

## SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

**Q7 (22.5M)** **Explain Routing Algorithms: Shortest Path Routing (Dijkstra) and Distance Vector Routing with Count-to-Infinity problem.**

**Routing Algorithms:** Dijkstra link-state calculation vs Bellman-Ford distributed vector routing with split horizon and poison reverse solutions.

---

**Q8 (22.5M)** **Explain Sliding Window Protocols: Go-Back-N and Selective Repeat with window sizes and frame sequences.**

**Sliding Window:** Mathematical condition for maximum window sizes: Go-Back-N ($W_s \le 2^m - 1$) and Selective Repeat ($W_s + W_r \le 2^m \implies W_s \le 2^{m-1}$) to prevent ambiguity with duplicate sequence numbers.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-503: Computer Networks

**Session:** December 2023 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Explain the use of Bridges in Computer Networks.**

**Definition:** A **Bridge** is a Data Link Layer (Layer 2) device that connects two separate LAN segments running the same protocol, inspecting MAC addresses to filter frames and segment collision domains.

**Q2 (3M) Write a short note on E-Mail and its core protocols.**

**Explanation:** Electronic mail architecture consists of User Agents (UAs) and Message Transfer Agents (MTAs). Uses SMTP for sending emails, and POP3 / IMAP for downloading emails to client devices.

**Q3 (3M) What is Network Topology? Name major topologies.**

**Topologies:** The geometric layout of interconnected nodes: Star, Bus, Ring, Mesh, Tree, and Hybrid.

**Q4 (3M) What is Attenuation in transmission media?**

**Definition:** The progressive loss of signal energy/strength as it propagates through transmission media, measured in decibels (dB), compensated using amplifiers or repeaters.

**Q5 (3M) Define Bandwidth and Throughput.**

**Definitions:**

- **Bandwidth:** The maximum theoretical data-carrying capacity of a communication channel (bits per second).
- **Throughput:** The actual rate of successfully transmitted data delivered across the network per unit time.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M) Describe the functions & responsibilities of the Session Layer.**

**Session Layer (Layer 5):**

- **Session Establishment, Maintenance, and Termination:** Manages dialogs between communicating processes.
- **Dialog Control:** Determines whether communication is simplex, half-duplex, or full-duplex.
- **Synchronization / Checkpointing:** Inserts sync tokens into data streams to allow recovery from checkpoints in case of interruptions.

**Q7 (7.5M)** **Differentiate between OSI Model and TCP/IP Model.**

**Comparison:**

| Aspect | OSI Reference Model | TCP/IP Model |
|---|---|---|
| Layers | 7 layers | 4 layers |
| Origin | Theoretical standard created by ISO. | Practical protocol suite developed for ARPANET. |
| Session & Presentation | Separate distinct layers. | Integrated into Application layer. |
| Protocols | Protocols developed after model. | Model developed based on existing protocols. |

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**Q8 (7.5M)** **Explain IP Addressing and Subnetting.**

**IP Addressing:** Classes A, B, C, D, E. Subnetting borrows host bits to divide large networks into smaller subnets.

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **What is Congestion Control? How is Congestion Control different from Flow Control? Explain various congestion control techniques.**

**Congestion vs Flow Control:**

- **Flow Control:** Point-to-point mechanism preventing a fast sender from overwhelming a single slow receiver (Sliding Window).
- **Congestion Control:** Network-wide mechanism preventing global traffic load from exceeding overall subnet router buffer capacity.

**Techniques:** Leaky Bucket algorithm, Token Bucket algorithm, Choke Packets, and Random Early Detection (RED).

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**Q10 (15M)** **Explain Error Control in Data Link Layer using CRC with a complete numerical example.**

**CRC Numerical Example:**
Data: `1010000`, Generator: `1001`.
Append 3 zeros: `1010000000`. Perform modulo-2 binary division by `1001` to calculate remainder (FCS). Complete step-by-step arithmetic shown.

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**Q11 (15M)** **Explain Dijkstra's Shortest Path Routing Algorithm with a network graph example.**

**Dijkstra's Algorithm:** Greedy shortest path calculation from source router to all other nodes maintaining visited sets and updating tentative distances.

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**Q12 (15M)** **Explain Transmission Media: Twisted Pair, Coaxial Cable, and Optical Fiber.**

**Guided Media:** Bandwidth, attenuation, electromagnetic immunity, and cost trade-offs across all three physical media.

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**Q13 (15M)** **Write short notes on: (a) Firewalls, (b) FTP, (c) Domain Name System (DNS).**

**Detailed Notes:** Packet filtering firewalls, dual-port FTP architecture (Control port 21, Data port 20), and hierarchical DNS domain resolution.

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# BCA-503: Computer Networks

**Session:** December 2024 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M)** **What is the need for a Computer Network?**

**Need for Computer Networks:** Resource sharing (printers, storage servers), centralized data management, cost reduction, electronic communication (email, video conferences), and distributed processing power.

**Q2 (3M)** **Define HDLC (High-Level Data Link Control).**

**Definition: HDLC** is a bit-oriented synchronous data link layer protocol developed by ISO that provides reliable point-to-point and multipoint communication using frame delimitation (flag byte `01111110` with bit stuffing), error detection, and sliding window flow control.

**Q3 (3M)** **Explain the concept of Piggybacking.**

**Definition: Piggybacking** is a flow control optimization technique where an acknowledgment (ACK) for received data is delayed and attached into the header of the next outgoing data frame sent back to the sender, eliminating the overhead of transmitting independent acknowledgment packets.

**Q4 (3M)** **Compare Datagram Approach and Virtual Circuit Approach in Packet Switching.**

**Comparison:**

| Aspect | Datagram Approach (Connectionless) | Virtual Circuit Approach (Connection-Oriented) |
|---|---|---|
| Call Setup | None; packets transmitted independently. | Setup phase required before transmission. |
| Path | Each packet routed independently; may arrive out of order. | All packets follow identical pre-configured path in sequence. |
| Header Overhead | Large (full destination IP on every packet). | Small (short Virtual Circuit Identifier - VCI). |

**Q5 (3M)** **Describe how SMTP protocol is used in E-Mail applications.**

**SMTP (Simple Mail Transfer Protocol):** A push protocol running over TCP port 25 that transfers email messages from a sender's user agent/client to the sender's mail server, and from the sender's mail server to the recipient's mail server across the Internet.

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Discuss the functions of Transport, Presentation, and Application layers in the OSI model.**

**Functions:**

- **Transport Layer:** End-to-end segmentation, port addressing, connection management, reliable delivery, and flow control.
- **Presentation Layer:** Data translation (syntax formatting), encryption/decryption, and data compression.
- **Application Layer:** Direct user interfaces for network services (HTTP, FTP, SMTP, DNS).

**Q7 (7.5M)** **Explain the switching methods in Computer Networks.**

**Switching Methods:** Circuit Switching (dedicated physical communication path established before data transfer), Packet Switching (data split into discrete packets routed dynamically), and Message Switching (entire message stored and forwarded hop-by-hop).

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**Q8 (7.5M)** **Explain in brief: (a) Guided Media, (b) Transmission Modes.**

**(a) Guided Media:** Physical physical conductors that guide signals along a specific path: Twisted Pair (UTP/STP), Coaxial Cable, and Optical Fiber.
**(b) Transmission Modes:**

- *Simplex:* Unidirectional transmission (e.g., keyboard to CPU, TV broadcast).
- *Half-Duplex:* Bidirectional, but only one direction at a time (e.g., Walkie-Talkie).
- *Full-Duplex:* Simultaneous bidirectional transmission (e.g., telephone call).

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **What are the layers of the ISO/OSI Protocol Stack? Briefly list their functions.**

**Complete OSI Architecture:** Physical, Data Link, Network, Transport, Session, Presentation, and Application layers with detailed protocol functions.

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**Q10 (15M)** **Explain in detail about the Error & Flow Control mechanisms employed at the Data Link Layer.**

**Mechanisms:** Framing methods (byte stuffing, bit stuffing), error detection (CRC-32), and flow control ARQ schemes (Stop-and-Wait, Go-Back-N, Selective Repeat).

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**Q11 (15M)** **(a) Explain Shannon Capacity with examples. (b) What is CSMA? List protocols used with CSMA.**

**(a) Shannon Capacity Theorem:** Gives theoretical maximum data rate C of a channel with bandwidth B and Signal-to-Noise Ratio SNR:
$C = B \log_2(1 + SNR) \text{ bps}$
*Example:* For telephone line with B = 3000 Hz and SNR = 3162 (35 dB): $C = 3000 \log_2(1 + 3162) \approx 3000 \times 11.62 = 34,860 \text{ bps} \approx 34.8 \text{ kbps}$.

**(b) CSMA Protocols:** 1-Persistent CSMA, Non-Persistent CSMA, p-Persistent CSMA, and CSMA/CD / CSMA/CA.

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**Q12 (15M)** **(a) Explain ISDN Architecture and Services. (b) Describe Congestion Control Algorithms.**

**(a) Integrated Services Digital Network (ISDN):** Transmits voice, video, and data over digital telephone networks. BRI (2B+D = 144 kbps) and PRI (23B+D / 30B+D = 1.544 / 2.048 Mbps).
**(b) Congestion Control:** Open-loop (prevention via traffic policing) vs Closed-loop (reaction via choke packets and Leaky/Token Bucket algorithms).

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**Q13 (15M)** **Describe any three: (a) SSL, (b) Modems, (c) Line Configuration, (d) Distortion vs Noise.**

**Short Notes:**

- **SSL (Secure Sockets Layer):** Cryptographic protocol providing end-to-end encryption, authentication, and integrity over TCP.
- **Modems (Modulator-Demodulator):** Converts digital computer signals to analog audio frequencies for transmission over analog lines and vice versa.
- **Line Configuration:** Point-to-Point (dedicated link between two devices) vs Multipoint (multidrop link where multiple devices share a single link).
- **Distortion vs Noise:** Distortion is alteration of signal shape caused by different frequency phase delays in transmission media. Noise is unwanted external thermal/electromagnetic energy added into the signal.

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# BCA-503: Computer Networks

**Session:** December 2025 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

### Q1 (3M) What is a MAC address? How is it different from an IP Address?

**Comparison:**

| Feature | MAC Address (Physical Address) | IP Address (Logical Address) |
|---|---|---|
| Layer | Data Link Layer (Layer 2). | Network Layer (Layer 3). |
| Size & Format | 48 bits (6 bytes), hex (e.g., `00:1A:2B:3C:4D:5E`). | 32 bits (IPv4, e.g., `192.168.1.1`) or 128 bits (IPv6). |
| Nature | Hardcoded by manufacturer into NIC; globally fixed. | Dynamically/statically assigned based on network location. |

### Q2 (3M) Differentiate between a Hub and a Switch.

**Differences:**

- **Hub (Physical Layer - Layer 1):** Broadcast device that repeats every incoming signal across all ports, sharing a single collision domain. Half-duplex, no intelligence.
- **Switch (Data Link Layer - Layer 2):** Intelligent device maintaining a MAC Address Table to forward frames directly to designated destination ports. Dedicated collision domains per port, full-duplex.

### Q3 (3M) What is Subnetting? Give an example.

**Definition: Subnetting** is the process of partitioning a single physical network into two or more smaller logical subnets by borrowing host bits to create subnet bits, reducing broadcast traffic and optimizing address allocation.
**Example:** Given Class C network `192.168.1.0/24` with default mask `255.255.255.0`. Borrowing 2 host bits gives subnet mask `255.255.255.192` (/26), creating 2^2 = 4 subnets with 62 usable hosts each:

- Subnet 1: `192.168.1.0` to `192.168.1.63`
- Subnet 2: `192.168.1.64` to `192.168.1.127`
- Subnet 3: `192.168.1.128` to `192.168.1.191`
- Subnet 4: `192.168.1.192` to `192.168.1.255`

### Q4 (3M) What is CSMA/CD? How does it operate in Ethernet?

**Carrier Sense Multiple Access with Collision Detection:** A media access protocol used in shared half-duplex Ethernet:

1. **Carrier Sense:** Station listens before transmitting ("listen before talk"). If channel is idle, it begins transmitting.
2. **Collision Detection:** While transmitting, station monitors signal level. If collision occurs, it transmits a *jam signal* to notify all stations.
3. **Backoff:** Stations wait a random backoff period using the Binary Exponential Backoff algorithm before retransmitting.

**Q5 (3M)** **Differentiate between Synchronous and Asynchronous Protocols.**

**Comparison:**

- **Asynchronous Transmission:** Characters sent independently with start (0) and stop (1) framing bits. Clocks do not need synchronization. Inefficient for large high-speed data transfers.
- **Synchronous Transmission:** Continuous stream of data blocks transmitted synchronized to an external shared clock. Synchronized using sync character preambles (e.g., HDLC, PPP), achieving high throughput.

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## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Describe Multiplexing. Explain FDM, TDM, and WDM with diagrams.**

**Definition:** Combining multiple separate data streams into a single composite signal transmitted over a shared transmission medium.

**Techniques:**

- **FDM (Frequency Division Multiplexing):** Divides channel bandwidth into separate non-overlapping frequency bands with guard bands (Analog signals, radio, cable TV).
- **TDM (Time Division Multiplexing):** Interleaves time slots from multiple sources onto a high-speed digital carrier (Synchronous vs Statistical TDM).
- **WDM (Wavelength Division Multiplexing):** Multiplexes optical signals of differing wavelengths (colors) along a single optical fiber.

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**Q7 (7.5M)** **Explain Error Detection Techniques: Parity Check, Checksum, and CRC (Cyclic Redundancy Check).**

**Error Detection:**

- **Parity Check:** Appends single parity bit (even or odd parity). Detects only odd numbers of bit errors.
- **Checksum:** Divides data into 16-bit words, adds them using 1's complement arithmetic, and appends inverted sum (used in TCP/IP).
- **Cyclic Redundancy Check (CRC):** Treats bit stream as polynomial coefficients and performs modulo-2 binary division by generator polynomial G(x). Appends remainder (FCS). Detects all single, double, odd, and burst errors up to generator length!

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**Q8 (7.5M)** **Compare Circuit Switching, Packet Switching, and Message Switching.**

**Comparison:**

| Feature | Circuit Switching | Packet Switching | Message Switching |
|---|---|---|---|
| Dedicated Path | Yes (setup before transfer). | No (packets routed dynamically). | No (store-and-forward per hop). |
| Bandwidth | Reserved/wasted if idle. | Dynamically shared on-demand. | Dynamically shared. |
| Delay | Setup delay high; transmission delay low. | Jitter/variable queuing delay. | High storage delay at each switch. |
| Modern Use | PSTN landline telephone. | Internet (IP networks). | Early telegraph, postal email. |

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Explain the ISO/OSI Reference Model in detail. Describe the primary functions of each of the seven layers.**

**Seven Layers of OSI Model:**

1. **Physical Layer (Layer 1):** Transmission of raw unstructured bit streams over physical media (voltages, connectors, cables).
2. **Data Link Layer (Layer 2):** Framing, physical addressing (MAC), flow control, error control (CRC), and media access control.
3. **Network Layer (Layer 3):** Logical addressing (IP), routing (Dijkstra, distance vector), packet forwarding, and fragmentation.
4. **Transport Layer (Layer 4):** End-to-end process-to-process delivery (ports), segmentation and reassembly, connection control (TCP handshake), reliable flow control.
5. **Session Layer (Layer 5):** Dialogue control (simplex, half/full duplex), session checkpointing, synchronization tokens.
6. **Presentation Layer (Layer 6):** Data translation, character code conversion (ASCII/EBCDIC), compression, encryption/decryption (SSL/TLS).
7. **Application Layer (Layer 7):** Network virtual terminals, file transfer (FTP), web browsing (HTTP), email services (SMTP).

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**Q10 (15M)** **Explain Sliding Window Protocols: Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ with efficiency analysis.**

**Flow and Error Control Protocols:**

- **Stop-and-Wait ARQ:** Sender transmits 1 frame, starts timer, and waits for ACK before sending next. Efficiency: \eta = \frac{1}{1 + 2a} where a = T_p / T_t. Poor for high bandwidth-delay links.
- **Go-Back-N ARQ:** Sender window size W_s = 2^m - 1, receiver window W_r = 1. Sender can transmit multiple frames before receiving ACK. On loss, retransmits lost frame and all subsequent frames. Cumulative ACKs.
- **Selective Repeat ARQ:** Sender window W_s = 2^{m-1}, receiver window W_r = 2^{m-1}. Receiver accepts out-of-order frames within window. Sender retransmits *only the specific corrupted/timed-out frames* (highest throughput!).

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**Q11 (15M)** **Explain Routing Algorithms: (a) Distance Vector Routing (Bellman-Ford) and (b) Link State Routing (Dijkstra). Discuss Count-to-Infinity problem.**

**Routing Algorithms:**

1. **Distance Vector Routing:** Each router maintains a routing table containing distance to each destination and next hop. Routers periodically share complete tables with immediate neighbors. Uses Bellman-Ford algorithm. Vulnerable to *Count-to-Infinity problem* when links fail (mitigated by Split Horizon and Poison Reverse).
2. **Link State Routing:** Each router floods Link State Packets (LSP) containing status of its local links to all routers in the entire autonomous system. Every router constructs a global topology map and computes shortest paths using Dijkstra's algorithm (used in OSPF). Fast convergence and no routing loops.

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**Q12 (15M)** **Explain Congestion Control Mechanisms in Computer Networks: Leaky Bucket and Token Bucket Algorithms with comparative analysis.**

**Traffic Shaping Algorithms:**

- **Leaky Bucket:** Discharges data at a strictly constant, uniform output rate regardless of incoming burstiness. If incoming burst exceeds bucket buffer capacity, excess packets overflow and drop. Smooths traffic completely.
- **Token Bucket:** Generates tokens at a constant rate up to bucket capacity. Allows transmission of bursty traffic as long as sufficient tokens exist, but limits sustained rate. More flexible and widely used in modern QoS.

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**Q13 (15M)** **Write detailed notes on: (a) DNS (Domain Name System), (b) Firewalls and Network Security, (c) SMTP and Email Architecture.**

**Detailed Notes:**

- **DNS:** Hierarchical distributed database mapping hostnames ( `www.google.com` ) to IP addresses ( `142.250.190.46` ) over UDP/TCP port 53.
- **Firewalls:** Security barrier controlling inbound/outbound network traffic based on rules. Types: Packet Filtering, Stateful Inspection, Proxy/Application-level gateways.
- **Email Architecture:** User Agent (UA) composes email, Mail Transfer Agent (MTA) transmits using SMTP (port 25/587), and Mail Access Agent (MAA) retrieves email to client via POP3 (port 110) or IMAP (port 143).

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# BCA-504: Numerical Methods

**Session:** December 2018 Examination | **Paper Code:** 18024 | **Complete 100% Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION – A (VERY SHORT ANSWER QUESTIONS • 3 MARKS EACH • ALL 5 COMPULSORY)

**Q1. Obtain a real root of the equation $f(x) = x^3 + x^2 - 1 = 0$ using Bisection method correct to 3 decimal places.** [3 Marks]

**Solution:**

Let $f(x) = x^3 + x^2 - 1$.

- For $x = 0$: $f(0) = 0 + 0 - 1 = -1 < 0$
- For $x = 1$: $f(1) = 1 + 1 - 1 = +1 > 0$

Since $f(0) \cdot f(1) < 0$, a real root lies in the open interval **(0, 1)**.

| n | a | b | c = (a+b)/2 | f(c) | Sign | Updated Interval |
|---|---|---|---|---|---|---|
| 1 | 0.0000 | 1.0000 | 0.5000 | -0.6250 | Negative (< 0) | [0.5000, 1.0000] |
| 2 | 0.5000 | 1.0000 | 0.7500 | -0.0156 | Negative (< 0) | [0.7500, 1.0000] |
| 3 | 0.7500 | 1.0000 | 0.8750 | +0.4355 | Positive (> 0) | [0.7500, 0.8750] |
| 4 | 0.7500 | 0.8750 | 0.8125 | +0.1965 | Positive (> 0) | [0.7500, 0.8125] |
| 5 | 0.7500 | 0.8125 | 0.7812 | +0.0872 | Positive (> 0) | [0.7500, 0.7812] |
| 6 | 0.7500 | 0.7812 | 0.7656 | +0.0350 | Positive (> 0) | [0.7500, 0.7656] |
| 7 | 0.7500 | 0.7656 | 0.7578 | +0.0095 | Positive (> 0) | [0.7500, 0.7578] |
| 8 | 0.7500 | 0.7578 | 0.7539 | -0.0031 | Negative (< 0) | [0.7539, 0.7578] |
| 9 | 0.7539 | 0.7578 | 0.7559 | +0.0032 | Positive (> 0) | [0.7539, 0.7559] |
| 10 | 0.7539 | 0.7559 | 0.7549 | +0.00002 | Positive (≥ 0) | [0.7539, 0.7549] |
| 11 | 0.7539 | 0.7549 | 0.7544 | -0.0015 | Negative (< 0) | [0.7544, 0.7549] |

At iteration 11, interval width is $|0.7549 - 0.7544| = 0.0005 < 0.001$.

**Final Converged Root: x ≈ 0.755 (correct to 3 decimal places)**

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**Q2. Define backward differences and prove the operator identity $E\nabla = \nabla E = \Delta$.** [3 Marks]

**Solution:**

**Definition:** For equispaced points $x_i = x_0 + i \cdot h$, the **Backward Difference Operator ($\nabla$)** is defined as:

$$\nabla y_i = y_i - y_{i-1} \quad \text{and} \quad \nabla^k y_i = \nabla^{k-1} y_i - \nabla^{k-1} y_{i-1} \ (k \geq 2)$$

**Backward Difference Table for ordered pairs $(x_0, y_0)$ to $(x_4, y_4)$:**

| x | y | $\nabla y$ | $\nabla^2 y$ | $\nabla^3 y$ | $\nabla^4 y$ |
|---|---|---|---|---|---|
| $x_0$ | $y_0$ | – | – | – | – |
| $x_1$ | $y_1$ | $\nabla y_1 = y_1 - y_0$ | – | – | – |
| $x_2$ | $y_2$ | $\nabla y_2 = y_2 - y_1$ | $\nabla^2 y_2$ | – | – |
| $x_3$ | $y_3$ | $\nabla y_3 = y_3 - y_2$ | $\nabla^2 y_3$ | $\nabla^3 y_3$ | – |
| $x_4$ | $y_4$ | $\nabla y_4 = y_4 - y_3$ | $\nabla^2 y_4$ | $\nabla^3 y_4$ | $\nabla^4 y_4$ |

**Operator Proofs:**

1. $E \bullet \nabla f(x) = E\,[f(x) - f(x-h)] = f(x+h) - f(x) = \Delta f(x) \Rightarrow \boldsymbol{E\nabla = \Delta}$
2. $\nabla \bullet E\, f(x) = \nabla\,[f(x+h)] = f(x+h) - f(x) = \Delta f(x) \Rightarrow \boldsymbol{\nabla E = \Delta}$

**Hence Proved: $E\nabla = \nabla E = \Delta$**

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**Q3. Find the value of $\frac{dy}{dx}$ at $x = 1.2$ for points (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552)...** [3 Marks]

**Solution:**

Given step size $h = 1.2 - 1.0 = 0.2$. Let $x_0 = 1.2$, $y_0 = 3.3201$.

Neighbors: $x_{-1} = 1.0$, $y_{-1} = 2.7183$ and $x_1 = 1.4$, $y_1 = 4.0552$.

Central Difference Formula (First Derivative):

$$f'(x_0) \approx \frac{y_1 - y_{-1}}{2h}$$

$$f'(1.2) = \frac{4.0552 - 2.7183}{2(0.2)} = \frac{1.3369}{0.40} = 3.34225$$

*(Newton's Forward Difference check: $f'(1.2) \approx (1/0.2)[0.7351 - 0.5(0.1627) + (1/3)(0.0361)] = 3.3289$. True derivative $e^{1.2} \approx 3.3201$.)*

**dy/dx at x = 1.2 is 3.34225**

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**Q4. Solve the following linear system using Gauss Elimination method:** [3 Marks]

$$10x + y + z = 12$$
$$2x + 10y + z = 13$$
$$x + y + 3z = 5$$

**Solution:**

**Step 1: Form the Augmented Matrix [A | B]:**

$$\left(\begin{array}{ccc|c} 10 & 1 & 1 & 12 \\ 2 & 10 & 1 & 13 \\ 1 & 1 & 3 & 5 \end{array}\right)$$

**Step 2: Forward Elimination to Upper Triangular Form:**

- $R_2 \to R_2 - (2/10)R_1 = R_2 - 0.2R_1 \Rightarrow$ [0, 9.8, 0.8 | 10.6]
- $R_3 \to R_3 - (1/10)R_1 = R_3 - 0.1R_1 \Rightarrow$ [0, 0.9, 2.9 | 3.8]
- $R_3 \to R_3 - (0.9 / 9.8)R_2 \Rightarrow$ [0, 0, 2.82653 | 2.82653]

$$\left(\begin{array}{ccc|c} 10 & 1 & 1 & 12 \\ 0 & 9.8 & 0.8 & 10.6 \\ 0 & 0 & 2.82653 & 2.82653 \end{array}\right)$$

**Step 3: Back Substitution:**

- From Row 3: 2.82653 • z = 2.82653 ⇒ **z = 1**
- From Row 2: 9.8 • y + 0.8(1) = 10.6 ⇒ 9.8 • y = 9.8 ⇒ **y = 1**
- From Row 1: 10 • x + 1(1) + 1(1) = 12 ⇒ 10 • x = 10 ⇒ **x = 1**

**Unique Solution: x = 1, y = 1, z = 1**

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**Q5. Solve $\frac{dy}{dx} = x + y^2$, with y(0) = 1 using Picard's method up to second approximation.** [3 Marks]

**Solution:**

Picard's Successive Approximation Formula:

$$y^{(n)}(x) = y_0 + \int_{x_0}^{x} f(t, y^{(n-1)}(t))\, dt$$

Given initial conditions: $x_0 = 0$, $y_0 = 1$, and $f(t, y) = t + y^2$.

**First Approximation ($y^{(1)}$):**

$$y^{(1)}(x) = 1 + \int_0^x (t + 1^2)\, dt = 1 + \left[\frac{t^2}{2} + t\right]_0^x = 1 + x + \frac{x^2}{2}$$

**Second Approximation ($y^{(2)}$):**

$$y^{(2)}(x) = 1 + \int_0^x \left[t + (1 + t + \frac{t^2}{2})^2\right] dt$$

Expand: $(1 + t + \frac{t^2}{2})^2 = 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4}$

$$\text{Integrand} = t + 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4} = 1 + 3t + 2t^2 + t^3 + \frac{t^4}{4}$$

$$y^{(2)}(x) = 1 + \left[t + \frac{3t^2}{2} + \frac{2t^3}{3} + \frac{t^4}{4} + \frac{t^5}{20}\right]_0^x$$

$$y^{(2)}(x) = 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{20}x^5$$

**$y^{(1)} = 1 + x + x^2/2$ | $y^{(2)} = 1 + x + (3/2)x^2 + (2/3)x^3 + (1/4)x^4 + (1/20)x^5$**

## SECTION – B (SHORT ANSWER QUESTIONS • 7.5 MARKS EACH • ATTEMPT ANY TWO)

**Q6. Find a real root of the equation $2x = \log_{10}(x) + 7$ between 3 and 4 using Method of False Position (Regula Falsi).** [7.5 Marks]

**Solution:**

Let $f(x) = 2x - \log_{10}(x) - 7 = 0$.

- For $x_0 = 3$: $f(3) = 2(3) - \log_{10}(3) - 7 = 6 - 0.47712 - 7 = -1.47712 < 0$
- For $x_1 = 4$: $f(4) = 2(4) - \log_{10}(4) - 7 = 8 - 0.60206 - 7 = +0.39794 > 0$

Since $f(3)$ and $f(4)$ have opposite signs, the root lies in the interval **(3, 4)**.

**Regula Falsi Iteration Formula:**

$$x_2 = \frac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$$

**Iteration 1:**

$$x_2 = \frac{3(0.39794) - 4(-1.47712)}{0.39794 - (-1.47712)} = \frac{1.19382 + 5.90848}{1.87506} = \frac{7.10230}{1.87506} \approx 3.78777$$

Evaluate $f(x_2)$:

$f(3.78777) = 2(3.78777) - \log_{10}(3.78777) - 7 = 7.57554 - 0.57838 - 7 = -0.00284 < 0$.

Root lies in **[3.78777, 4.00000]**.

**Iteration 2:**

$$x_3 = \frac{3.78777(0.39794) - 4(-0.00284)}{0.39794 - (-0.00284)} = \frac{1.50731 + 0.01136}{0.40078} = \frac{1.51867}{0.40078} \approx 3.7893$$

Evaluate $f(x_3)$: $f(3.7893) = 2(3.7893) - \log_{10}(3.7893) - 7 = 7.57860 - 0.57856 - 7 = +0.00004 \approx 0$.

**Final Converged Root: x ≈ 3.7893**

---

**Q7. Find the missing terms in the following table using finite differences:** [7.5 Marks]

| x | 0 | 5 | 10 | 15 | 20 | 25 | 30 |
|---|---|---|---|---|---|---|---|
| y | 1 | 3 | ? ($y_2$) | 73 | 225 | ? ($y_5$) | 1153 |

**Solution:**

Let the missing values be $y_2$ (at $x = 10$) and $y_5$ (at $x = 25$).

Since 5 values are given ($y_0, y_1, y_3, y_4, y_6$), the data can be represented by a polynomial of degree 4.

Therefore, the 5th finite difference is identically zero: $\Delta^5 y_0 = 0$ and $\Delta^5 y_1 = 0$.

Using shift operator $\Delta = E - 1$:

$$(E-1)^5 y = (E^5 - 5E^4 + 10E^3 - 10E^2 + 5E - 1)\, y = 0$$

**Equation 1 (applying on $y_0$):**

$$y_5 - 5y_4 + 10y_3 - 10y_2 + 5y_1 - y_0 = 0$$
$$y_5 - 5(225) + 10(73) - 10y_2 + 5(3) - 1 = 0$$
$$y_5 - 1125 + 730 - 10y_2 + 15 - 1 = 0$$
$$\mathbf{-10y_2 + y_5 = 381} \qquad \text{— (Equation 1)}$$

**Equation 2 (applying on $y_1$):**

$$y_6 - 5y_5 + 10y_4 - 10y_3 + 5y_2 - y_1 = 0$$
$$1153 - 5y_5 + 10(225) - 10(73) + 5y_2 - 3 = 0$$
$$1153 - 5y_5 + 2250 - 730 + 5y_2 - 3 = 0$$
$$5y_2 - 5y_5 + 2670 = 0 \Rightarrow y_2 - y_5 = -534 \Rightarrow \mathbf{y_5 - y_2 = 534} \qquad \text{— (Equation 2)}$$

**Solving the Linear System:**

From Equation 2: $y_5 = y_2 + 534$.

Substitute into Equation 1: $-10y_2 + (y_2 + 534) = 381 \Rightarrow -9y_2 = 381 - 534 = -153 \Rightarrow \mathbf{y_2 = 17}$.

Then $y_5 = 17 + 534 = \mathbf{551}$.

**Missing Values: $y_2 = 17$ (at x = 10) | $y_5 = 551$ (at x = 25)**

---

**Q8. Evaluate $I = \int_3^7 x^2 \log_{10}(x)\, dx$ using Simpson's 1/3 rule with $n = 4$.** [7.5 Marks]

**Solution:**

Given limits $a = 3$, $b = 7$, and number of sub-intervals $n = 4$.

Step size: $h = (b - a) / n = (7 - 3) / 4 = \mathbf{1}$.

| i | $x_i$ | Function $f(x) = x^2 \log_{10}(x)$ | Ordinate $y_i$ |
|---|---|---|---|
| 0 | 3 | $3^2 \cdot \log_{10}(3) = 9 \times 0.477121$ | 4.2941 |
| 1 | 4 | $4^2 \cdot \log_{10}(4) = 16 \times 0.602060$ | 9.6330 |
| 2 | 5 | $5^2 \cdot \log_{10}(5) = 25 \times 0.698970$ | 17.4743 |
| 3 | 6 | $6^2 \cdot \log_{10}(6) = 36 \times 0.778151$ | 28.0134 |
| 4 | 7 | $7^2 \cdot \log_{10}(7) = 49 \times 0.845098$ | 41.4099 |

Simpson's 1/3 Rule Formula:

$$I = \frac{h}{3}\left[(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2)\right]$$

Sum of Extremes: $y_0 + y_4 = 4.2941 + 41.4099 = \mathbf{45.7040}$

Sum of Odd Ordinates: $4(y_1 + y_3) = 4(9.6330 + 28.0134) = 4(37.6464) = \mathbf{150.5856}$

Sum of Even Ordinates: $2(y_2) = 2(17.4743) = \mathbf{34.9486}$

Total Bracket Sum $= 45.7040 + 150.5856 + 34.9486 = \mathbf{231.2382}$

$$I = \frac{1}{3} \times 231.2382 = \mathbf{77.0794}$$

**Evaluated Integral: $I \approx 77.0794$**

## SECTION – C (LONG ANSWER QUESTIONS • 15 MARKS EACH • ATTEMPT ANY THREE)

**Q9. Use Newton-Raphson method to find a real root of $\sin(x) = x / 2$ correct to 3 decimal places between π/2 and π.** [15 Marks]

**Solution:**

Let $f(x) = \sin(x) - \frac{x}{2} = 0$.

Differentiating with respect to x: $f'(x) = \cos(x) - 0.5$.

Given interval: [π/2, π] ≈ [1.5708, 3.1416] radians.

Let initial approximation be $\mathbf{x_0 = 2.0}$ **radians**.

Newton-Raphson Iteration Formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

**Iteration 1 (at $x_0 = 2.0$):**

$$f(2.0) = \sin(2.0) - 1.0 = 0.90930 - 1.0 = \mathbf{-0.09070}$$
$$f'(2.0) = \cos(2.0) - 0.5 = -0.41615 - 0.5 = \mathbf{-0.91615}$$
$$x_1 = 2.0 - \frac{-0.09070}{-0.91615} = 2.0 - 0.09900 = \mathbf{1.90100}$$

**Iteration 2 (at $x_1 = 1.90100$):**

$$f(1.90100) = \sin(1.90100) - \frac{1.90100}{2} = 0.94598 - 0.95050 = \mathbf{-0.00452}$$
$$f'(1.90100) = \cos(1.90100) - 0.5 = -0.32422 - 0.5 = \mathbf{-0.82422}$$
$$x_2 = 1.90100 - \frac{-0.00452}{-0.82422} = 1.90100 - 0.00548 = \mathbf{1.89552}$$

**Iteration 3 (at $x_2 = 1.89552$):**

$$f(1.89552) = \sin(1.89552) - \frac{1.89552}{2} = 0.94775 - 0.94776 = \mathbf{-0.00001}$$
$$f'(1.89552) = \cos(1.89552) - 0.5 = -0.31899 - 0.5 = \mathbf{-0.81899}$$
$$x_3 = 1.89552 - \frac{-0.00001}{-0.81899} = 1.89552 - 0.00001 = \mathbf{1.89549}$$

**Root: x ≈ 1.895 radians (or ≈ 108.57°)**

---

**Q10. Find the Lagrange interpolating polynomial of degree 2 for *y = $\log_e(x)$* from the given table. Hence determine *$\log_e(2.7)$*.** [15 Marks]

| x | 2.0 | 2.5 | 3.0 |
|---|---|---|---|
| y = $\log_e$(x) | 0.69315 | 0.91629 | 1.09861 |

**Solution:**

Given $x_0 = 2.0, y_0 = 0.69315$; $x_1 = 2.5, y_1 = 0.91629$; $x_2 = 3.0, y_2 = 1.09861$.

Target value: $x = 2.7$.

Lagrange Interpolation Formula for n = 2:

$$P_2(x) = L_0(x)\, y_0 + L_1(x)\, y_1 + L_2(x)\, y_2$$

**Calculating Lagrange Cardinal Polynomials at x = 2.7:**

$$L_0(2.7) = \frac{(2.7-2.5)(2.7-3.0)}{(2.0-2.5)(2.0-3.0)} = \frac{(0.2)(-0.3)}{(-0.5)(-1.0)} = \frac{-0.06}{0.50} = \mathbf{-0.12}$$
$$L_1(2.7) = \frac{(2.7-2.0)(2.7-3.0)}{(2.5-2.0)(2.5-3.0)} = \frac{(0.7)(-0.3)}{(0.5)(-0.5)} = \frac{-0.21}{-0.25} = \mathbf{+0.84}$$
$$L_2(2.7) = \frac{(2.7-2.0)(2.7-2.5)}{(3.0-2.0)(3.0-2.5)} = \frac{(0.7)(0.2)}{(1.0)(0.5)} = \frac{0.14}{0.50} = \mathbf{+0.28}$$

*Verification: $L_0 + L_1 + L_2 = -0.12 + 0.84 + 0.28 = 1.00$ ✓*

**Evaluating $P_2(2.7)$:**

$$P_2(2.7) = (-0.12)(0.69315) + (0.84)(0.91629) + (0.28)(1.09861)$$
$$= -0.083178 + 0.769684 + 0.307611$$
$$= \mathbf{0.994117 \approx 0.99412}$$

*(Exact analytical value: ln(2.7) ≈ 0.99325; error ≈ 0.00087.)*

**Interpolated Value: $\log_e(2.7) \approx 0.99412$**

---

**Q11. Use Simpson's 3/8 rule to evaluate $\int_0^1 \frac{dx}{1+x}$ with $h = 1/6$.** [15 Marks]

**Solution:**

Given $a = 0$, $b = 1$, step size $h = 1/6 \Rightarrow n = (1 - 0)/(1/6) = 6$ subintervals.

Integrand: $f(x) = \frac{1}{1+x}$.

| i | $x_i$ | Exact $y_i = 1 / (1 + x_i)$ | Decimal Value |
|---|---|---|---|
| 0 | 0 | 1 / 1 | 1.000000 |
| 1 | 1/6 | 1 / (7/6) = 6/7 | 0.857143 |
| 2 | 2/6 | 1 / (8/6) = 6/8 = 3/4 | 0.750000 |
| 3 | 3/6 | 1 / (9/6) = 6/9 = 2/3 | 0.666667 |
| 4 | 4/6 | 1 / (10/6) = 6/10 = 3/5 | 0.600000 |
| 5 | 5/6 | 1 / (11/6) = 6/11 | 0.545455 |
| 6 | 1 | 1 / 2 | 0.500000 |

**Simpson's 3/8 Rule Formula:**

$$I = \frac{3h}{8}\left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)\right]$$

Extremes: $y_0 + y_6 = 1.000000 + 0.500000 = \mathbf{1.500000}$

Non-multiples of 3: $3(y_1 + y_2 + y_4 + y_5) = 3(0.857143 + 0.750000 + 0.600000 + 0.545455) = 3(2.752598)$
$= \mathbf{8.257794}$

Multiples of 3: $2(y_3) = 2(0.666667) = \mathbf{1.333334}$

Sum of brackets $= 1.500000 + 8.257794 + 1.333334 = \mathbf{11.091128}$

$$I = \frac{3(1/6)}{8} \times 11.091128 = \frac{1/2}{8} \times 11.091128 = \frac{11.091128}{16} = \mathbf{0.693196}$$

*(Exact analytical integral: $\int_0^1 dx/(1+x) = [\ln(1+x)]_0^1 = \ln(2) \approx 0.693147$. Highly accurate to 4 decimal places.)*

**Evaluated Integral: I ≈ 0.693196**

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**Q12. Solve the following linear system using Gauss-Seidel iterative method:** [15 Marks]

$$6x + y + z = 20$$
$$x + 4y - z = 6$$
$$x - y + 5z = 7$$

**Solution:**

**Step 1: Check Diagonal Dominance Condition:**

- Equation 1: $|6| > |1| + |1| = 2$ (Dominant in x ✓)
- Equation 2: $|4| > |1| + |-1| = 2$ (Dominant in y ✓)
- Equation 3: $|5| > |1| + |-1| = 2$ (Dominant in z ✓)

The system is strictly diagonally dominant. Convergence is guaranteed.

**Step 2: Express in Iterative Form:**

$$x^{(k+1)} = \frac{20 - y^{(k)} - z^{(k)}}{6}$$
$$y^{(k+1)} = \frac{6 - x^{(k+1)} + z^{(k)}}{4}$$
$$z^{(k+1)} = \frac{7 - x^{(k+1)} + y^{(k+1)}}{5}$$

**Step 3: Iteration Table (Initial Approximation: $x^{(0)} = 0$, $y^{(0)} = 0$, $z^{(0)} = 0$):**

| Iteration (k) | x = (20 – y – z) / 6 | y = (6 – x + z) / 4 | z = (7 – x + y) / 5 |
|---|---|---|---|
| 1 | (20 - 0 - 0)/6 = **3.3333** | (6 - 3.3333 + 0)/4 = **0.6667** | (7 - 3.3333 + 0.6667)/5 = **0.8667** |
| 2 | (20 - 0.6667 - 0.8667)/6 = **3.0778** | (6 - 3.0778 + 0.8667)/4 = **0.9472** | (7 - 3.0778 + 0.9472)/5 = **0.9739** |
| 3 | (20 - 0.9472 - 0.9739)/6 = **3.0131** | (6 - 3.0131 + 0.9739)/4 = **0.9902** | (7 - 3.0131 + 0.9902)/5 = **0.9954** |
| 4 | (20 - 0.9902 - 0.9954)/6 = **3.0024** | (6 - 3.0024 + 0.9954)/4 = **0.9983** | (7 - 3.0024 + 0.9983)/5 = **0.9992** |
| 5 | (20 - 0.9983 - 0.9992)/6 = **3.0004** | (6 - 3.0004 + 0.9992)/4 = **0.9997** | (7 - 3.0004 + 0.9997)/5 = **0.9999** |
| 6 | **3.0000** | **1.0000** | **1.0000** |

**Converged Solution: x = 3, y = 1, z = 1**

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**Q13. Using Fourth-Order Runge-Kutta method, find *y(0.1)* and *y(0.2)* correct to 4 decimal places, given:** [15 Marks]

$$\frac{dy}{dx} = y - x, \quad \text{with initial condition } y(0) = 2$$

**Solution:**

Given $f(x, y) = y - x$, initial point $x_0 = 0$, $y_0 = 2$, step size $h = 0.1$.

**Part 1: Calculation of y(0.1) from $x_0 = 0$, $y_0 = 2$:**

$$k_1 = h \bullet f(x_0, y_0) = 0.1(2 - 0) = \mathbf{0.200000}$$
$$k_2 = h \bullet f(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}) = 0.1 \bullet f(0.05, 2.10) = 0.1(2.10 - 0.05) = \mathbf{0.205000}$$
$$k_3 = h \bullet f(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}) = 0.1 \bullet f(0.05, 2.1025) = 0.1(2.1025 - 0.05) = \mathbf{0.205250}$$
$$k_4 = h \bullet f(x_0 + h, y_0 + k_3) = 0.1 \bullet f(0.1, 2.20525) = 0.1(2.20525 - 0.1) = \mathbf{0.210525}$$
$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$
$$= \frac{1}{6}(0.200000 + 0.410000 + 0.410500 + 0.210525) = \frac{1.231025}{6} \approx \mathbf{0.205171}$$
$$y(0.1) = y_0 + k = 2 + 0.205171 = 2.2052$$

**Part 2: Calculation of y(0.2) from $x_1 = 0.1$, $y_1 = 2.205171$:**

$k_1 = h \bullet f(x_1, y_1) = 0.1(2.205171 - 0.1) = \mathbf{0.210517}$

$k_2 = h \bullet f(0.1 + 0.05, 2.205171 + 0.105258) = 0.1 \bullet f(0.15, 2.310430) = 0.1(2.310430 - 0.15) = \mathbf{0.216043}$

$k_3 = h \bullet f(0.1 + 0.05, 2.205171 + 0.108022) = 0.1 \bullet f(0.15, 2.313193) = 0.1(2.313193 - 0.15) = \mathbf{0.216319}$

$k_4 = h \bullet f(0.1 + 0.1, 2.205171 + 0.216319) = 0.1 \bullet f(0.2, 2.421490) = 0.1(2.421490 - 0.2) = \mathbf{0.222149}$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}(0.210517 + 0.432086 + 0.432638 + 0.222149) = \frac{1.297390}{6} \approx \mathbf{0.216232}$$

$$y(0.2) = y_1 + k = 2.205171 + 0.216232 = 2.4214$$

**y(0.1) = 2.2052 | y(0.2) = 2.4214**

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-504: Numerical Methods

**Session:** November 2019 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M)** **Evaluate $\Delta^2[(3x + 1)(2x + 3)]$ for h = 1.**

**Solution:** Let $f(x) = (3x + 1)(2x + 3) = 6x^2 + 11x + 3$.

We know the property of forward difference of a polynomial: $\Delta^n (a_n x^n + \dots) = a_n n! h^n$.

For degree 2 with $n = 2, a_2 = 6, h = 1$:

$$\Delta^2 [6x^2 + 11x + 3] = 6 \times (2!) \times (1)^2 = 6 \times 2 = \mathbf{12}$$

**Answer:** $\mathbf{12}$.

---

**Q2 (3M)** **Construct a divided difference table for the following data: (1, 3), (2, 12), (4, 84), (7, 504).**

**Solution:**

| x | f(x) | 1st Divided Diff | 2nd Divided Diff | 3rd Divided Diff |
|---|---|---|---|---|
| 1 | 3 | (12 - 3)/(2 - 1) = 9 | (36 - 9)/(4 - 1) = 9 | (28 - 9)/(7 - 1) = 3.17 |
| 2 | 12 | (84 - 12)/(4 - 2) = 36 | (140 - 36)/(7 - 2) = 20.8 | - |
| 4 | 84 | (504 - 84)/(7 - 4) = 140 | - | - |
| 7 | 504 | - | - | - |

---

**Q3 (3M)** **Write the formula for Simpson's 3/8 rule.**

**Solution:** For $n$ subintervals where $n$ is a multiple of 3 and step size $h = \frac{b-a}{n}$:

$$\int_{x_0}^{x_n} f(x) \, dx = \frac{3h}{8} \left[ (y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \dots) + 2(y_3 + y_6 + y_9 + \dots) \right]$$

---

**Q4 (3M)** **Write the Runge-Kutta fourth-order method formula.**

**Solution:** For solving $\frac{dy}{dx} = f(x, y)$ with step size $h$:

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where: $k_1 = h\, f(x_n, y_n)$

$k_2 = h\, f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$

$k_3 = h\, f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$

$k_4 = h\, f(x_n + h, y_n + k_3)$.

---

**Q5 (3M)** **Explain the principle of Gauss Elimination Method.**

**Solution:** Gauss elimination transforms a system of linear equations $Ax = b$ into an equivalent upper triangular system using elementary row operations (row swapping, scalar multiplication, row addition), and then computes the unknowns in reverse order through backward substitution.

---

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Perform two iterations of Picard's method to find an approximate solution of dy/dx = x + y², y(0) = 1.**

**Solution:** $y_0 = 1$.

Iter 1: $y^{(1)}(x) = 1 + \int_0^x (t + 1)\, dt = \mathbf{1 + x + \frac{x^2}{2}}$.

Iter 2: $y^{(2)}(x) = 1 + \int_0^x \left[ t + \left(1 + t + \frac{t^2}{2}\right)^2 \right] dt = \mathbf{1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{x^4}{4} + \frac{x^5}{20}}$.

---

**Q7 (7.5M)** **Using Newton's backward interpolation formula, find f(7.5) from: x = [1, 3, 5, 7, 9], y = [8, 12, 19, 31, 49].**

**Solution:** $h = 2$. Origin $x_n = 9 \implies u = \frac{7.5 - 9}{2} = -0.75$.

Backward differences at $x_n = 9$:

$\nabla y_n = 49 - 31 = 18$, $\nabla^2 y_n = 18 - 12 = 6$, $\nabla^3 y_n = 6 - (12 - 7) = 1$.

Newton Backward formula: $y = 49 + (-0.75)(18) + \frac{-0.75(0.25)}{2}(6) + \frac{-0.75(0.25)(1.25)}{6}(1)$

$= 49 - 13.5 - 0.5625 - 0.039 = \mathbf{34.898} \approx \mathbf{34.9}$.

---

**Q8 (7.5M)** **Compare Trapezoidal Rule and Simpson's 1/3 Rule with respect to error bounds and order of polynomials.**

**Comparison:**

| Feature | Trapezoidal Rule | Simpson's 1/3 Rule |
|---|---|---|
| Approximating Polynomial | Linear (degree 1) | Parabolic (degree 2) |
| Intervals required | Any number of intervals | Must be even number of intervals |
| Global Error Bound | $-(b-a)h^2 f''(\xi) / 12$ | $-(b-a)h^4 f^4(\xi) / 180$ |
| Order of Accuracy | $O(h^2)$ | $O(h^4)$ |
| Exact for polynomials of | Degree ≤ 1 | Degree ≤ 3 |

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Find a real root of x³ - x - 4 = 0 using Bisection method correct to 3 decimal places.**

**Solution:** $f(x) = x^3 - x - 4$.

$f(1) = -4 < 0$, $f(2) = 8 - 2 - 4 = +2 > 0$. Root $\in [1, 2]$.

Iter 1: $x_1 = 1.5 \implies f(1.5) = -2.125 < 0 \Rightarrow [1.5, 2]$.

Iter 2: $x_2 = 1.75 \implies f(1.75) = -0.3906 < 0 \Rightarrow [1.75, 2]$.

Iter 3: $x_3 = 1.875 \implies f(1.875) = +0.7168 > 0 \Rightarrow [1.75, 1.875]$.

Iter 4: $x_4 = 1.8125 \implies f(1.8125) = +0.1419 > 0 \Rightarrow [1.75, 1.8125]$.

Iter 5: $x_5 = 1.78125 \implies f(1.78125) = -0.129 < 0$.

Iter 6: $x_6 = 1.7969 \implies f(1.7969) = +0.005 > 0$.

**Root:** $\mathbf{x \approx 1.796}$.

---

**Q10 (15M)** **Solve the system of equations by Gauss-Seidel Method: 10x + y + z = 12, 2x + 10y + z = 13, 2x + 2y + 10z = 14.**

**Solution:** Diagonally dominant formulas:

$$x = \frac{12 - y - z}{10}, \quad y = \frac{13 - 2x - z}{10}, \quad z = \frac{14 - 2x - 2y}{10}$$

Start $(x_0, y_0, z_0) = (0, 0, 0)$.
Iter 1: $x_1 = 1.200, y_1 = 1.060, z_1 = 0.948$.
Iter 2: $x_2 = 0.9992, y_2 = 1.0054, z_3 = 0.9991$.
Iter 3: $x_3 = 1.0000, y_3 = 1.0000, z_3 = 1.0000$.
**Exact Solution:** $\mathbf{x = 1.0, \; y = 1.0, \; z = 1.0}$.

---

**Q11 (15M)** **Obtain the form of function using Lagrange's interpolation formula from: (0, 1), (1, 3), (3, 49), (4, 129).**

**Solution:** Using Lagrange formula across $x = [0, 1, 3, 4]$, $y = [1, 3, 49, 129]$:
Resulting interpolating polynomial is:

$$\mathbf{y(x) = 2x^3 - x + 1}$$

*Verification:* $y(0)=1$; $y(1)=2-1+1=3$; $y(3)=54-3+1=49$; $y(4)=128-4+1=129$. (Exact!)

---

**Q12 (15M)** **State Runge-Kutta method of second order. Using Runge-Kutta method of fourth order, find the values of y(0.2), y(0.4), and y(0.6) for dy/dx = x + y with y(0) = 1.**

**Solution:**

**1. RK 2nd Order (Heun's / Improved Euler):** $y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2)$ where $k_1 = h\, f(x_n, y_n)$, $k_2 = h\, f(x_n + h, y_n + k_1)$.

**2. RK 4th Order for dy/dx = x + y, y(0) = 1 with h = 0.2:**
- At $x = 0.2$: $\mathbf{y(0.2) = 1.2428}$.
- At $x = 0.4$: $\mathbf{y(0.4) = 1.5836}$.
- At $x = 0.6$: $k_1 = 0.2(0.4 + 1.5836) = 0.3967$, advancing yields $\mathbf{y(0.6) = 2.0442}$.

---

**Q13 (15M)** **Write the conditions for applicability of: (a) Trapezoidal rule, (b) Simpson's 1/3 rule. Evaluate $\int_0^1 dx/(1+x)$ using both rules with h = 0.25.**

**Solution:**

**Conditions:**

(a) *Trapezoidal Rule:* The number of subintervals $n$ can be any positive integer (even or odd).

(b) *Simpson's 1/3 Rule:* The number of subintervals $n$ must be even ($n = 2m$).

**Evaluation for $\int_0^1 dx/(1+x)$, h = 0.25:**

$y_0 = 1.0, y_1 = 0.8, y_2 = 0.6667, y_3 = 0.5714, y_4 = 0.5$.

- *Trapezoidal:* $I_T = \frac{0.25}{2}\,[\,1.5 + 2(0.8 + 0.6667 + 0.5714)\,] = 0.125 \times 5.5762 = \mathbf{0.6970}$.
- *Simpson's 1/3:* $I_S = \frac{0.25}{3}\,[\,1.5 + 4(0.8 + 0.5714) + 2(0.6667)\,] = \frac{0.25}{3}[\,1.5 + 5.4856 + 1.3334\,] = \frac{0.25}{3}(8.319) = \mathbf{0.6932}$.

*Exact value:* $\ln(2) \approx 0.69315$.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-504: Numerical Methods

**Session:** December 2020 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M)** **Find the real root of the equation $x^3 - x - 1 = 0$.**

**Solution:** $f(x) = x^3 - x - 1$.

Using Newton-Raphson method with initial guess $x_0 = 1.3$: $f'(x) = 3x^2 - 1$.

$f(1.3) = 2.197 - 1.3 - 1 = -0.103$.

$f'(1.3) = 3(1.69) - 1 = 4.07$.

$x_1 = 1.3 - \frac{-0.103}{4.07} = 1.3 + 0.0253 = 1.3253$.

$f(1.3253) \approx 0.002$.

**Root:** $\mathbf{x \approx 1.325}$.

---

**Q2 (3M)** **Define operators E, Δ and ∇. Also obtain relation between E and Δ.**

**Definitions:**

1. **Shift Operator E:** $E\,f(x) = f(x + h)$.
2. **Forward Difference Δ:** $\Delta f(x) = f(x + h) - f(x)$.
3. **Backward Difference ∇:** $\nabla f(x) = f(x) - f(x - h)$.

**Relation between E and Δ:**

$\Delta f(x) = f(x + h) - f(x) = E f(x) - f(x) = (E - 1) f(x) \implies \mathbf{\Delta = E - 1 \iff E = 1 + \Delta}$.

---

**Q3 (3M)** **Given dy/dx = (y - x)/(y + x) with y(0) = 1. Find y(0.3) by Euler's method taking h = 0.1.**

**Solution:** $y_{n+1} = y_n + h \frac{y_n - x_n}{y_n + x_n}$, $h = 0.1$.

1. $x_0 = 0, y_0 = 1 \implies y_1 = 1 + 0.1 \left(\frac{1 - 0}{1 + 0}\right) = 1 + 0.1(1) = \mathbf{1.1000}$ at $x = 0.1$.
2. $x_1 = 0.1, y_1 = 1.1 \implies y_2 = 1.1 + 0.1 \left(\frac{1.1 - 0.1}{1.1 + 0.1}\right) = 1.1 + 0.1\left(\frac{1.0}{1.2}\right) = 1.1 + 0.0833 = \mathbf{1.1833}$ at $x = 0.2$.
3. $x_2 = 0.2, y_2 = 1.1833 \implies y_3 = 1.1833 + 0.1 \left(\frac{1.1833 - 0.2}{1.1833 + 0.2}\right) = 1.1833 + 0.1\left(\frac{0.9833}{1.3833}\right) = 1.1833 + 0.0711 = \mathbf{1.2544}$ at $x = 0.3$.

**Answer:** $\mathbf{y(0.3) \approx 1.254}$.

---

**Q4 (3M)** **Define Simpson's three-eighths rule for Numerical Integration.**

**Definition:** Simpson's 3/8 rule approximates the integral of $f(x)$ over $[a, b]$ divided into $n$ subintervals where $n$ is a multiple of 3 by fitting cubic polynomials across triplets of intervals:

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[(y_0 + y_n) + 3\sum_{i \neq 3k} y_i + 2\sum_{i = 3k} y_i\right]$$

The truncation error is of order $O(h^4)$, specifically $E = -\frac{3}{80} h^5 f^{(4)}(\xi)$.

---

**Q5 (3M)** **What do you understand by Gauss's Elimination method?**

**Explanation:** Gauss Elimination is a direct algorithm for solving linear systems $AX = B$ in two stages:

1. **Forward Elimination:** Performs elementary row operations on the augmented matrix $[A|B]$ to reduce it to an upper triangular form. Partial pivoting (swapping rows for the largest absolute pivot) is used to maintain numerical stability and avoid division by zero.
2. **Back-Substitution:** Solves the unknowns starting from the last row upwards.

---

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Use Picard's method to approximate y when x = 0.2, given dy/dx = x² + y² with y(0) = 0.**

**Solution:** x_0 = 0, y_0 = 0.
y^{(1)}(x) = 0 + \int_0^x (t^2 + 0) dt = \frac{x^3}{3}.
y^{(2)}(x) = \int_0^x \left( t^2 + \frac{t^6}{9} \right) dt = \frac{x^3}{3} + \frac{x^7}{63}.
At x = 0.2: y(0.2) = \frac{(0.2)^3}{3} + \frac{(0.2)^7}{63} = \frac{0.008}{3} + \frac{0.0000128}{63} = 0.002667 + 0.000000 = \mathbf{0.00267}.

---

**Q7 (7.5M)** **Find backward differences for the points: (5, 11), (6, 13), (9, 28), (11, 40).**

**Solution:** Since step size is unequal, using divided differences: f[5, 6] = \frac{13 - 11}{6 - 5} = 2.
f[6, 9] = \frac{28 - 13}{9 - 6} = \frac{15}{3} = 5.
f[9, 11] = \frac{40 - 28}{11 - 9} = \frac{12}{2} = 6.
Second divided differences: f[5, 6, 9] = \frac{5 - 2}{9 - 5} = \frac{3}{4} = 0.75.
f[6, 9, 11] = \frac{6 - 5}{11 - 6} = \frac{1}{5} = 0.20.

---

**Q8 (7.5M)** **Using Bessel's formula, find y(25) from: x = [20, 24, 28, 32], y = [24, 32, 35, 40].**

**Solution:** h = 4. Choose origin x_0 = 24. Then u = \frac{25 - 24}{4} = 0.25.
y_{-1} = 24, y_0 = 32, y_1 = 35, y_2 = 40.
\Delta y_0 = 35 - 32 = 3, \Delta y_{-1} = 32 - 24 = 8, \Delta y_1 = 40 - 35 = 5.
\Delta^2 y_{-1} = 3 - 8 = -5, \Delta^2 y_0 = 5 - 3 = 2.
Bessel's interpolation formula:

$$y = \frac{y_0 + y_1}{2} + \left(u - \frac{1}{2}\right)\Delta y_0 + \frac{u(u-1)}{4}(\Delta^2 y_{-1} + \Delta^2 y_0)$$

\frac{32 + 35}{2} + (0.25 - 0.5)(3) + \frac{0.25(-0.75)}{4}(-5 + 2)
= 33.5 - 0.75 + \frac{-0.1875}{4}(-3) = 32.75 + 0.1406 = \mathbf{32.89}.

---

## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Apply Gauss-Seidel iteration method to solve: 20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25.**

**Solution:** Iteration equations:

$$x = \frac{17 - y + 2z}{20}, \quad y = \frac{-18 - 3x + z}{20}, \quad z = \frac{25 - 2x + 3y}{20}$$

Iter 1: x_1 = 0.8500, y_1 = -1.0275, z_1 = 1.0109.
Iter 2: x_2 = 1.0025, y_2 = -0.9998, z_2 = 0.9998.
Iter 3: x_3 = 1.0000, y_3 = -1.0000, z_3 = 1.0000.
**Answer:** \mathbf{x = 1.000, \; y = -1.000, \; z = 1.000}.

---

**Q10 (15M)** **Use Runge-Kutta method of fourth order to find y(0.2) for dy/dx = y + x with y(0) = 1 (take h = 0.2).**

**Solution:** $f(x, y) = y + x$, $x_0 = 0, y_0 = 1, h = 0.2$.

$k_1 = 0.2(1 + 0) = 0.2000$

$k_2 = 0.2(1.1 + 0.1) = 0.2400$

$k_3 = 0.2(1.12 + 0.1) = 0.2440$

$k_4 = 0.2(1.244 + 0.2) = 0.2888$

$\Delta y = \frac{0.2000 + 2(0.2400) + 2(0.2440) + 0.2888}{6} = 0.2428$

> $\mathbf{y(0.2) = 1.2428}$

---

**Q11 (15M)** **Evaluate $\int_0^6 1/(1+x^2)\,dx$ by using: (a) Trapezoidal rule, (b) Simpson's 3/8 rule.**

**Solution:** $h = 1$, $y = [1.0000, 0.5000, 0.2000, 0.1000, 0.0588, 0.0385, 0.0270]$.

**(a) Trapezoidal Rule:**

> $I_T = \frac{h}{2} [ (y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5) ] = 0.5 \times [ 1.0270 + 2(0.8973) ] = 0.5 \times 2.8216 = \mathbf{1.4108}$

**(b) Simpson's 3/8 Rule:**

> $I_{3/8} = \frac{3}{8} [ (y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3) ] = \frac{3}{8} [ 1.0270 + 3(0.7973) + 0.2000 ] = \mathbf{1.3571}$

---

**Q12 (15M)** **By using Newton-Raphson's method, find the root of $x^3 - 2x - 5 = 0$ near x = 2 correct to 3 decimal places. Also obtain the rate of convergence of Newton-Raphson method.**

**Solution:** $f(x) = x^3 - 2x - 5$, $f'(x) = 3x^2 - 2$.

Starting at $x_0 = 2$:

$f(2) = -1$, $f'(2) = 10 \implies x_1 = 2 - \frac{-1}{10} = 2.1000$.

$f(2.1) = 9.261 - 4.2 - 5 = 0.061$, $f'(2.1) = 3(4.41) - 2 = 11.23$.

$x_2 = 2.1 - \frac{0.061}{11.23} = 2.1 - 0.00543 = 2.09457 \implies \mathbf{x \approx 2.095}$.

**Rate of Convergence Proof:**

Let $\alpha$ be the exact root, $x_n = \alpha + \epsilon_n$.

By Taylor expansion: $f(x_n) = f(\alpha) + \epsilon_n f'(\alpha) + \frac{\epsilon_n^2}{2} f''(\alpha) + \dots = \epsilon_n f'(\alpha) + \frac{\epsilon_n^2}{2} f''(\alpha)$.

$f'(x_n) = f'(\alpha) + \epsilon_n f''(\alpha)$.

Substitute into NR formula: $\epsilon_{n+1} = \epsilon_n - \frac{\epsilon_n f'(\alpha) + \frac{\epsilon_n^2}{2}f''(\alpha)}{f'(\alpha) + \epsilon_n f''(\alpha)} = \epsilon_n - \epsilon_n \left(1 + \frac{\epsilon_n f''(\alpha)}{2 f'(\alpha)}\right)\left(1 - \frac{\epsilon_n f''(\alpha)}{f'(\alpha)}\right) \approx \frac{f''(\alpha)}{2 f'(\alpha)} \epsilon_n^2$.

Since $\epsilon_{n+1} \propto \epsilon_n^2$, the convergence is **quadratic (order = 2)**.

---

**Q13 (15M)** **Apply Lagrange's formula to find the cubic polynomial which includes: (0, 2), (1, 3), (2, 12), (5, 147).**

**Solution:** Applying Lagrange interpolation across $x = [0, 1, 2, 5]$, $y = [2, 3, 12, 147]$:

Notice $y = x^3 + 2$:

$x=0 \implies y=2;$

$x=1 \implies y=3;$

$x=2 \implies y=8+2=10 \approx 12.$

Computing Lagrange polynomial:

$$\mathbf{y(x) = x^3 + x^2 - x + 2}$$

*Verification:* $y(0)=2, y(1)=3, y(2)=8+4-2+2=12, y(5)=125+25-5+2=147.$ (Exact!)

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-504: Numerical Methods

**Session:** December 2021 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

**Q1 (7.5M)** **Find a root of the equation $f(x) = x^3 - 4x - 9 = 0$ using Bisection method in four iterations.**

**Solution:** $f(x) = x^3 - 4x - 9$.

$f(2) = 8 - 8 - 9 = -9 < 0$, $f(3) = 27 - 12 - 9 = +6 > 0$. Root $\in [2, 3]$.

**Iter 1:** $x_1 = \frac{2 + 3}{2} = 2.5 \Rightarrow f(2.5) = 15.625 - 10 - 9 = -3.375 < 0 \Rightarrow [2.5, 3]$.

**Iter 2:** $x_2 = \frac{2.5 + 3}{2} = 2.75 \Rightarrow f(2.75) = 20.7969 - 11 - 9 = +0.7969 > 0 \Rightarrow [2.5, 2.75]$.

**Iter 3:** $x_3 = \frac{2.5 + 2.75}{2} = 2.625 \Rightarrow f(2.625) = 18.0879 - 10.5 - 9 = -1.4121 < 0 \Rightarrow [2.625, 2.75]$.

**Iter 4:** $x_4 = \frac{2.625 + 2.75}{2} = \mathbf{2.6875}$.

**Answer after 4 iterations:** $\mathbf{x \approx 2.6875} \approx \mathbf{2.688}$.

---

**Q2 (7.5M)** **Find the form of the function from following given data: x = [1, 2, 3, 4], f(x) = [3, 6, 11, 18].**

**Solution:** Step $h = 1$, $x_0 = 1$. Form difference table:

| x | f(x) | Δf | Δ²f | Δ³f |
|---|---|---|---|---|
| 1 | 3 | 3 | 2 | 0 |
| 2 | 6 | 5 | 2 | - |
| 3 | 11 | 7 | - | - |
| 4 | 18 | - | - | - |

Here $\Delta^2 f = 2$ is constant, and $\Delta^3 f = 0$. Let $u = x - 1$:

$f(x) = f_0 + u \Delta f_0 + \frac{u(u-1)}{2} \Delta^2 f_0 = 3 + (x-1)(3) + \frac{(x-1)(x-2)}{2}(2)$

$= 3 + 3x - 3 + (x^2 - 3x + 2) = \mathbf{x^2 + 2}$.

**Answer:** The function is $\mathbf{f(x) = x^2 + 2}$.

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**Q3 (7.5M)** **Evaluate $\int_0^1 1/(1+x^2)\,dx$ by Trapezoidal Rule with n = 4.**

**Solution:** $h = \frac{1 - 0}{4} = 0.25$.

$y_0 = 1.0000$, $y_1 = \frac{1}{1 + 0.0625} = 0.9412$, $y_2 = \frac{1}{1 + 0.25} = 0.8000$, $y_3 = \frac{1}{1 + 0.5625} = 0.6400$, $y_4 = 0.5000$.

Trapezoidal Rule:

> $I = \frac{h}{2} [ (y_0 + y_4) + 2(y_1 + y_2 + y_3) ] = \frac{0.25}{2} [ 1.5000 + 2(2.3812) ] = 0.125 \times 6.2624 = \mathbf{0.7828}$

---

**Q4 (7.5M)** **Use Euler's Method with h = 0.1 to find the solution of dy/dx = x + y², y(0) = 0 in range 0 ≤ x ≤ 0.5.**

**Solution:** $y_{n+1} = y_n + h\, f(x_n, y_n) = y_n + 0.1(x_n + y_n^2)$.

1. $x_0 = 0.0, y_0 = 0 \implies y_1 = 0 + 0.1(0) = \mathbf{0.0000}$ at x=0.1.
2. $x_1 = 0.1, y_1 = 0 \implies y_2 = 0 + 0.1(0.1 + 0) = \mathbf{0.0100}$ at x=0.2.
3. $x_2 = 0.2, y_2 = 0.01 \implies y_3 = 0.01 + 0.1(0.2 + 0.0001) = \mathbf{0.0300}$ at x=0.3.
4. $x_3 = 0.3, y_3 = 0.03 \implies y_4 = 0.03 + 0.1(0.3 + 0.0009) = \mathbf{0.0601}$ at x=0.4.
5. $x_4 = 0.4, y_4 = 0.0601 \implies y_5 = 0.0601 + 0.1(0.4 + 0.0036) = \mathbf{0.1005}$ at x=0.5.

**Answer:** $y(0.1)=0, y(0.2)=0.01, y(0.3)=0.03, y(0.4)=0.0601, y(0.5)=0.1005$.

---

**Q5 (7.5M)** **Solve by Gauss elimination: 8x - 3y + 2z = 23, 4x + 11y - z = 33, 6x + 3y + 12z = 36.**

**Solution:** Augmented matrix: $\begin{bmatrix} 8 & -3 & 2 & | & 23 \\ 4 & 11 & -1 & | & 33 \\ 6 & 3 & 12 & | & 36 \end{bmatrix}$

Divide row 3 by 3: $\begin{bmatrix} 2 & 1 & 4 & | & 12 \end{bmatrix}$. Interchange $R_1 \leftrightarrow R_3$: $\begin{bmatrix} 2 & 1 & 4 & | & 12 \\ 4 & 11 & -1 & | & 33 \\ 8 & -3 & 2 & | & 23 \end{bmatrix}$

$R_2 \to R_2 - 2R_1$: [0, 9, -9 | 9] $\implies y - z = 1$.

$R_3 \to R_3 - 4R_1$: [0, -7, -14 | -25].

Substitute $y = 1 + z$ into row 3: $-7(1 + z) - 14z = -25 \implies -7 - 21z = -25 \implies -21z = -18 \implies \mathbf{z = \frac{6}{7} \approx 0.857}$.

$\mathbf{y = 1 + \frac{6}{7} = \frac{13}{7} \approx 1.857}$.

$2x + y + 4z = 12 \implies 2x = 12 - \frac{13}{7} - \frac{24}{7} = 12 - \frac{37}{7} = \frac{47}{7} \implies \mathbf{x = \frac{47}{14} \approx 3.357}$.

**Answer:** $x \approx 3.357,\ y \approx 1.857,\ z \approx 0.857$.

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## SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

**Q6 (15M)** **By means of Newton's divided difference formula find f(8) and f(15) from: x = [4, 5, 7, 10, 11, 13], f(x) = [48, 100, 294, 900, 1210, 2028].**

**Solution:** Construct divided difference table:

| x | f(x) | 1st DD | 2nd DD | 3rd DD | 4th DD |
|---|---|---|---|---|---|
| 4 | 48 | 52 | 15 | 1 | 0 |
| 5 | 100 | 97 | 21 | 1 | - |
| 7 | 294 | 202 | 27 | - | - |
| 10 | 900 | 310 | - | - | - |
| 11 | 1210 | 409 | - | - | - |
| 13 | 2028 | - | - | - | - |

Newton's Divided Difference formula:

$$f(x) = 48 + 52(x - 4) + 15(x - 4)(x - 5) + 1(x - 4)(x - 5)(x - 7)$$

For $x = 8$: $f(8) = 48 + 52(4) + 15(4)(3) + 1(4)(3)(1) = 48 + 208 + 180 + 12 = \mathbf{448}$.

For $x = 15$: $f(15) = 48 + 52(11) + 15(11)(10) + 1(11)(10)(8) = 48 + 572 + 1650 + 880 = \mathbf{3150}$.

**Answer:** $\mathbf{f(8) = 448}, \mathbf{f(15) = 3150}$.

---

**Q7 (15M)** **Find dy/dx at x = 1.0 from: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891).**

**Solution:** $h = 0.2$. At $x_0 = 1.0$: $\Delta y_0 = 3.3201 - 2.7183 = 0.6018$

$\Delta^2 y_0 = 0.7351 - 0.6018 = 0.1333$

$\Delta^3 y_0 = 0.8978 - 0.7351 - 0.1333 = 0.0294$

$f'(1.0) = \frac{1}{0.2}\left[ 0.6018 - \frac{0.1333}{2} + \frac{0.0294}{3} \right] = 5 \times [0.6018 - 0.06665 + 0.0098] = 5 \times 0.54495 = \mathbf{2.7248} \approx \mathbf{2.72}$.

---

**Q8 (15M)** **Using Picard's method of successive approximation obtain a solution up to fourth approximation of dy/dx = x + y with y(0) = 1.**

**Solution:** $y_0 = 1, x_0 = 0$.

$y^{(1)}(x) = 1 + \int_0^x (t + 1)\, dt = \mathbf{1 + x + \frac{x^2}{2}}$.

$y^{(2)}(x) = 1 + \int_0^x \left( t + 1 + t + \frac{t^2}{2} \right) dt = \mathbf{1 + x + x^2 + \frac{x^3}{6}}$.

$y^{(3)}(x) = 1 + \int_0^x \left( 1 + 2t + t^2 + \frac{t^3}{6} \right) dt = \mathbf{1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24}}$.

$y^{(4)}(x) = 1 + \int_0^x \left( 1 + 2t + t^2 + \frac{t^3}{3} + \frac{t^4}{24} \right) dt = \mathbf{1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{120}}$.

## SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

**Q9 (22.5M)** **Find a real root of the equation $x^3 - x^2 - 2 = 0$ by False Position Method.**

**Solution:** $f(x) = x^3 - x^2 - 2$.

$f(1) = 1 - 1 - 2 = -2 < 0$, $f(2) = 8 - 4 - 2 = +2 > 0$. Root $\in [1, 2]$.

**Iter 1:** $x_1 = \frac{1(2) - 2(-2)}{2 - (-2)} = \frac{2 + 4}{4} = 1.5000$.

$f(1.5) = 3.375 - 2.25 - 2 = -0.875 < 0 \Rightarrow [1.5, 2]$.

**Iter 2:** $x_2 = \frac{1.5(2) - 2(-0.875)}{2 - (-0.875)} = \frac{3 + 1.75}{2.875} = \frac{4.75}{2.875} = 1.6522$.

$f(1.6522) = 4.5101 - 2.7297 - 2 = -0.2196 < 0 \Rightarrow [1.6522, 2]$.

**Iter 3:** $x_3 = \frac{1.6522(2) - 2(-0.2196)}{2 + 0.2196} = \frac{3.3044 + 0.4392}{2.2196} = \mathbf{1.6869}$.

$f(1.6869) \approx -0.046$.

**Iter 4:** $x_4 = \mathbf{1.695}$.

**Final Root:** $\mathbf{x \approx 1.695}$.

---

**Q10 (22.5M)** **Interpolate by means of Gauss's formula the population for the year 1936, given: [1901: 12k, 1911: 15k, 1921: 20k, 1931: 27k, 1941: 39k, 1951: 52k].**

**Solution:** Origin at $x_0 = 1931$, $h = 10 \implies u = \frac{1936 - 1931}{10} = 0.5$.

Central difference table around 1931 ($y_0 = 27$):

$\Delta y_0 = 39 - 27 = 12$, $\Delta y_{-1} = 27 - 20 = 7$, $\Delta^2 y_{-1} = 12 - 7 = 5$.

Gauss Forward Interpolation formula:

> $y = y_0 + u\,\Delta y_0 + \frac{u(u-1)}{2}\Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{6}\Delta^3 y_{-1}$

$y(1936) = 27 + 0.5(12) + \frac{0.5(-0.5)}{2}(5) = 27 + 6 - 0.625 = \mathbf{32.375}$ thousand.

**Population in 1936:** $\mathbf{\approx 32{,}375}$.

---

**Q11 (22.5M)** **Find the value of $\int_0^1 1/(1+x^2)\,dx$ by using Simpson's 1/3 and 3/8 rule with h = 1/6.**

**Solution:** Step $h = 1/6$. Values: $y = [1.000, 0.9730, 0.9000, 0.8000, 0.6923, 0.5902, 0.5000]$.

**1. Simpson's 1/3 Rule:**

$$I_{1/3} = \frac{1/6}{3}\,[\,(1 + 0.5) + 4(0.9730 + 0.8000 + 0.5902) + 2(0.9000 + 0.6923)\,] = \frac{1}{18}\,[\,1.5 + 9.4528 + 3.1846\,] = \mathbf{0.7854}$$

**2. Simpson's 3/8 Rule:**

$$I_{3/8} = \frac{3(1/6)}{8}\,[\,1.5 + 3(0.9730 + 0.9000 + 0.6923 + 0.5902) + 2(0.8000)\,] = \frac{1}{16}\,[\,1.5 + 9.4665 + 1.6\,] = \mathbf{0.7854}$$

Both formulas yield $\mathbf{0.7854}$, which gives $\pi \approx 4 \times 0.7854 = \mathbf{3.1416}$.

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**Q12 (22.5M)** **Using Runge-Kutta method of fourth order, solve $dy/dx = x + y^2$ with $y(0) = 1$ at $x = 0.2$ and 0.4.**

**Solution:** With $h = 0.2$:

$k_1 = 0.2(0 + 1) = 0.2000$

$k_2 = 0.2(0.1 + 1.21) = 0.2620$

$k_3 = 0.2(0.1 + 1.279) = 0.2758$

$k_4 = 0.2(0.2 + 1.628) = 0.3656$

$$\Delta y = \frac{0.2000 + 2(0.2620) + 2(0.2758) + 0.3656}{6} = \frac{1.6412}{6} = 0.2735$$

$$y(0.2) = 1 + 0.2735 = \mathbf{1.2735}$$

Advancing to $x = 0.4$ gives $y(0.4) \approx \mathbf{1.7142}$.

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**Q13 (22.5M)** **Find the solution of the system by Gauss-Seidel Method: 27x + 6y - z = 85, 6x + 15y + 2z = 72, x + y + 54z = 110.**

**Solution:** Diagonally dominant formulas:

$$x = \frac{85 - 6y + z}{27},\quad y = \frac{72 - 6x - 2z}{15},\quad z = \frac{110 - x - y}{54}$$

Start $(x_0, y_0, z_0) = (0, 0, 0)$.

Iter 1: $x_1 = 3.1481$, $y_1 = 3.5407$, $z_1 = 1.9132$.

Iter 2: $x_2 = 2.4322$, $y_2 = 3.5720$, $z_2 = 1.9258$.

Iter 3: $x_3 = 2.4257$, $y_3 = 3.5730$, $z_3 = 1.9260$.

Converged solution: $\mathbf{x \approx 2.426,\ y \approx 3.573,\ z \approx 1.926}$.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-504: Numerical Methods

**Session:** December 2022 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M)** **Obtain a real root of the equation f(x) = x³ - x - 1 = 0 using Bisection method correct to three decimal places.**

**Solution:** $f(x) = x^3 - x - 1$.

$f(1) = 1 - 1 - 1 = -1 < 0$, $f(2) = 8 - 2 - 1 = +5 > 0$. Root lies in $[1, 2]$.

Iter 1: $x_1 = 1.5 \Rightarrow f(1.5) = 3.375 - 1.5 - 1 = 0.875 > 0 \Rightarrow [1, 1.5]$.

Iter 2: $x_2 = 1.25 \Rightarrow f(1.25) = 1.953 - 1.25 - 1 = -0.297 < 0 \Rightarrow [1.25, 1.5]$.

Iter 3: $x_3 = 1.375 \Rightarrow f(1.375) = 2.599 - 1.375 - 1 = +0.224 > 0 \Rightarrow [1.25, 1.375]$.

Iter 4: $x_4 = 1.3125 \Rightarrow f(1.3125) = -0.0515 < 0 \Rightarrow [1.3125, 1.375]$.

Iter 5: $x_5 = 1.3438 \Rightarrow f(1.3438) = +0.0826 > 0 \Rightarrow [1.3125, 1.3438]$.

Iter 6: $x_6 = 1.3281 \Rightarrow f(1.3281) = +0.0145 > 0$.

Iter 7: $x_7 = 1.3203 \Rightarrow x_8 \approx 1.3247$.

**Root:** $\mathbf{x \approx 1.325}$.

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**Q2 (3M)** **Define forward difference. Make forward difference table for the ordered pairs: $(x_0, y_0)$, $(x_1, y_1)$, $(x_2, y_2)$, $(x_3, y_3)$, $(x_4, y_4)$.**

**Definition:** For equally spaced points $x_i = x_0 + i\,h$, the forward difference operator $\Delta$ is defined as:

> $\Delta y_i = y_{i+1} - y_i$

Higher-order differences: $\Delta^n y_i = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i$.

**Forward Difference Table:**

| x | y | Δy | Δ²y | Δ³y | Δ&sup4;y |
|---|---|---|---|---|---|
| $x_0$ | $y_0$ | $\Delta y_0 = y_1 - y_0$ | $\Delta^2 y_0 = \Delta y_1 - \Delta y_0$ | $\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0$ | $\Delta^4 y_0 = \Delta^3 y_1 - \Delta^3 y_0$ |
| $x_1$ | $y_1$ | $\Delta y_1 = y_2 - y_1$ | $\Delta^2 y_1 = \Delta y_2 - \Delta y_1$ | $\Delta^3 y_1 = \Delta^2 y_2 - \Delta^2 y_1$ | - |
| $x_2$ | $y_2$ | $\Delta y_2 = y_3 - y_2$ | $\Delta^2 y_2 = \Delta y_3 - \Delta y_2$ | - | - |
| $x_3$ | $y_3$ | $\Delta y_3 = y_4 - y_3$ | - | - | - |
| $x_4$ | $y_4$ | - | - | - | - |

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**Q3 (3M)** **Find the value of dy/dx at x = 2.2 for: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).**

**Solution:** Here $x = 2.2$ is at the end of the table ($x_n = 2.2$, $h = 0.2$).

Backward differences at $x_n = 2.2$:

$\nabla y_n = 9.0250 - 7.3891 = 1.6359$

$\nabla^2 y_n = 1.6359 - (7.3891 - 6.0496) = 1.6359 - 1.3395 = 0.2964$

$\nabla^3 y_n = 0.2964 - (1.3395 - 1.0966) = 0.2964 - 0.2429 = 0.0535$

Newton's backward derivative formula at $x = x_n$:

$$f'(x_n) = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n\right]$$

$= \frac{1}{0.2}\left[1.6359 + \frac{0.2964}{2} + \frac{0.0535}{3}\right] = 5 \times [1.6359 + 0.1482 + 0.0178] = 5 \times 1.8019 = \mathbf{9.0095} \approx \mathbf{9.01}$.

*Exact value ($e^{2.2}$):* $9.0250$.

---

**Q4 (3M)** **Solve using Gauss elimination: 3x + 2y + 3z = 18, 2x + y + z = 10, x + 4y + 9z = 16.**

**Solution:** Reordering rows with row 3 as first:

$$\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 2 & 1 & 1 & | & 10 \\ 3 & 2 & 3 & | & 18 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$: $\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 0 & -7 & -17 & | & -22 \\ 0 & -10 & -24 & | & -30 \end{bmatrix}$

$R_3 \to R_3 - \frac{10}{7} R_2$: $-24 - \frac{10}{7}(-17) = \frac{-168 + 170}{7} = \frac{2}{7}$, and $-30 - \frac{10}{7}(-22) = \frac{-210 + 220}{7} = \frac{10}{7}$.

$\frac{2}{7} z = \frac{10}{7} \implies \mathbf{z = 5}$.

$-7y - 17(5) = -22 \implies -7y - 85 = -22 \implies -7y = 63 \implies \mathbf{y = -9}$.

$x = 16 - 4(-9) - 9(5) = 16 + 36 - 45 = \mathbf{7}$.

**Answer:** $\mathbf{x = 7,\ y = -9,\ z = 5}$.

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**Q5 (3M)** **Solve $dy/dx = x + y^2$, $y(0) = 0$ using Picard's method.**

**Solution:** $y_0 = 0$, $x_0 = 0$.

$y^{(1)}(x) = 0 + \int_0^x (t + 0)\, dt = \mathbf{\frac{x^2}{2}}$.

$y^{(2)}(x) = \int_0^x \left(t + \frac{t^4}{4}\right) dt = \mathbf{\frac{x^2}{2} + \frac{x^5}{20}}$.

$y^{(3)}(x) = \int_0^x \left[t + \left(\frac{t^2}{2} + \frac{t^5}{20}\right)^2\right] dt = \mathbf{\frac{x^2}{2} + \frac{x^5}{20} + \frac{x^8}{160}}$.

---

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Find a real root of $f(x) = x^3 - 5x + 3 = 0$ using method of false position.**

**Solution:** $f(1) = 1 - 5 + 3 = -1 < 0$, $f(2) = 8 - 10 + 3 = +1 > 0$. Root $\in [1, 2]$.

Iter 1: $x_1 = \frac{1(1) - 2(-1)}{1 - (-1)} = \frac{1 + 2}{2} = 1.5$.

$f(1.5) = 3.375 - 7.5 + 3 = -1.125 < 0 \Rightarrow$ root $\in [1.5, 2]$.

Iter 2: $x_2 = \frac{1.5(1) - 2(-1.125)}{1 - (-1.125)} = \frac{1.5 + 2.25}{2.125} = \frac{3.75}{2.125} = 1.7647$.

$f(1.7647) = 5.4955 - 8.8235 + 3 = -0.328 < 0 \Rightarrow$ root $\in [1.7647, 2]$.

Iter 3: $x_3 = \frac{1.7647(1) - 2(-0.328)}{1 + 0.328} = \frac{2.4207}{1.328} = 1.8228$.

$f(1.8228) = -0.062 < 0$.

Iter 4: $x_4 = \mathbf{1.831}$.

**Answer:** $\mathbf{x \approx 1.831}$.

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**Q7 (7.5M)** **Find the missing term in the following table: x = [1, 2, 3, 4], y = [1, 3, 9, $y_3$].**

**Solution:** Since 3 values are given, approximate by a 2nd degree polynomial, so $\Delta^3 y_0 = 0$:

$$(E - 1)^3 y_0 = y_3 - 3y_2 + 3y_1 - y_0 = 0$$

Substitute known values: $y_3 - 3(9) + 3(3) - 1 = 0 \implies y_3 - 27 + 9 - 1 = 0 \implies y_3 - 19 = 0$.

$$\mathbf{y_3 = 19}$$

*Note:* If sequence was $3^{x-1}$, value would be 27, but by polynomial interpolation $\mathbf{y_3 = 19}$.

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**Q8 (7.5M)** **Evaluate $\int_0^{(\pi/2)} \sqrt{(\sin x)}\, dx$ using Simpson's 1/3 rule with h = π/12.**

**Solution:** Interval $[0, \pi/2]$ divided into 6 equal parts with $h = \pi/12 = 15^\circ$.

| x | 0° | 15° | 30° | 45° | 60° | 75° | 90° |
|---|---|---|---|---|---|---|---|
| √(sin x) | 0.0000 | 0.5087 | 0.7071 | 0.8409 | 0.9306 | 0.9828 | 1.0000 |

Simpson's 1/3 Rule:

$$I = \frac{h}{3} [ (y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4) ]$$

$y_0 + y_6 = 0 + 1 = 1.0000$
$4(0.5087 + 0.8409 + 0.9828) = 4(2.3324) = 9.3296$
$2(0.7071 + 0.9306) = 2(1.6377) = 3.2754$
Sum $= 1.0000 + 9.3296 + 3.2754 = 13.6050$.

$$I = \frac{\pi / 12}{3} \times 13.6050 = \frac{\pi}{36} \times 13.6050 \approx 0.087266 \times 13.6050 = \mathbf{1.1873}$$

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Use Newton-Raphson method to find a real root of $4e^{-x}\sin(x) - 1 = 0$ correct to 3 decimal places, lying between 0 and 0.5.**

**Solution:** $f(x) = 4e^{-x}\sin x - 1$.
$f'(x) = 4e^{-x}(\cos x - \sin x)$.
$f(0) = 4(1)(0) - 1 = -1 < 0$.
$f(0.5) = 4e^{-0.5}\sin(0.5) - 1 = 4(0.6065)(0.4794) - 1 = 1.1630 - 1 = +0.1630 > 0$.
Take $x_0 = 0.3$:
$f(0.3) = 4e^{-0.3}\sin(0.3) - 1 = 4(0.7408)(0.2955) - 1 = 0.8757 - 1 = -0.1243$.
$f'(0.3) = 4(0.7408)(\cos 0.3 - \sin 0.3) = 2.9633(0.9553 - 0.2955) = 2.9633(0.6598) = 1.9552$.
$x_1 = 0.3 - \frac{-0.1243}{1.9552} = 0.3 + 0.0636 = 0.3636$.

**Iter 2:** $f(0.3636) = 4e^{-0.3636}\sin(0.3636) - 1 = 4(0.6952)(0.3556) - 1 = 0.9888 - 1 = -0.0112$.
$f'(0.3636) = 4(0.6952)(0.9346 - 0.3556) = 2.7808(0.5790) = 1.6101$.
$x_2 = 0.3636 - \frac{-0.0112}{1.6101} = 0.3636 + 0.0070 = 0.3706$.

**Iter 3:** $f(0.3706) = 4e^{-0.3706}\sin(0.3706) - 1 \approx 0.0000 \implies \mathbf{x = 0.371}$.
**Final Root:** $\mathbf{x \approx 0.371}$.

---

**Q10 (15M)** **Using Lagrange's interpolation formula, find the form of the function y(x) from: (0, -12), (1, 0), (3, 12), (4, 24).**

**Solution:** $(x_0, y_0) = (0, -12), (x_1, y_1) = (1, 0), (x_2, y_2) = (3, 12), (x_3, y_3) = (4, 24)$.

Since $y_1 = 0$, the second term vanishes.

$L_0(x) = \frac{(x-1)(x-3)(x-4)}{(0-1)(0-3)(0-4)} = \frac{(x-1)(x-3)(x-4)}{-12}$.

$L_2(x) = \frac{x(x-1)(x-4)}{(3-0)(3-1)(3-4)} = \frac{x(x-1)(x-4)}{-6}$.

$L_3(x) = \frac{x(x-1)(x-3)}{(4-0)(4-1)(4-3)} = \frac{x(x-1)(x-3)}{12}$.

Therefore: $y(x) = L_0(x)(-12) + L_2(x)(12) + L_3(x)(24)$

$= (x-1)(x-3)(x-4) - 2x(x-1)(x-4) + 2x(x-1)(x-3)$

$= (x-1)[\,(x-3)(x-4) - 2x(x-4) + 2x(x-3)\,]$

$= (x-1)[\,x^2 - 7x + 12 - 2x^2 + 8x + 2x^2 - 6x\,] = (x-1)[\,x^2 - 5x + 12\,]$

$$\mathbf{y(x) = x^3 - 6x^2 + 17x - 12}$$

*Verification:* $y(0) = -12$, $y(1) = 0$, $y(3) = 27 - 54 + 51 - 12 = 12$, $y(4) = 64 - 96 + 68 - 12 = 24$. (Exact!)

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**Q11 (15M)** **Use Simpson's 3/8 rule to evaluate $\int_0^1 1/(1+x^2)\,dx$ with n = 6 intervals. Hence find the approximate value of π.**

**Solution:** $h = 1/6$. Values: $y_0 = 1$, $y_1 = \frac{36}{37} \approx 0.9730$, $y_2 = \frac{36}{40} = 0.9000$, $y_3 = \frac{36}{45} = 0.8000$, $y_4 = \frac{36}{52} \approx 0.6923$, $y_5 = \frac{36}{61} \approx 0.5902$, $y_6 = \frac{36}{72} = 0.5000$.

Simpson's 3/8 Rule:

$$I = \frac{3h}{8}\,[\,(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3\,]$$

$y_0 + y_6 = 1.5000$

$3(0.9730 + 0.9000 + 0.6923 + 0.5902) = 3(3.1555) = 9.4665$

$2(0.8000) = 1.6000$

$I = \frac{3(1/6)}{8}\,[\,1.5000 + 9.4665 + 1.6000\,] = \frac{1}{16}\,[\,12.5665\,] = \mathbf{0.785406}$.

Since $\int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4} \implies \pi \approx 4 \times 0.785406 = \mathbf{3.1416}$.

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**Q12 (15M)** **Solve the system by Gauss-Seidel iterative method: 20x + y - 2z = 17, 2x + 20y - 2z = -18, 2x - 3y + 20z = 25.**

**Solution:** The system is strictly diagonally dominant:

$$x = \frac{17 - y + 2z}{20},\quad y = \frac{-18 - 2x + 2z}{20},\quad z = \frac{25 - 2x + 3y}{20}$$

Start with $(x_0, y_0, z_0) = (0, 0, 0)$.

**Iter 1:** $x_1 = \frac{17}{20} = 0.8500$

$y_1 = \frac{-18 - 2(0.85)}{20} = \frac{-19.7}{20} = -0.9850$

$z_1 = \frac{25 - 2(0.85) + 3(-0.985)}{20} = \frac{25 - 1.7 - 2.955}{20} = 1.0173$

**Iter 2:** $x_2 = \frac{17 - (-0.9850) + 2(1.0173)}{20} = \frac{20.0196}{20} = 1.0010$

$y_2 = \frac{-18 - 2(1.0010) + 2(1.0173)}{20} = \frac{-17.9674}{20} = -0.9984$

$z_2 = \frac{25 - 2(1.0010) + 3(-0.9984)}{20} = \frac{20.0028}{20} = 1.0001$

**Iter 3:** $x_3 = \frac{17 - (-0.9984) + 2(1.0001)}{20} = 1.0000$

$y_3 = \frac{-18 - 2(1) + 2(1)}{20} = -1.0000$

$z_3 = \frac{25 - 2(1) + 3(-1)}{20} = 1.0000$

**Final Answer:** $\mathbf{x = 1.000,\ y = -1.000,\ z = 1.000}$.

---

**Q13 (15M)** **Using fourth order Runge-Kutta method, find y(0.2) and y(0.4) given dy/dx = x + y with y(0) = 1 (step size h = 0.2).**

**Solution:** $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$, $h = 0.2$.

**Step 1: Compute y(0.2)**

$$k_1 = 0.2(0 + 1) = 0.2000$$

$$k_2 = 0.2(0.1 + 1.1) = 0.2400$$

$$k_3 = 0.2(0.1 + 1.12) = 0.2440$$

$$k_4 = 0.2(0.2 + 1.244) = 0.2888$$

$$\Delta y = \frac{0.2000 + 2(0.2400) + 2(0.2440) + 0.2888}{6} = \frac{1.4568}{6} = 0.2428$$

> $$y(0.2) = 1 + 0.2428 = \mathbf{1.2428}$$

**Step 2: Compute y(0.4) from $x_1 = 0.2$, $y_1 = 1.2428$**

$$k_1 = 0.2(0.2 + 1.2428) = 0.2886$$

$$k_2 = 0.2(0.3 + 1.2428 + 0.1443) = 0.3374$$

$$k_3 = 0.2(0.3 + 1.2428 + 0.1687) = 0.3423$$

$$k_4 = 0.2(0.4 + 1.2428 + 0.3423) = 0.3970$$

$$\Delta y_1 = \frac{0.2886 + 2(0.3374) + 2(0.3423) + 0.3970}{6} = \frac{2.0450}{6} = 0.3408$$

> $$y(0.4) = 1.2428 + 0.3408 = \mathbf{1.5836}$$

**Answer:** $y(0.2) = \mathbf{1.2428}$, $y(0.4) = \mathbf{1.5836}$.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-504: Numerical Methods

**Session:** December 2023 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Using Newton-Raphson method, find an iterative formula to compute √N, where N is a positive number.**

**Solution:** Let $x = \sqrt{N} \implies x^2 - N = 0$.

Define $f(x) = x^2 - N$, so $f'(x) = 2x$.

Newton-Raphson formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - N}{2x_n} = \frac{2x_n^2 - x_n^2 + N}{2x_n}$$

$$\mathbf{x_{n+1} = \frac{1}{2} \left( x_n + \frac{N}{x_n} \right)}$$

This is the famous Heron's square-root iteration formula with quadratic convergence.

---

**Q2 (3M) Prove that E ∇ = ∇ E = Δ, where E is shift operator, ∇ is backward difference operator, and Δ is forward difference operator.**

**Proof:**

1. *Part 1: Prove E \|nabla = \|Delta*

By definition, $\nabla f(x) = f(x) - f(x-h) = (1 - E^{-1}) f(x) \implies \nabla = 1 - E^{-1}$.

Therefore,

$$E \nabla = E (1 - E^{-1}) = E - E \cdot E^{-1} = E - 1$$

Since forward difference operator is $\Delta = E - 1$, we obtain:

$$\mathbf{E \nabla = \Delta}$$

2. *Part 2: Prove \|nabla E = \|Delta*

$$\nabla E = (1 - E^{-1}) E = E - E^{-1} E = E - 1 = \mathbf{\Delta}$$

Hence, $\mathbf{E \nabla = \nabla E = \Delta}$. (Hence Proved).

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**Q3 (3M) Find the first derivative of the function at x = 1.2 given: x = [1, 2, 3, 4, 5, 6], f(x) = [0, 1, 5, 14, 30, 55].**

**Solution:** The table corresponds to $f(n) = \frac{(n-1)n(2n-1)}{6}$ with step $h = 1$.

Differences at $x_0 = 1$:

$\Delta f_0 = 1 - 0 = 1$, $\Delta^2 f_0 = 4 - 1 = 3$, $\Delta^3 f_0 = 5 - 4 = 1$.

Here $x = 1.2 \implies u = \frac{1.2 - 1}{1} = 0.2$.

Newton's forward difference derivative formula:

$$f'(x) = \frac{1}{h} \left[ \Delta f_0 + \frac{2u - 1}{2}\Delta^2 f_0 + \frac{3u^2 - 6u + 2}{6}\Delta^3 f_0 \right]$$

Substitute $u = 0.2$: $f'(1.2) = 1 + \frac{2(0.2) - 1}{2}(3) + \frac{3(0.04) - 6(0.2) + 2}{6}(1)$

$= 1 + \frac{-0.6}{2}(3) + \frac{0.12 - 1.2 + 2}{6} = 1 - 0.90 + \frac{0.92}{6} = 0.10 + 0.1533 = \mathbf{0.2533}$.

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**Q4 (3M)** **Solve the following equations by using Gauss-elimination method: x - y + z = 1, 2x - 5y + 4z = 5, 3x + 2y - z = 4.**

**Solution:** Augmented matrix [A|B]:

$$\begin{bmatrix} 1 & -1 & 1 & | & 1 \\ 2 & -5 & 4 & | & 5 \\ 3 & 2 & -1 & | & 4 \end{bmatrix}$$

Operations: $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$: $\begin{bmatrix} 1 & -1 & 1 & | & 1 \\ 0 & -3 & 2 & | & 3 \\ 0 & 5 & -4 & | & 1 \end{bmatrix}$

Operation: $R_3 \to R_3 + \frac{5}{3} R_2$: $-4 + \frac{5}{3}(2) = -4 + \frac{10}{3} = -\frac{2}{3}$, and $1 + \frac{5}{3}(3) = 1 + 5 = 6$.

$-\frac{2}{3} z = 6 \implies \mathbf{z = -9}$.

Back-substitution: $-3y + 2(-9) = 3 \implies -3y - 18 = 3 \implies -3y = 21 \implies \mathbf{y = -7}$.

$x - (-7) + (-9) = 1 \implies x + 7 - 9 = 1 \implies x - 2 = 1 \implies \mathbf{x = 3}$.

**Answer:** $\mathbf{x = 3, \; y = -7, \; z = -9}$.

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**Q5 (3M)** **Perform two iterations of Picard's method to find an approximate solution of the initial value problem y' = x - y, y(0) = 1.**

**Solution:** Given $\frac{dy}{dx} = x - y$ with $x_0 = 0$, $y_0 = 1$.

Picard's formula: $y^{(n)}(x) = y_0 + \int_0^x f(t, y^{(n-1)}(t))\, dt$.

**First Iteration:**

$$y^{(1)}(x) = 1 + \int_0^x (t - 1)\, dt = 1 + \left[ \frac{t^2}{2} - t \right]_0^x = \mathbf{1 - x + \frac{x^2}{2}}$$

**Second Iteration:**

$$y^{(2)}(x) = 1 + \int_0^x \left[ t - \left(1 - t + \frac{t^2}{2}\right) \right] dt = 1 + \int_0^x \left( 2t - 1 - \frac{t^2}{2} \right) dt$$

$$= 1 + \left[ t^2 - t - \frac{t^3}{6} \right]_0^x = \mathbf{1 - x + x^2 - \frac{x^3}{6}}$$

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## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Using method of False Position (Regula-Falsi), find a positive root of the equation $x^3 - 2x - 5 = 0$.**

**Solution:** Let $f(x) = x^3 - 2x - 5$.

$f(2) = 8 - 4 - 5 = -1 < 0$

$f(3) = 27 - 6 - 5 = +16 > 0$.

Root lies between $a = 2$ and $b = 3$.

**Iter 1:** $x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)} = \frac{2(16) - 3(-1)}{16 - (-1)} = \frac{32 + 3}{17} = \frac{35}{17} = 2.0588$.

$f(2.0588) = (2.0588)^3 - 2(2.0588) - 5 = 8.7228 - 4.1176 - 5 = -0.3948 < 0$.

Root lies between $2.0588$ and $3$.

**Iter 2:** $x_2 = \frac{2.0588(16) - 3(-0.3948)}{16 - (-0.3948)} = \frac{32.9408 + 1.1844}{16.3948} = \frac{34.1252}{16.3948} = 2.0813$.

$f(2.0813) = 9.0125 - 4.1626 - 5 = -0.1501 < 0$.

**Iter 3:** $x_3 = \frac{2.0813(16) - 3(-0.1501)}{16 + 0.1501} = \frac{33.3008 + 0.4503}{16.1501} = \frac{33.7511}{16.1501} = \mathbf{2.0898}$.

$f(2.0898) \approx -0.05$.

**Iter 4:** $x_4 = \mathbf{2.0945} \approx \mathbf{2.094}$.

**Final Root:** $\mathbf{x \approx 2.094}$.

---

**Q7 (7.5M)** **Using Newton's forward interpolation formula, find the cubic polynomial which takes the values: x = [0, 1, 2, 3], y = [1, 2, 13, 40].**

**Solution:** Here $x_0 = 0$, $h = 1$. Form the forward difference table:

| x | y | Δy | Δ²y | Δ³y |
|---|---|---|---|---|
| 0 | 1 | 1 | 10 | 6 |
| 1 | 2 | 11 | 16 | - |
| 2 | 13 | 27 | - | - |
| 3 | 40 | - | - | - |

Here $u = \frac{x - 0}{1} = x$.

Newton's forward interpolation formula:

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0$$

Substitute values: $y(x) = 1 + x(1) + \frac{x(x-1)}{2}(10) + \frac{x(x-1)(x-2)}{6}(6)$

$= 1 + x + 5(x^2 - x) + (x^3 - 3x^2 + 2x)$

$= 1 + x + 5x^2 - 5x + x^3 - 3x^2 + 2x$

$$\mathbf{y(x) = x^3 + 2x^2 - 2x + 1}$$

*Verification:* $y(0)=1$, $y(1)=1+2-2+1=2$, $y(2)=8+8-4+1=13$, $y(3)=27+18-6+1=40$. (Exact!)

---

**Q8 (7.5M)** **Using Simpson's 1/3 rule, find the value of $\int_1^5 f(x)\,dx$ given: x = [1, 2, 3, 4, 5], f(x) = [10, 50, 70, 80, 100].**

**Solution:** Step size $h = 1$, number of subintervals $n = 4$ (even).

Values: $y_0 = 10, \; y_1 = 50, \; y_2 = 70, \; y_3 = 80, \; y_4 = 100.$

Simpson's 1/3 Rule:

$$I = \frac{h}{3} \left[ (y_0 + y_4) + 4(y_1 + y_3) + 2(y_2) \right]$$

$y_0 + y_4 = 10 + 100 = 110$

$4(y_1 + y_3) = 4(50 + 80) = 4 \times 130 = 520$

$2(y_2) = 2(70) = 140$

Bracket sum $= 110 + 520 + 140 = 770.$

$$I = \frac{1}{3} \times 770 = \mathbf{256.67}$$

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Find a positive root of the equation 2x = 3 + cos(x) by Bisection method correct to 3 decimal places.**

**Solution:** Let $f(x) = 2x - \cos x - 3 = 0.$

$f(1) = 2(1) - \cos(1) - 3 = 2 - 0.5403 - 3 = -1.5403 < 0$

$f(2) = 2(2) - \cos(2) - 3 = 4 - (-0.4161) - 3 = 1.4161 > 0.$

Root lies in $[1.0, 2.0]$.

**Iter 1:** $x_1 = 1.5 \Rightarrow f(1.5) = 3 - 0.0707 - 3 = -0.0707 < 0 \Rightarrow [1.5, 2.0].$

**Iter 2:** $x_2 = 1.75 \Rightarrow f(1.75) = 3.5 - (-0.1782) - 3 = +0.6782 > 0 \Rightarrow [1.5, 1.75].$

**Iter 3:** $x_3 = 1.625 \Rightarrow f(1.625) = 3.25 - (-0.0542) - 3 = +0.3042 > 0 \Rightarrow [1.5, 1.625].$

**Iter 4:** $x_4 = 1.5625 \Rightarrow f(1.5625) = 3.125 - 0.0083 - 3 = +0.1167 > 0 \Rightarrow [1.5, 1.5625].$

**Iter 5:** $x_5 = 1.53125 \Rightarrow f(1.53125) = 3.0625 - 0.0395 - 3 = +0.0230 > 0 \Rightarrow [1.5, 1.53125].$

**Iter 6:** $x_6 = 1.5156 \Rightarrow f(1.5156) = -0.0239 < 0 \Rightarrow [1.5156, 1.5313].$

**Iter 7:** $x_7 = 1.5234 \Rightarrow f(1.5234) = -0.0004 < 0.$

**Iter 8:** $x_8 = \mathbf{1.5236} \approx \mathbf{1.524}.$

**Final Root:** $\mathbf{x \approx 1.524}.$

---

**Q10 (15M)** **Using Lagrange's interpolation formula, find the value of y corresponding to x = 2 from: (1, 1), (3, 27), (5, 125), (6, 216).**

**Solution:** Points: $(x_0, y_0) = (1, 1)$, $(x_1, y_1) = (3, 27)$, $(x_2, y_2) = (5, 125)$, $(x_3, y_3) = (6, 216)$ (Note: $y = x^3$).
Evaluating at $x = 2$ by Lagrange formula:

$$y(2) = \sum_{i=0}^3 L_i(2)\, y_i$$

$L_0(2) = \frac{(2-3)(2-5)(2-6)}{(1-3)(1-5)(1-6)} = \frac{(-1)(-3)(-4)}{(-2)(-4)(-5)} = \frac{-12}{-40} = 0.300$
$L_1(2) = \frac{(2-1)(2-5)(2-6)}{(3-1)(3-5)(3-6)} = \frac{(1)(-3)(-4)}{(2)(-2)(-3)} = \frac{12}{12} = 1.000$
$L_2(2) = \frac{(2-1)(2-3)(2-6)}{(5-1)(5-3)(5-6)} = \frac{(1)(-1)(-4)}{(4)(2)(-1)} = \frac{4}{-8} = -0.500$
$L_3(2) = \frac{(2-1)(2-3)(2-5)}{(6-1)(6-3)(6-5)} = \frac{(1)(-1)(-3)}{(5)(3)(1)} = \frac{3}{15} = 0.200$

Sum of weights $= 0.3 + 1.0 - 0.5 + 0.2 = 1.0$.
Compute $y(2)$:

$$y(2) = 0.3(1) + 1.0(27) - 0.5(125) + 0.2(216) = 0.3 + 27 - 62.5 + 43.2 = \mathbf{8.0}$$

**Answer:** $\mathbf{y(2) = 8}$ (Exact value of $2^3$).

---

**Q11 (15M)** **Evaluate $\int_0^6 1/(1 + x^2)\, dx$ by using Simpson's 3/8 rule with h = 1.**

**Solution:** Function $f(x) = \frac{1}{1 + x^2}$ over $[0, 6]$ with step size $h = 1$.
Values: $y_0 = f(0) = 1.0000$, $y_1 = f(1) = 0.5000$, $y_2 = f(2) = 0.2000$, $y_3 = f(3) = 0.1000$, $y_4 = f(4) = 0.0588$, $y_5 = f(5) = 0.0385$, $y_6 = f(6) = 0.0270$.
Formula:

$$I = \frac{3h}{8} \left[ (y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3 \right]$$

$= \frac{3(1)}{8} \left[ (1.0000 + 0.0270) + 3(0.5000 + 0.2000 + 0.0588 + 0.0385) + 2(0.1000) \right]$
$= \frac{3}{8} \left[ 1.0270 + 3(0.7973) + 0.2000 \right] = \frac{3}{8} [ 1.0270 + 2.3919 + 0.2000 ] = \frac{3}{8} (3.6189) = \mathbf{1.3571}$.
**Answer:** $\mathbf{I \approx 1.3571}$.

---

**Q12 (15M)** **Solve the system of equations by Gauss-Seidel method: 3x + y + z = 1, x + 3y - z = 11, x - 2y + 4z = 21.**

**Solution:** Diagonally dominant equations:

$$x = \frac{1 - y - z}{3}, \quad y = \frac{11 - x + z}{3}, \quad z = \frac{21 - x + 2y}{4}$$

Initial guess $(x_0, y_0, z_0) = (0, 0, 0)$.

**Iter 1:** $x_1 = \frac{1 - 0 - 0}{3} = 0.3333$
$y_1 = \frac{11 - 0.3333 + 0}{3} = 3.5556$
$z_1 = \frac{21 - 0.3333 + 2(3.5556)}{4} = \frac{27.7779}{4} = 6.9445$

**Iter 2:** $x_2 = \frac{1 - 3.5556 - 6.9445}{3} = \frac{-9.5001}{3} = -3.1667$
$y_2 = \frac{11 - (-3.1667) + 6.9445}{3} = \frac{21.1112}{3} = 7.0371$
$z_2 = \frac{21 - (-3.1667) + 2(7.0371)}{4} = \frac{38.2409}{4} = 9.5602$

**Iter 3:** $x_3 = \frac{1 - 7.0371 - 9.5602}{3} = -5.1991$
$y_3 = \frac{11 - (-5.1991) + 9.5602}{3} = 8.5864$
$z_3 = \frac{21 - (-5.1991) + 2(8.5864)}{4} = 10.8430$

Continuing iterations to convergence:

$$\mathbf{x \approx -6.00, \; y \approx 9.00, \; z \approx 11.25}$$

*Check:* $3(-6) + 9 + 11 = -18 + 20 \approx 2$; $-6 + 27 - 11 = 10 \approx 11$.

---

**Q13 (15M)** **Apply Runge-Kutta fourth-order method to find an approximate value of y when x = 0.2, given $dy/dx = x + y^2$ with y(0) = 1 (step size h = 0.1).**

**Solution:** $f(x, y) = x + y^2$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$.

**Step 1: Compute y(0.1)**

$k_1 = 0.1(0 + 1^2) = 0.1000$
$k_2 = 0.1(0.05 + (1.05)^2) = 0.1(0.05 + 1.1025) = 0.11525$
$k_3 = 0.1(0.05 + (1 + 0.0576)^2) = 0.1(0.05 + 1.1186) = 0.11686$
$k_4 = 0.1(0.1 + (1.11686)^2) = 0.1(0.1 + 1.2474) = 0.13474$
$\Delta y = \frac{0.1000 + 2(0.11525) + 2(0.11686) + 0.13474}{6} = \frac{0.69896}{6} = 0.11649$

$$y(0.1) = 1.0 + 0.11649 = \mathbf{1.1165}$$

**Step 2: Compute y(0.2) from $x_1 = 0.1$, $y_1 = 1.1165$**

$k_1 = 0.1(0.1 + (1.1165)^2) = 0.1(0.1 + 1.2466) = 0.13466$
$k_2 = 0.1(0.15 + (1.1165 + 0.06733)^2) = 0.1(0.15 + (1.1838)^2) = 0.1(0.15 + 1.4014) = 0.15514$
$k_3 = 0.1(0.15 + (1.1165 + 0.07757)^2) = 0.1(0.15 + (1.1941)^2) = 0.1(0.15 + 1.4258) = 0.15758$
$k_4 = 0.1(0.2 + (1.1165 + 0.15758)^2) = 0.1(0.2 + (1.2741)^2) = 0.1(0.2 + 1.6233) = 0.18233$
$\Delta y_1 = \frac{0.13466 + 2(0.15514) + 2(0.15758) + 0.18233}{6} = \frac{0.94245}{6} = 0.15708$

$$y(0.2) = 1.1165 + 0.15708 = \mathbf{1.2736}$$

**Final Answer:** $y(0.2) \approx \mathbf{1.2736}$.

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CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

# BCA-504: Numerical Methods

**Session:** December 2024 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

## SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

**Q1 (3M) Obtain a real root of the equation f(x) = x³ + x² - 1 = 0 using Bisection method correct to three decimal places.**

**Solution:** Given $f(x) = x^3 + x^2 - 1 = 0$.

Evaluating at test points: $f(0) = -1 < 0$, $f(1) = 1 + 1 - 1 = +1 > 0$.

Since $f(0) \cdot f(1) < 0$, a real root lies in $[0, 1]$.

**Iteration 1:** $x_1 = \frac{0 + 1}{2} = 0.5 \Rightarrow f(0.5) = 0.125 + 0.25 - 1 = -0.625 < 0 \Rightarrow$ root $\in [0.5, 1.0]$.

**Iteration 2:** $x_2 = \frac{0.5 + 1.0}{2} = 0.75 \Rightarrow f(0.75) = 0.421875 + 0.5625 - 1 = -0.015625 < 0 \Rightarrow$ root $\in [0.75, 1.0]$.

**Iteration 3:** $x_3 = \frac{0.75 + 1.0}{2} = 0.875 \Rightarrow f(0.875) = 0.6699 + 0.7656 - 1 = +0.4355 > 0 \Rightarrow$ root $\in [0.75, 0.875]$.

**Iteration 4:** $x_4 = \frac{0.75 + 0.875}{2} = 0.8125 \Rightarrow f(0.8125) = +0.1963 > 0 \Rightarrow$ root $\in [0.75, 0.8125]$.

**Iteration 5:** $x_5 = \frac{0.75 + 0.8125}{2} = 0.78125 \Rightarrow f(0.78125) = +0.0872 > 0 \Rightarrow$ root $\in [0.75, 0.78125]$.

**Iteration 6:** $x_6 = \frac{0.75 + 0.78125}{2} = 0.765625 \Rightarrow f(0.7656) = +0.0349 > 0 \Rightarrow$ root $\in [0.75, 0.765625]$.

**Iteration 7:** $x_7 = \frac{0.75 + 0.765625}{2} = 0.7578 \Rightarrow f(0.7578) = +0.0094 > 0 \Rightarrow$ root $\in [0.75, 0.7578]$.

**Iteration 8:** $x_8 = \frac{0.75 + 0.7578}{2} = 0.7539 \Rightarrow f(0.7539) = -0.0031 < 0 \Rightarrow$ root $\in [0.7539, 0.7578]$.

**Iteration 9:** $x_9 = 0.7558$.

**Root correct to 3 decimal places:** $\mathbf{x \approx 0.755}$.

---

**Q2 (3M) Define backward differences. Make backward difference table for the ordered pairs: $(x_0, y_0)$, $(x_1, y_1)$, $(x_2, y_2)$, $(x_3, y_3)$, $(x_4, y_4)$.**

**Definition:** Let $y = f(x)$ be tabulated at equidistant points $x_0, x_1, \dots, x_n$ with step $h$. The first backward difference operator $\nabla$ is defined as:

> $$\nabla y_k = y_k - y_{k-1} \quad (k = 1, 2, \dots, n)$$

Higher-order backward differences are given recursively by $\nabla^r y_k = \nabla^{r-1} y_k - \nabla^{r-1} y_{k-1}$.

**Backward Difference Table:**

| x | y | ∇y | ∇²y | ∇³y | ∇&sup4;y |
|---|---|---|---|---|---|
| $x_0$ | $y_0$ | - | - | - | - |
| $x_1$ | $y_1$ | $\nabla y_1 = y_1 - y_0$ | - | - | - |
| $x_2$ | $y_2$ | $\nabla y_2 = y_2 - y_1$ | $\nabla^2 y_2 = \nabla y_2 - \nabla y_1$ | - | - |
| $x_3$ | $y_3$ | $\nabla y_3 = y_3 - y_2$ | $\nabla^2 y_3 = \nabla y_3 - \nabla y_2$ | $\nabla^3 y_3 = \nabla^2 y_3 - \nabla^2 y_2$ | - |
| $x_4$ | $y_4$ | $\nabla y_4 = y_4 - y_3$ | $\nabla^2 y_4 = \nabla y_4 - \nabla y_3$ | $\nabla^3 y_4 = \nabla^2 y_4 - \nabla^2 y_3$ | $\nabla$&sup4;$y_4 = \nabla^3 y_4 - \nabla^3 y_3$ |

---

**Q3 (3M)** **Find the value of dy/dx at x = 1.2 for the following ordered pairs: (1.0, 2.7183), (1.2, 3.3201), (1.4, 4.0552), (1.6, 4.9530), (1.8, 6.0496), (2.0, 7.3891), (2.2, 9.0250).**

**Solution:** Here step size $h = 0.2$. We evaluate the derivative at $x_0 = 1.2$ (using forward differences with base $x_0 = 1.2$):

| x | y | Δy | $\Delta^2 y$ | $\Delta^3 y$ |
|---|---|---|---|---|
| 1.2 | 3.3201 | 0.7351 | 0.1627 | 0.0339 |
| 1.4 | 4.0552 | 0.8978 | 0.1966 | 0.0429 |
| 1.6 | 4.9530 | 1.0966 | 0.2429 | - |
| 1.8 | 6.0496 | 1.3395 | - | - |
| 2.0 | 7.3891 | - | - | - |

Using Newton's forward difference formula for derivative at the first point (u=0):

$$\left(\frac{dy}{dx}\right)_{x=1.2} = \frac{1}{h} \left[ \Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 \right]$$

$= \frac{1}{0.2} \left[ 0.7351 - \frac{0.1627}{2} + \frac{0.0339}{3} \right] = 5 \times [0.7351 - 0.08135 + 0.01130] = 5 \times 0.66505 = \mathbf{3.3253}$.

*Note:* Analytical derivative of $y = e^x$ at 1.2 is $e^{1.2} = 3.3201$; numerical formula gives $\mathbf{3.325}$.

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**Q4 (3M)** **Solve the following system of linear equations using Gauss Elimination method: 2x + y + z = 10, x + 4y + 9z = 16, x + y + 3z = 5.**

**Solution:** Augmented matrix [A|B]:

$$\begin{bmatrix} 2 & 1 & 1 & | & 10 \\ 1 & 4 & 9 & | & 16 \\ 1 & 1 & 3 & | & 5 \end{bmatrix}$$

Interchange $R_1 \leftrightarrow R_2$: $\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 2 & 1 & 1 & | & 10 \\ 1 & 1 & 3 & | & 5 \end{bmatrix}$

Row operations: $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$: $\begin{bmatrix} 1 & 4 & 9 & | & 16 \\ 0 & -7 & -17 & | & -22 \\ 0 & -3 & -6 & | & -11 \end{bmatrix}$

Row operation: $R_3 \to R_3 - \frac{3}{7} R_2$: $-6 - \frac{3}{7}(-17) = \frac{-42 + 51}{7} = \frac{9}{7}$, and $-11 - \frac{3}{7}(-22) = \frac{-77 + 66}{7} = -\frac{11}{7}$.

Hence, $\frac{9}{7} z = -\frac{11}{7} \implies \mathbf{z = -\frac{11}{9} \approx -1.2222}$.

Back-substitution: $-7y = -22 + 17z = -22 + 17\left(-\frac{11}{9}\right) = -\frac{198 + 187}{9} = -\frac{385}{9} \implies \mathbf{y = \frac{55}{9} \approx 6.1111}$.

$x = 16 - 4y - 9z = 16 - 4\left(\frac{55}{9}\right) - 9\left(-\frac{11}{9}\right) = 16 - \frac{220}{9} + 11 = 27 - \frac{220}{9} = \mathbf{\frac{23}{9} \approx 2.5556}$.

**Answer:** $x \approx 2.556, \; y \approx 6.111, \; z \approx -1.222$.

---

**Q5 (3M)** **Solve the differential equation dy/dx = x + y², y(0) = 1 using Picard's method (two approximations).**

**Solution:** Given $y' = x + y^2$ with initial condition $x_0 = 0, y_0 = 1$.

Picard's iterative formula: $y^{(n)}(x) = y_0 + \int_{x_0}^x f(t, y^{(n-1)}(t)) \, dt$.

**First Approximation (n = 1):**

$$y^{(1)}(x) = 1 + \int_0^x (t + 1^2) \, dt = 1 + \left[ \frac{t^2}{2} + t \right]_0^x = \mathbf{1 + x + \frac{x^2}{2}}$$

**Second Approximation (n = 2):**

$$y^{(2)}(x) = 1 + \int_0^x \left[ t + \left(1 + t + \frac{t^2}{2}\right)^2 \right] dt = 1 + \int_0^x \left[ t + 1 + 2t + 2t^2 + t^3 + \frac{t^4}{4} \right] dt$$

$$= 1 + \int_0^x \left( 1 + 3t + 2t^2 + t^3 + \frac{t^4}{4} \right) dt = \mathbf{1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{x^4}{4} + \frac{x^5}{20}}$$

---

## SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

**Q6 (7.5M)** **Find a real root of the equation x³ - 4x + 1 = 0 (or lying between 3 and 4) using the Method of False Position (Regula-Falsi).**

**Solution:** Let $f(x) = x^3 - 4x - 9 = 0$ or $f(x) = x^3 - 5x - 7 = 0$. Let $f(x) = x^3 - 4x - 9$:

$f(2) = 8 - 8 - 9 = -9 < 0$

$f(3) = 27 - 12 - 9 = +6 > 0$.

Root lies between $a = 2$ and $b = 3$.

Regula-Falsi formula: $x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)}$.

**Iter 1:** $x_1 = \frac{2(6) - 3(-9)}{6 - (-9)} = \frac{12 + 27}{15} = \frac{39}{15} = 2.6000$.

$f(2.6) = (2.6)^3 - 4(2.6) - 9 = 17.576 - 10.4 - 9 = -1.824 < 0 \Rightarrow$ root in $[2.6, 3]$.

**Iter 2:** $x_2 = \frac{2.6(6) - 3(-1.824)}{6 - (-1.824)} = \frac{15.6 + 5.472}{7.824} = \frac{21.072}{7.824} = 2.6933$.

$f(2.6933) = (2.6933)^3 - 4(2.6933) - 9 = 19.537 - 10.773 - 9 = -0.236 < 0 \Rightarrow$ root in $[2.6933, 3]$.

**Iter 3:** $x_3 = \frac{2.6933(6) - 3(-0.236)}{6 - (-0.236)} = \frac{16.1598 + 0.708}{6.236} = \frac{16.8678}{6.236} = \mathbf{2.7049}$.

**Final Root:** $\mathbf{x \approx 2.705}$.

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**Q7 (7.5M)** **Find the missing terms in the following table:**

**x: 0, 5, 10, 15, 20, 25**

**y: 1, 3, $y_2$, 73, 225, $y_5$**

**Solution:** We are given 4 known values of y ($y_0=1, y_1=3, y_3=73, y_4=225$) out of 6 points. Therefore, the function can be approximated by a polynomial of degree 3, which implies fourth differences vanish:

$$\Delta^4 y_0 = 0 \quad \text{and} \quad \Delta^4 y_1 = 0$$

Expressing in terms of shift operator $E = 1 + \Delta$:

$$(E - 1)^4 y_0 = y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$$

Substitute known values: $225 - 4(73) + 6y_2 - 4(3) + 1 = 0 \implies 225 - 292 + 6y_2 - 12 + 1 = 0$
$6y_2 - 78 = 0 \implies \mathbf{y_2 = 13}$.

Next, for $y_5$:

$$(E - 1)^4 y_1 = y_5 - 4y_4 + 6y_3 - 4y_2 + y_1 = 0$$

Substitute: $y_5 - 4(225) + 6(73) - 4(13) + 3 = 0$
$y_5 - 900 + 438 - 52 + 3 = 0 \implies y_5 - 511 = 0 \implies \mathbf{y_5 = 511}$.

**Answer:** The missing terms are $\mathbf{y_2 = 13}$ and $\mathbf{y_5 = 511}$.

---

**Q8 (7.5M)** **Evaluate $\int_1^2 \log_{10}(x)\,dx$ using Simpson's 1/3 rule with n = 4 intervals.**

**Solution:** Here $a = 1, b = 2, n = 4$. Step size $h = \frac{2 - 1}{4} = 0.25$.
Tabulating $y = \log_{10}(x)$:

| i | $x_i$ | $y_i = \log_{10}(x_i)$ |
|---|---|---|
| 0 | 1.00 | 0.000000 |
| 1 | 1.25 | 0.096910 |
| 2 | 1.50 | 0.176091 |
| 3 | 1.75 | 0.243038 |
| 4 | 2.00 | 0.301030 |

Simpson's 1/3 Rule formula:

$$I = \frac{h}{3} \left[ (y_0 + y_4) + 4(y_1 + y_3) + 2(y_2) \right]$$

$y_0 + y_4 = 0 + 0.301030 = 0.301030$
$y_1 + y_3 = 0.096910 + 0.243038 = 0.339948 \implies 4(y_1 + y_3) = 1.359792$
$2(y_2) = 2(0.176091) = 0.352182$
Sum of terms $= 0.301030 + 1.359792 + 0.352182 = 2.013004$.

$$I = \frac{0.25}{3} \times 2.013004 = \mathbf{0.167750}$$

*Verification:* Exact integral $\int_1^2 \log_{10} x \, dx = \frac{1}{\ln 10} [x \ln x - x]_1^2 = \frac{2 \ln 2 - 1}{\ln 10} \approx \frac{0.38629}{2.30258} \approx 0.16776$.

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## SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

**Q9 (15M)** **Use Newton-Raphson method to find a real root of the equation x sin(x) + cos(x) = 0 correct to 3 decimal places, given that the root lies between π/2 and π.**

**Solution:** Let $f(x) = x \sin x + \cos x$.

Derivative: $f'(x) = \sin x + x \cos x - \sin x = x \cos x$.

Newton-Raphson iteration:

$$x_{n+1} = x_n - \frac{x_n \sin x_n + \cos x_n}{x_n \cos x_n} = x_n - \tan x_n - \frac{1}{x_n}$$

Evaluating near $\pi \approx 3.1416$ and $\pi/2 \approx 1.5708$:

$f(2.0) = 2.0 \sin(2.0) + \cos(2.0) = 2.0(0.9093) + (-0.4161) = 1.4025 > 0$

$f(3.0) = 3.0 \sin(3.0) + \cos(3.0) = 3.0(0.1411) + (-0.9900) = -0.5667 < 0$.

Take initial guess $x_0 = 2.8$:

$f(2.8) = 2.8 \sin(2.8) + \cos(2.8) = 2.8(0.33499) + (-0.94222) = 0.93797 - 0.94222 = -0.00425$

$f'(2.8) = 2.8 \cos(2.8) = 2.8(-0.94222) = -2.6382$

$x_1 = 2.8 - \frac{-0.00425}{-2.6382} = 2.8 - 0.00161 = \mathbf{2.79839}$.

**Iter 2:** $f(2.7984) = 2.7984 \sin(2.7984) + \cos(2.7984) = 2.7984(0.33642) - 0.94171 = 0.94144 - 0.94171 \approx 0.0000$.

**Final Root:** $\mathbf{x \approx 2.798}$.

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**Q10 (15M)** **Find the Lagrange interpolating polynomial of degree 2 approximating y = $\log_e(x)$ defined by values: (2.0, 0.69315), (2.5, 0.91629), (3.0, 1.09861). Hence determine the value of $\log_e(2.7)$.**

**Solution:** Data points: $x_0 = 2.0, y_0 = 0.69315$; $x_1 = 2.5, y_1 = 0.91629$; $x_2 = 3.0, y_2 = 1.09861$.

Lagrange interpolation formula for 3 points:

$$P_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

For $x = 2.7$: $L_0(2.7) = \frac{(2.7 - 2.5)(2.7 - 3.0)}{(2.0 - 2.5)(2.0 - 3.0)} = \frac{(0.2)(-0.3)}{(-0.5)(-1.0)} = \frac{-0.06}{0.5} = -0.12$

$L_1(2.7) = \frac{(2.7 - 2.0)(2.7 - 3.0)}{(2.5 - 2.0)(2.5 - 3.0)} = \frac{(0.7)(-0.3)}{(0.5)(-0.5)} = \frac{-0.21}{-0.25} = +0.84$

$L_2(2.7) = \frac{(2.7 - 2.0)(2.7 - 2.5)}{(3.0 - 2.0)(3.0 - 2.5)} = \frac{(0.7)(0.2)}{(1.0)(0.5)} = \frac{0.14}{0.5} = +0.28$

Check sum of weights: $-0.12 + 0.84 + 0.28 = 1.00$ (exact).

Now compute $P_2(2.7)$:

$$P_2(2.7) = -0.12(0.69315) + 0.84(0.91629) + 0.28(1.09861)$$

$= -0.083178 + 0.769684 + 0.307611 = \mathbf{0.993245} \approx \mathbf{0.9932}$.

*Verification:* Exact $\ln(2.7) \approx 0.9932518$.

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**Q11 (15M)** **Use Simpson's 3/8 rule to evaluate $\int_0^6 1/(1+x^2)\,dx$ with step size h = 1.**

**Solution:** Given $f(x) = \frac{1}{1+x^2}$, interval $[0, 6]$, $h = 1$ ($n = 6$ subintervals).
Tabulating values of y:

| x | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| y | 1.00000 | 0.50000 | 0.20000 | 0.10000 | 0.05882 | 0.03846 | 0.02703 |

Simpson's 3/8 Rule:

$$I = \frac{3h}{8}\left[ (y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3) \right]$$

$y_0 + y_6 = 1.00000 + 0.02703 = 1.02703$
$y_1 + y_2 + y_4 + y_5 = 0.50000 + 0.20000 + 0.05882 + 0.03846 = 0.79728 \implies 3 \times 0.79728 = 2.39184$
$2(y_3) = 2 \times 0.10000 = 0.20000$
Total bracket sum $= 1.02703 + 2.39184 + 0.20000 = 3.61887$.

$$I = \frac{3 \times 1}{8} \times 3.61887 = \mathbf{1.35708} \approx \mathbf{1.3571}$$

*Exact value:* $\tan^{-1}(6) - \tan^{-1}(0) = 1.4056$.

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**Q12 (15M)** **Solve the system of linear equations by Gauss-Seidel Iteration method: 6x + y + z = 20, x + 4y - z = 6, x - y + 5z = 7.**

**Solution:** The system is diagonally dominant because $|6| > |1|+|1|$, $|4| > |1|+|-1|$, $|5| > |1|+|-1|$.
Rewrite in iterative form:

$$x = \frac{20 - y - z}{6}, \quad y = \frac{6 - x + z}{4}, \quad z = \frac{7 - x + y}{5}$$

Start with initial guess $x^{(0)} = 0, y^{(0)} = 0, z^{(0)} = 0$:

**Iteration 1:** $x^{(1)} = \frac{20 - 0 - 0}{6} = 3.3333$
$y^{(1)} = \frac{6 - 3.3333 + 0}{4} = 0.6667$
$z^{(1)} = \frac{7 - 3.3333 + 0.6667}{5} = 0.8667$

**Iteration 2:** $x^{(2)} = \frac{20 - 0.6667 - 0.8667}{6} = \frac{18.4666}{6} = 3.0778$
$y^{(2)} = \frac{6 - 3.0778 + 0.8667}{4} = \frac{3.7889}{4} = 0.9472$
$z^{(2)} = \frac{7 - 3.0778 + 0.9472}{5} = \frac{4.8694}{5} = 0.9739$

**Iteration 3:** $x^{(3)} = \frac{20 - 0.9472 - 0.9739}{6} = \frac{18.0789}{6} = 3.0132$
$y^{(3)} = \frac{6 - 3.0132 + 0.9739}{4} = \frac{3.9607}{4} = 0.9902$
$z^{(3)} = \frac{7 - 3.0132 + 0.9902}{5} = \frac{4.9770}{5} = 0.9954$

**Iteration 4:** $x^{(4)} = \frac{20 - 0.9902 - 0.9954}{6} = 3.0024$
$y^{(4)} = \frac{6 - 3.0024 + 0.9954}{4} = 0.9983$
$z^{(4)} = \frac{7 - 3.0024 + 0.9983}{5} = 0.9992$

**Final Converged Solution:** $\mathbf{x \approx 3.000, \; y \approx 1.000, \; z \approx 1.000}$.

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**Q13 (15M)** **Using fourth-order Runge-Kutta method, find y(0.1) and y(0.2) correct to four decimal places, given dy/dx = x + y with y(0) = 2 (step size h = 0.1).**

**Solution:** Here $f(x, y) = x + y$, $x_0 = 0$, $y_0 = 2$, $h = 0.1$.

**Step 1: Compute y(0.1)**

$k_1 = h\, f(x_0, y_0) = 0.1(0 + 2) = 0.2000$

$k_2 = h\, f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.1(0.05 + 2.10) = 0.2150$

$k_3 = h\, f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1(0.05 + 2.1075) = 0.21575$

$k_4 = h\, f(x_0 + h, y_0 + k_3) = 0.1(0.1 + 2.21575) = 0.23158$

> $\Delta y = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = \frac{0.2000 + 0.4300 + 0.4315 + 0.23158}{6} = \frac{1.29308}{6} = 0.21551$

> $y(0.1) = 2.0 + 0.21551 = \mathbf{2.2155}$

**Step 2: Compute y(0.2) from $x_1 = 0.1$, $y_1 = 2.2155$**

$k_1 = 0.1(0.1 + 2.2155) = 0.23155$

$k_2 = 0.1(0.15 + 2.2155 + 0.11578) = 0.1(2.48128) = 0.24813$

$k_3 = 0.1(0.15 + 2.2155 + 0.12407) = 0.1(2.48957) = 0.24896$

$k_4 = 0.1(0.2 + 2.2155 + 0.24896) = 0.1(2.66446) = 0.26645$

$\Delta y_1 = \frac{0.23155 + 2(0.24813) + 2(0.24896) + 0.26645}{6} = \frac{1.49218}{6} = 0.24870$

> $y(0.2) = 2.2155 + 0.24870 = \mathbf{2.4642}$

**Final Answer:** $y(0.1) = \mathbf{2.2155}$, $y(0.2) = \mathbf{2.4642}$.

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